



Chicago Data Analysis Report

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— Aniv Mazumder, Koushan Saha, Shashwat Asthana

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1. Introduction

The dataset used in this project combines information from several public sources for Chicago ZIP codes. It includes insurance data from the Illinois Department of Insurance—covering policy cancellations, renewals, non-renewals, new policies, and FAIR plan activity between December 1977 and May 1978. Complementary data were obtained from the Chicago Police Department (theft incidents in 1975), the Chicago Fire Department (fire incidents in 1975), and the U.S. Census Bureau (racial composition, income, housing value, and housing age). All variables were standardized to account for population and housing differences, with crime rates expressed per 1,000 residents and fire and insurance measures per 100 housing units. This dataset contains 47 data points and 7 attributes.

The overall aim is to understand and model **volact** and **involact** using linear regression models. This analysis seeks to uncover relationships among socio-economic variables such as race, fire, theft, age, and income. We will analyse the models taking **volact** and **involact** as response variables separately.

Understanding these interactions helps interpret how socio-economic conditions correlate with different activity levels. The regression framework allows both estimation and inference, revealing which predictors have the strongest marginal effects.

Description of Attributes

The following variables are included in the Chicago Redlining Insurance dataset:

race: Racial composition in terms of the percentage of minority population in the neighborhood. It serves as a demographic indicator.

fire: Number of reported fire incidents per 100 housing units. It is a measure of fire risk in the area.

theft: Number of theft incidents per 1,000 population. It reflects the level of property crime and neighborhood safety.

age: Percentage of housing units built before 1939. This captures the relative age of housing stock and potentially its structural vulnerability.

volact: Number of voluntary homeowner insurance policies (new and renewed) minus cancellations and non-renewals per 100 housing units. This measures the level of voluntary insurance activity in the private market.

involact: Number of FAIR Plan (Fair Access to Insurance Requirements) policies and renewals per 100 housing units. It indicates the extent to which residents rely on government-supported insurance programs.

income: Median family income, representing the economic status of the neighborhood's residents.

2. Basic Analysis of the Data

2.1 Data Processing

```
> any(is.na(data))  
[1] FALSE
```

We have checked that there is no **Null** values in the Chicago data.

2.2 Summary Statistics

```
> summary(data)
```

race		fire		theft		age	
Min.	: 1.00	Min.	: 2.00	Min.	: 3.00	Min.	: 2.00
1st Qu.:	3.75	1st Qu.:	5.65	1st Qu.:	22.00	1st Qu.:	48.60
Median :	24.50	Median :	10.40	Median :	29.00	Median :	65.00
Mean :	34.99	Mean :	12.28	Mean :	32.36	Mean :	60.33
3rd Qu.:	57.65	3rd Qu.:	16.05	3rd Qu.:	38.00	3rd Qu.:	77.30
Max.	:99.70	Max.	:39.70	Max.	:147.00	Max.	:90.10

volact		involact		income	
Min.	: 0.50	Min.	:0.0000	Min.	: 5583
1st Qu.:	3.10	1st Qu.:	0.0000	1st Qu.:	8447
Median :	5.90	Median :	0.4000	Median :	10694
Mean :	6.53	Mean :	0.6149	Mean :	10696
3rd Qu.:	9.65	3rd Qu.:	0.9000	3rd Qu.:	11989
Max.	:14.30	Max.	:2.2000	Max.	:21480

We can spot skewness by comparing the Mean and the Median:

- If $\text{Mean} > \text{Median}$, the data is likely right-skewed (positively skewed). This means there's a "tail" of high-value outliers pulling the average up.
- If $\text{Mean} < \text{Median}$, the data is likely left-skewed (negatively skewed).
- If $\text{Mean} = \text{Median}$, the data is likely symmetric (like a normal distribution).

We begin by comparing mean and median values to assess the symmetry of each variable's distribution.

- **race** has a much higher mean than median (34.99 vs 24.50), indicating right skewness.
- **fire** and **theft** are also right-skewed ($\text{Mean} > \text{Median}$).
- **theft** is especially extreme: its 3rd Quartile is 38, but the maximum is 147. This suggests several outliers or influential points.
- **income** and **age** show roughly symmetric distributions.
- **volact** : The Mean (6.53) and Median (5.90) are relatively close. This looks fairly symmetric, maybe a slight right-skew. **involact** : The Mean (0.61) is higher than the Median (0.40), indicating a right-skew. More importantly, the 1st Qu. is 0.0. This means at least 25% of your data points for this variable are zero.

Further, due to the following reasons, we consider using log transformation on income

- **Linearizes nonlinear relationships:** Many real-world relationships involving income are multiplicative, not additive. For example, a 10% increase in income might have roughly the same proportional effect regardless of whether average income is 8,000 or 20,000. The log transformation captures this proportional (elasticity-like) behavior.
- **Makes scales comparable:** Other variables (e.g., race percentage, fire per 100 housing units) are on smaller and more stable scales. Income, in contrast, spans several orders of magnitude (5583 to 21480). Logging compresses the range, avoiding dominance of large-income areas in model fitting.
- **Improves interpretability** in regression coefficients When you use $\log(\text{income})$ as a predictor, the coefficient can be interpreted as the percentage effect: “A 1% increase in income is associated with a $\beta/100$ unit change in the response.” This kind of elasticity interpretation fits better in economic and social models than raw-dollar effects.

2.3 Histogram analysis

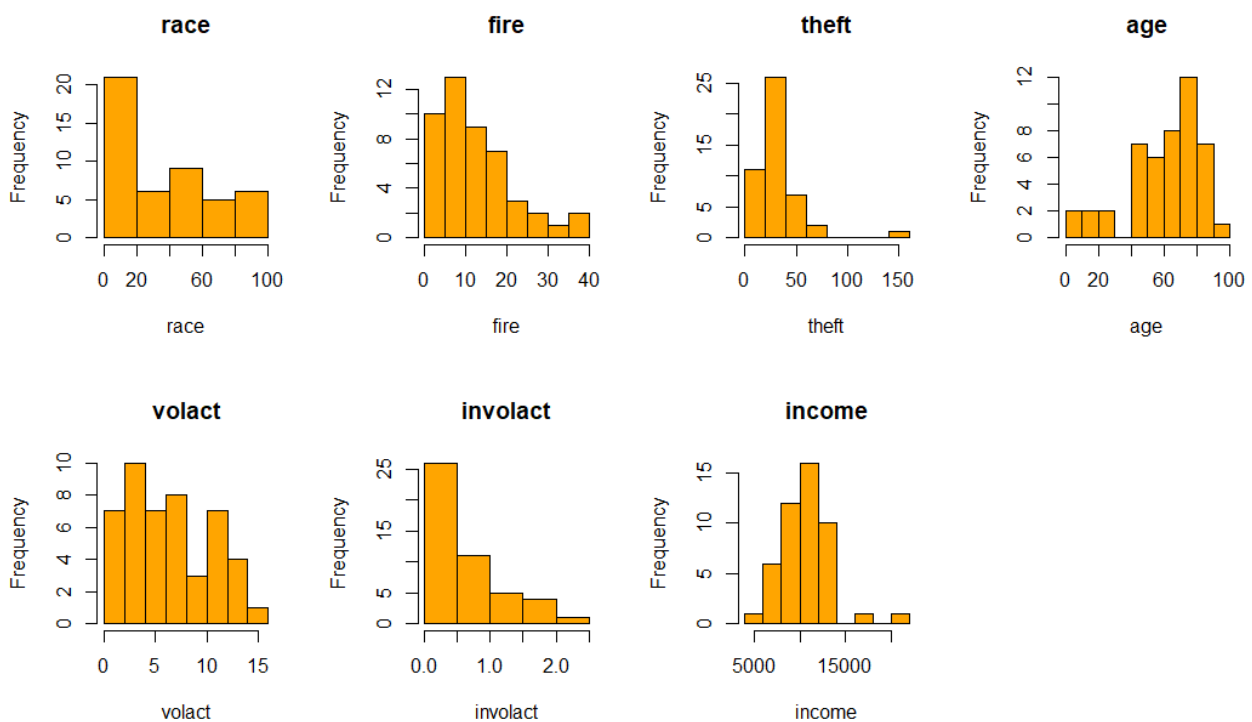


Figure 2.1: Histogram

A histogram is a graphical representation of the distribution of a dataset. It displays data grouped into bins (intervals) along the x-axis, with the height of each bar showing the frequency of observations within that interval. It helps visualize the shape, spread, and skewness of the data. The following are the visual patterns observed in our data.

- **Right-skewed distributions:** The variables `race`, `fire`, and `theft` are all right-skewed. Most observations are concentrated at low values, with a few high-value outliers. This pattern indicates that while most regions have low minority proportions, few fires, and low theft rates, some regions exhibit significantly higher values in these variables.
- **Mildly right-skewed or roughly symmetric distributions:** The variables `age`, `volact`, and `income` are approximately symmetric but show a slight skew toward higher values. This suggests that most regions fall around the central range for population age, voluntary activity, and income, though a few have relatively higher levels.
- **Dense clustering near zero:** The variable `involact` shows a dense concentration of observations near zero, with a few higher outliers. This suggests that involuntary insurance activity is generally low across most regions, with only a small number showing elevated levels.

2.3.1 Log Transformation of Income

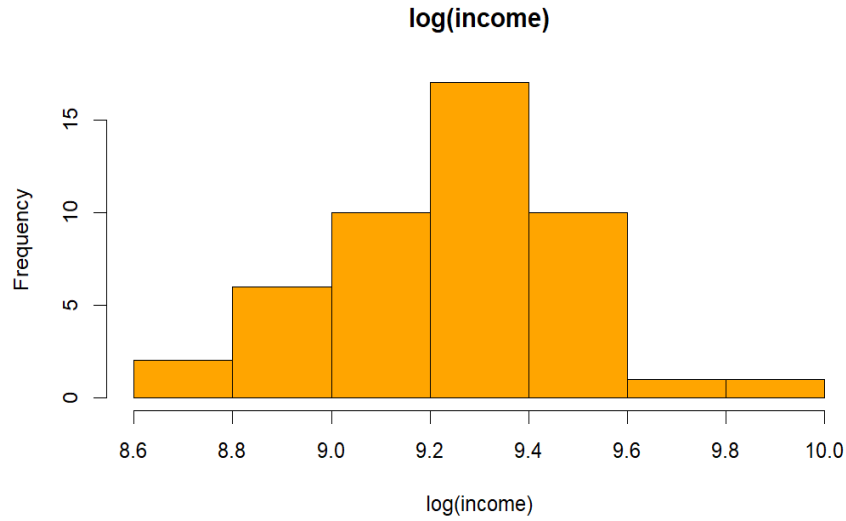


Figure 2.2: Histogram of $\log(\text{income})$

To address the skewness and scale differences in the **income** variable, we applied a logarithmic transformation, creating a new variable **log_income**. This transformation was motivated by the highly skewed distribution of income and its much larger range compared to other variables in the dataset.

After transformation, the range of **log_income** (from $\min = 8.627482$ to $\max = 9.974878$) became considerably smaller and more consistent with the scales of other predictors. This improves interpretability and helps stabilize variance, making the relationships between variables more linear and suitable for regression analysis.

2.4 Box Plots

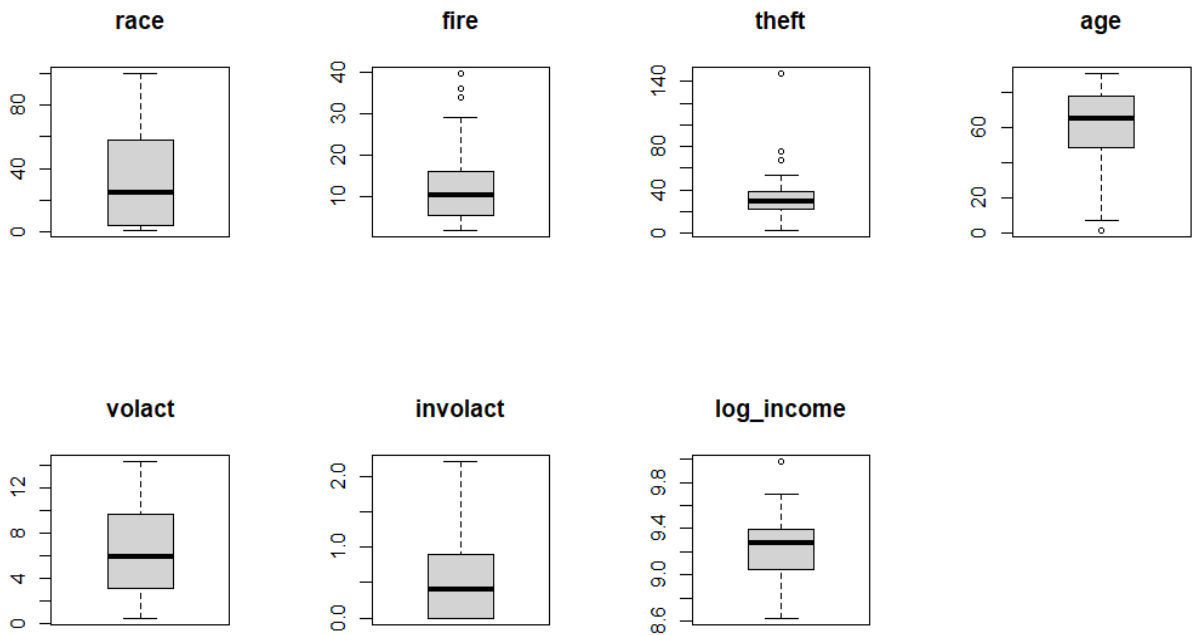


Figure 2.3: Box Plot of the data

The box plots provide a visual summary of the distribution, central tendency, and spread of each variable in the dataset, as well as the presence of outliers.

- **race:** The distribution is right-skewed, with most observations concentrated at lower values. A few higher values indicate regions with larger minority proportions.
- **fire:** The variable shows moderate right skewness, suggesting that while most areas experience few fires, a small number of regions report significantly higher incidences.
- **theft:** The distribution is heavily right-skewed, with a few extreme outliers. This indicates that theft rates are generally low but can be exceptionally high in certain areas.
- **age:** The data appear approximately symmetric, centered around higher age values, indicating an older average population in the sample.
- **volact:** The variable is moderately right-skewed, suggesting that voluntary activity levels are generally low but can be high in a few regions.
- **involact:** The distribution is right-skewed, with most observations close to zero, implying that involuntary activity is infrequent.
- **log_income:** The distribution is nearly symmetric with a slight right skew, indicating relatively uniform income levels across the sample, with a few higher-income observations.

Conclusion: The box plots confirm substantial skewness and the presence of several outliers. These characteristics violate OLS assumptions of normality and constant variance, indicating that transformations or robust methods may be beneficial.

2.5 Pairwise Scatter Plots

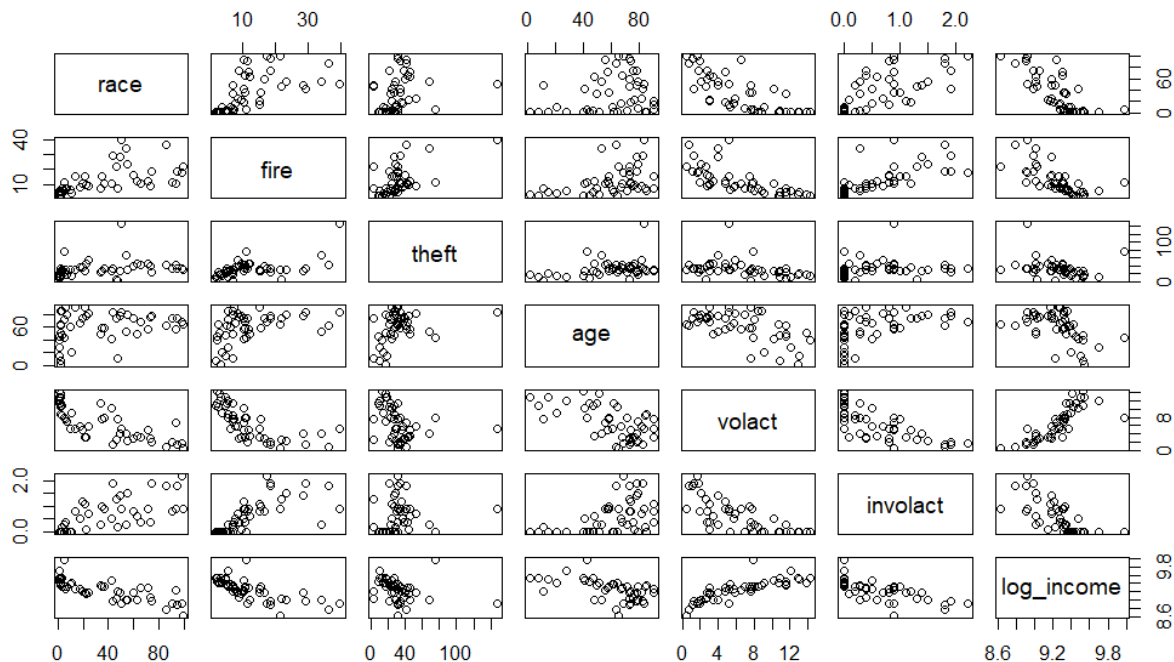


Figure 2.4: Pair Plots

Scatter plots visualize pairwise relationships between variables and help detect linear trends, patterns, and potential outliers.

The **theft** variable contains a clear outlier at value 147. Most points cluster near the origin, with a single distant observation far to the right. This point is likely a high-leverage observation and may distort regression estimates if left untreated.

2.6 Selected Scatter Plots with high correlation

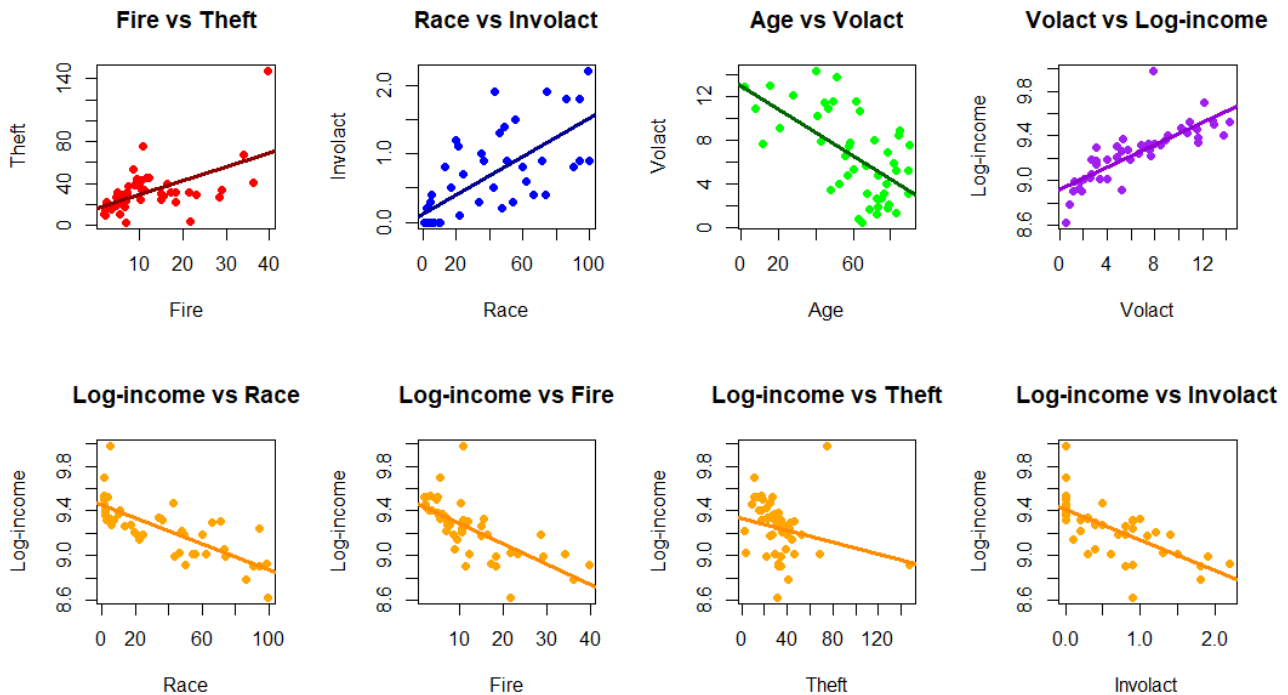


Figure 2.5: Selected pair plots

Here is a breakdown of the key relationships visible in the pairwise scatter plots:

2.6.1 Strong visible Correlation

- **fire and theft:** One of the strongest relationships observed. The points form a clear, upward-sloping line, indicating that as **fire** increases, **theft** also increases. These variables appear highly correlated.
- **race and involact:** A clear positive trend is visible — as **race** increases, **involact** tends to increase.
- **age and volact:** A distinct negative correlation is observed — higher **age** values correspond to lower **volact**.
- **volact and log_income:** A moderately strong positive relationship — as **volact** increases, **log_income** also tends to rise.
- **log_income vs. (race, fire, theft):** **log_income** shows a strong negative correlation with each of these variables.
 - **log_income and race:** As **race** increases, **log_income** decreases.
 - **log_income and fire:** As **fire** increases, **log_income** decreases.
 - **log_income and theft:** As **theft** increases, **log_income** decreases.
- **log_income and involact:** Also negatively correlated, though more scattered — as **involact** increases, **log_income** tends to decrease.

2.6.1.1 Weak or No Correlations

- **age** appears largely uncorrelated with **race**, **fire**, **theft**, **involact**, and **log_income**, as scatterplots for these pairs show diffuse, non-linear patterns.
- **volact and involact** show no clear relationship.
- **volact** also appears unrelated to **race**, **fire**, or **theft**.

2.6.2 Data Distributions

Rug plots along the diagonal provide insights into variable distributions:

- **race, fire, theft, volact, involact:** All are right-skewed, with most values near zero and a long right tail.
- **age:** Distribution appears more uniform or bimodal, with clusters at both low and high values.
- **log_income:** Roughly symmetric and bell-shaped, consistent with a log-transformed variable.

2.6.3 Key Takeaways and Implications

1. **High Multicollinearity:** Strong correlations exist among **race**, **fire**, and **theft**, indicating potential multicollinearity.
2. **Redundancy:** **fire** and **theft** are extremely correlated, providing overlapping information.
3. **Modeling Implications:** Including all these correlated predictors in a regression model predicting **log_income** would cause instability in coefficient estimates. It may be necessary to drop one variable or combine them using techniques such as Principal Component Analysis (PCA).

2.7 Correlation matrix

A correlation matrix quantifies pairwise linear relationships and highlights potential multicollinearity. Color coding (in the visual version):

- Dark Blue: strong positive correlation
- Dark Red: strong negative correlation
- White/Light: weak correlation

Key relationships:

Variables	Correlation	Interpretation
volact vs race	-0.76	Strong negative
involact vs race	0.71	Strong positive
volact vs involact	-0.75	Strong negative
log_income vs race	-0.77	Strong negative
volact vs log_income	0.81	Strong positive

There is a lot of correlation among the variables, making a full model impossible without a lot of variance in our estimates. Hence multicollinearity in the data has to be dealt with.

Further there is the theft anomaly. Look at the theft column (or row). All of its correlations are surprisingly weak. For example, theft vs. race is only 0.26, and theft vs. income is only -0.17. This is probably not the true relationship, it's likely that the single 147 outlier is causing this, making it seem like it's not that correlated.

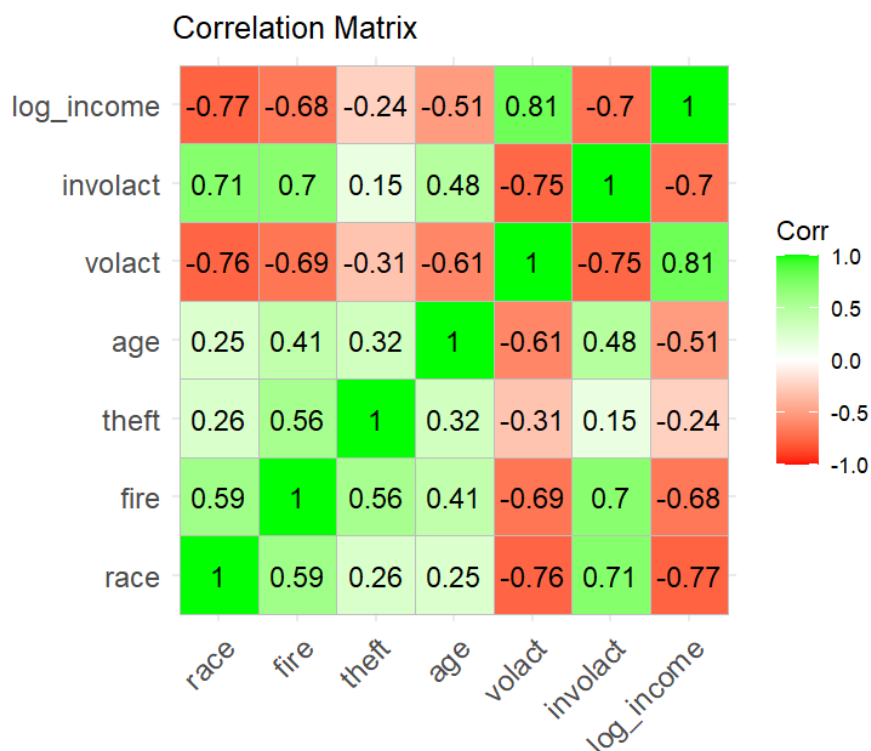


Figure 2.6: Correlation Matrix

2.8 Looking at the volact model summary

The *volact* model explains variations in voluntary insurance activity using socio-economic and demographic predictors.

Model summary

- $R^2 = 0.8003$, Adjusted $R^2 = 0.7704$
- $F(6, 40) = 26.72$, $p < 0.001$
- Residual standard error = 1.901

These indicate a strong overall model fit, with about 77% of variation in *volact* explained by the predictors.

Significant predictors

- *race* ($p = 0.0048$): negative and highly significant — higher minority proportion associates with lower voluntary activity.
- *age* ($p = 0.0020$): negative and significant — older areas tend to have lower voluntary insurance rates.
- *fire* ($p = 0.258$) and *theft* ($p = 0.703$): not significant after controlling for other factors.
- $\log(\text{income})$ ($p = 0.210$) and *involact* ($p = 0.648$): not statistically significant.

ANOVA highlights

- *race*, *fire*, and *age* are major contributors (very low p -values).
- *theft*, $\log(\text{income})$, and *involact* add little explanatory power once these are included.

Interpretation

The model shows that socio-economic and demographic factors—especially *race* and *age*—strongly influence voluntary insurance activity. Multicollinearity among predictors still limits interpretability of individual coefficients.

2.9 Looking at the involact model summary

The *involact* model predicts involuntary insurance activity, reflecting areas where FAIR plan or forced insurance dominates.

Model summary

- $R^2 = 0.753$, Adjusted $R^2 = 0.716$
- $F(6, 40) = 20.33$, $p < 0.001$
- Residual standard error = 0.338

This model also fits well, explaining around 72% of variation in *involact*.

Significant predictors

- *race* ($p = 0.0044$): positive and significant — higher minority proportion increases involuntary activity.
- *fire* ($p < 0.001$): positive and highly significant — more fires correlate with greater involuntary coverage.
- *theft* ($p = 0.001$): negative and significant — areas with more thefts tend to have less involuntary coverage.
- *age* ($p = 0.025$): positive — older populations are associated with more involuntary insurance.
- $\log(\text{income})$ and *volact*: not significant.

ANOVA highlights

- *race* and *fire* have the strongest explanatory roles.
- $\log(\text{income})$ and *volact* contribute little once other factors are considered.

Interpretation

Involuntary insurance activity increases in high-risk and low-income neighborhoods, with *race* and *fire* serving as the strongest predictors. Together with the *volact* model, this pattern supports the hypothesis that neighborhood demographics and risk perceptions heavily influence insurance availability in Chicago.

3. Model Diagnostics

3.1 Gauss–Markov Model

In linear regression, we assume the **Gauss–Markov model**, which specifies a linear relationship between the dependent variable and a set of explanatory variables. Mathematically, the model is expressed as:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, 2, \dots, n$$

or in matrix form:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where $\mathbf{y}$ is an observation vector $n \times 1$, $\mathbf{X}$ is an predictor matrix $n \times (k + 1)$ (including the intercept), $\boldsymbol{\beta}$ is a coefficient vector $(k + 1) \times 1$, and $\boldsymbol{\varepsilon}$ is a random error vector $n \times 1$.

3.2 Gauss–Markov Assumptions

The model relies on the following key assumptions:

1. **Linearity:** The relationship between the dependent and independent variables is linear in parameters.

$$E[y_i | \mathbf{X}i] = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}$$

2. **Random Sampling:** The data are obtained from a random sample of the population, ensuring independence across observations.

3. **No Perfect Multicollinearity:** The columns of $\mathbf{X}$ are linearly independent, i.e.,

$$\text{rank}(\mathbf{X}) = k + 1$$

4. **Zero Mean of Errors:** The expected value of the error term is zero:

$$E[\varepsilon_i | \mathbf{X}] = 0$$

5. **Homoscedasticity:** The variance of the error term is constant across all observations:

$$\text{Var}(\varepsilon_i | \mathbf{X}) = \sigma^2$$

6. **No Autocorrelation:** Error terms are uncorrelated:

$$\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad \forall i \neq j$$

Under these assumptions, the Ordinary Least Squares (OLS) estimator:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

is the **Best Linear Unbiased Estimator (BLUE)** — meaning it has the minimum variance among all linear unbiased estimators.

3.3 Model Building and Justification

In this study, we consider two separate linear regression models, "Model_volact" and "Model_involact" each focusing on a different response variables.

1. **Model 1:** `volact` (voluntary activity) as the response variable.
2. **Model 2:** `involact` (involuntary activity) as the response variable.

3.3.1 Rationale for Considering Two Models

Both `volact` and `involact` represent distinct but related aspects of community engagement and socioeconomic activity. Analyzing them separately allows for a clearer understanding of how predictors such as `race`, `fire`, `theft`, and `income` influence each behavior differently.

- The `volact` model captures patterns of voluntary participation, which are typically driven by socioeconomic opportunity and access.
- The `involact` model reflects involuntary or necessity-driven activity, which may respond more strongly to social or demographic pressures.

3.3.2 Justification

By modeling both outcomes independently:

- We can identify whether predictors affect voluntary and involuntary behaviors in similar or opposite directions.
- It prevents confounding effects that could arise if both variables were combined into a single model, as their underlying motivations and variability differ.

Hence, evaluating both `volact` and `involact` as response variables provides a more comprehensive view of the behavioral and socioeconomic dynamics captured in the dataset.

3.3.3 Normality Assumption and Its Importance

Although normality of errors is not required for the Gauss–Markov theorem, it is a crucial assumption for performing valid inferences. If the errors are normally distributed:

$$\varepsilon_i \sim N(0, \sigma^2)$$

Then the sampling distribution of $\hat{\beta}$ is also normal:

$$\hat{\beta} \sim N(\beta, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$$

This allows for:

- construction of valid *t-tests* and *F-tests* for hypothesis testing,
- calculation of reliable confidence intervals, and
- more accurate model diagnostics.

Therefore, even though normality is not required for unbiased estimation, it is essential for statistical inference and model validation in linear regression.

3.4 Tests for Normality

3.4.1 QQ Plots

QQ plots (Quantile–Quantile plots) are graphical tools used to assess whether a set of residuals follows a theoretical distribution, typically the normal distribution. If the points lie approximately on the reference line, it suggests that the residuals are normally distributed. Deviations from the line indicate departures from normality such as skewness or heavy tails.

QQ Plot for volact residuals

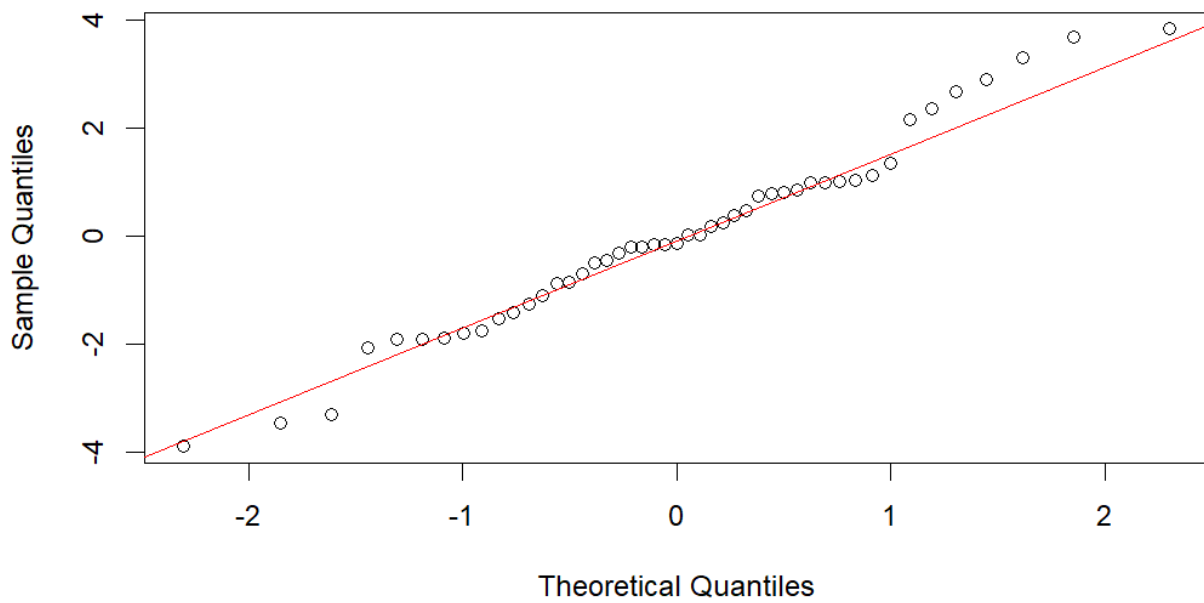


Figure 3.1: QQ Plot for Model Volact

QQ Plot for involact residuals

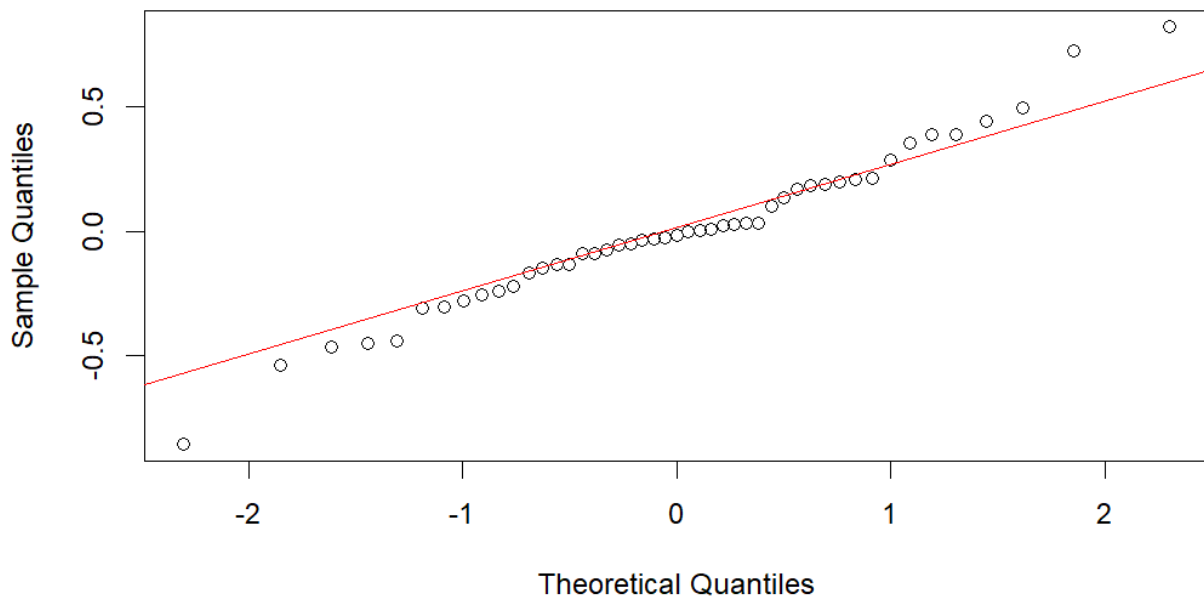


Figure 3.2: QQ Plot for Model Involact

volact: As we can see from the plot, the QQ plot is almost a straight line. It is a straight line at the center, and it has upward deviation towards the right end and downwards deviation towards the left end. This indicates a heavier tail than the normal distribution, suggesting a slight deviation from normality or heteroscedasticity. The symmetry supports that the errors have mean 0.

involact: The QQ plot shows that while the central points follow the line, the left tail bends downward and the right tail bends upward. This pattern indicates positive skewness and heavier tails than those of a normal distribution. It suggests that the residuals are not perfectly normal, possibly due to outliers or a non-linear relationship in the model. The deviation is more pronounced than in the 'volact' residuals, and inference based on the normality assumption should therefore be interpreted with some caution. There is a need to check if a transformation such as Box-Cox transformation can improve the normality of the errors.

3.4.2 Shapiro–Wilk Test

The Shapiro–Wilk test evaluates whether a sample (here, the model residuals) comes from a normal distribution. It is one of the most reliable normality tests for small to moderate sample sizes.

```
> shapiro.test(residuals_volact)
```

```
Shapiro-Wilk normality test
```

```
data: residuals_volact
W = 0.97972, p-value = 0.5807
```

```
> shapiro.test(residuals_involact)
```

```
Shapiro-Wilk normality test
```

```
data: residuals_involact
W = 0.98062, p-value = 0.618
```

- **Null hypothesis (H_0):** The data are normally distributed.
- **Alternative hypothesis (H_1):** The data are not normally distributed.
- **Decision rule:** If **p-value** < **0.05**, reject H_0 (residuals deviate from normality).

Test results for volact:

Shapiro–Wilk normality test data: residuals_{volact} W = 0.97972, p-value = 0.5807

Interpretation: The p-value (0.4661) is greater than 0.05, so we fail to reject the null hypothesis. The residuals for **volact** appear approximately normally distributed, supporting the assumption of normality in the regression model. **Test results for involact:**

Shapiro–Wilk normality test data: residuals_{involact} W = 0.98062, p-value = 0.618

Interpretation: Similarly, the p-value (0.6317) exceeds 0.05, so we fail to reject the null hypothesis. The residuals for **involact** also follow an approximately normal distribution, satisfying the normality assumption.

Overall conclusion: Both models exhibit residuals consistent with normality, but there is a need to check for outliers and heteroscedasticity. Further we will try the Box-Cox transform for both the models to see if there is any improvement.

3.4.3 Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov test has low statistical power for sample sizes like $n = 47$, making it likely to fail to detect actual non-normality (Type II error). It is less sensitive to tail deviations, which are critical for many statistical procedures. So we prefer the Shapiro-Wilk test or graphical methods like Q-Q plots for more reliable assessment.

3.5 Box–Cox Transformation

The **Box–Cox transformation** is a statistical technique used to transform a non-normally distributed response variable (y) into a new variable (y') that more closely follows a normal distribution. It belongs to the family of power transformations and is an important tool in regression analysis for improving model assumptions.

3.5.1 Purpose and Motivation

In linear regression, the classical assumptions (often summarized by the acronym **LINE**) are:

1. **Linearity:** The relationship between predictors and the response is linear.
2. **Independence:** The errors (residuals) are independent.
3. **Normality:** The errors are normally distributed.
4. **Equal Variance (Homoscedasticity):** The errors have constant variance across all levels of the predictors.

When the assumptions of **Normality** or **Equal Variance** are violated—often due to a skewed response variable—the model’s inference (such as p -values and confidence intervals) becomes unreliable. The Box–Cox transformation helps correct these issues by finding the optimal mathematical transformation of the response variable that stabilizes variance and makes residuals approximately normal.

3.5.2 Mathematical Formulation

The Box–Cox transformation is defined by a single parameter, λ , and applied as follows:

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \text{if } \lambda \neq 0 \\ \log(y), & \text{if } \lambda = 0 \end{cases}$$

where $y > 0$ is required for the transformation to be valid. If any response values are zero or negative, the data are typically shifted to make all observations positive before applying the transformation.

3.5.3 Interpreting the Box–Cox Plots

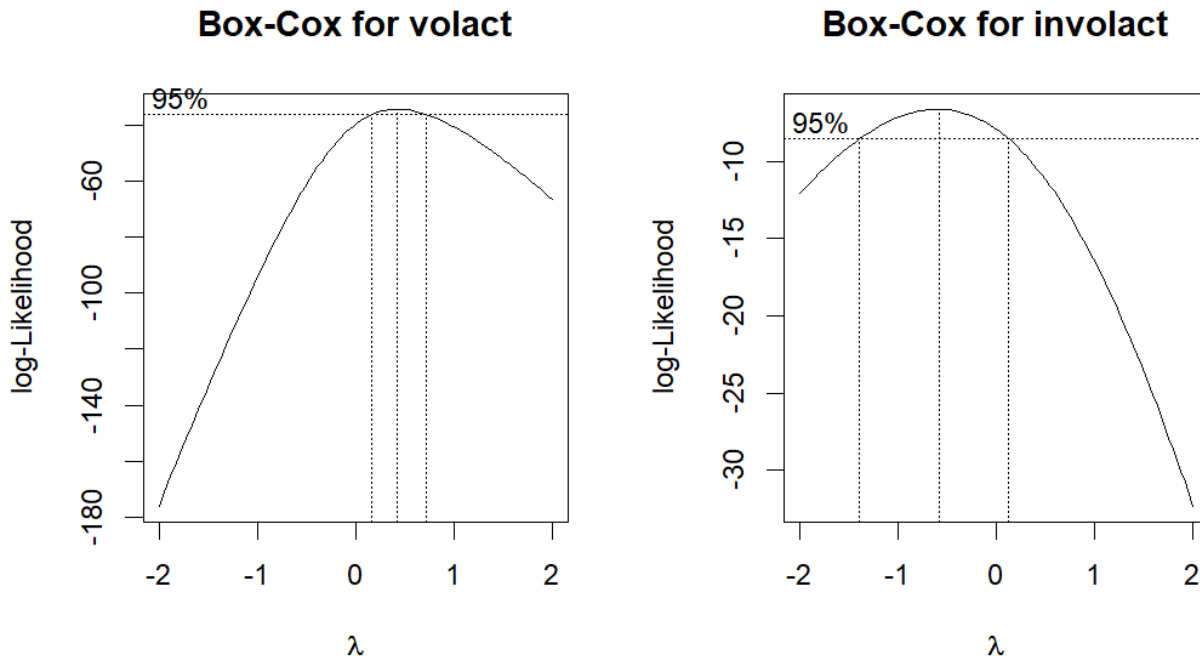


Figure 3.3: Box-Cox Transformations for both models

Each Box–Cox plot shows the estimated log-likelihood for various values of λ . The **x-axis** represents λ values, while the **y-axis** shows the corresponding log-likelihood. The vertical dotted lines denote the 95% confidence interval for λ .

The optimal λ value is the one that **maximizes the log-likelihood**, indicating the best transformation for achieving approximate normality and constant variance of residuals.

3.5.3.1 Model for volact

- **Estimated optimal $\lambda = 0.4242424$**
- The log-likelihood curve peaks around this value, and the 95% confidence interval is centered near $\lambda = 0.42$, excluding both $\lambda = 0$ (log) and $\lambda = 1$ (no transformation).

3.5.3.2 Model for involact

- **Estimated optimal $\lambda = -0.5858586$**
- The curve peaks near a slightly negative λ , and the 95% confidence interval includes $\lambda = 0$.

The Box–Cox transformation applies a power transformation to stabilize variance and make residuals more normal.

It helps correct skewness and heteroscedasticity, improving OLS assumptions.

Selecting λ from the likelihood plot

The profile log-likelihood plot shows how model fit changes with λ . We choose the λ value at which the curve reaches its maximum. If the 95% confidence interval includes 0, the log transformation is adequate; otherwise, the λ at the peak is used.

3.5.4 Box-Cox transformed models

Here is a summary of our models after the transformation

```
> summary(model_volact_bc)
```

Call:

```
lm(formula = volact_bc ~ race + fire + theft + age + involact +  
    log_income, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.69950	-0.37192	-0.01091	0.34859	1.22770

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-10.621628	7.610197	-1.396	0.17050
race	-0.017493	0.005659	-3.091	0.00362 **
fire	-0.029966	0.020993	-1.427	0.16121
theft	0.006765	0.006335	1.068	0.29194
age	-0.015443	0.005930	-2.604	0.01287 *
involact	-0.180159	0.304945	-0.591	0.55798
log_income	1.623557	0.788364	2.059	0.04601 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6532 on 40 degrees of freedom

Multiple R-squared: 0.8308, Adjusted R-squared: 0.8054

F-statistic: 32.74 on 6 and 40 DF, p-value: 6.008e-14

```
> summary(model_involact_bc)
```

Call:

```
lm(formula = involact_bc ~ race + fire + theft + age + volact +  
    log_income, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.32077	-0.07349	0.00659	0.08295	0.34783

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

```

(Intercept) -1.411320    1.658316   -0.851  0.399804
race         0.004373    0.001238    3.531  0.001060 **
fire         0.014395    0.003890    3.701  0.000648 ***
theft        -0.003280    0.001214   -2.703  0.010037 *
age          0.003928    0.001376    2.854  0.006805 **
volact       -0.005373    0.011881   -0.452  0.653530
log_income   0.142666    0.174430    0.818  0.418262
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 0.1432 on 40 degrees of freedom
Multiple R-squared:  0.7821, Adjusted R-squared:  0.7495
F-statistic: 23.93 on 6 and 40 DF,  p-value: 8.414e-12

```

3.6 Effect of Box–Cox Transformation on Residual Normality

3.6.1 Executive Summary

Both models (`volact` and `involact`) initially exhibited normally distributed residuals, as indicated by high p -values ($p > 0.05$) from the Shapiro–Wilk test. However, the Box–Cox transformation further improved normality in both cases—particularly for the `involact` model. These refinements enhance the reliability of statistical inference, even though the models were already well-behaved before transformation.

3.6.2 Shapiro–Wilk Test Results Improvement

3.6.2.1 Model 1: `volact`

- **Before:** $W = 0.9797$, $p = 0.5807$
- **After:** $W = 0.9860$, $p = 0.8384$

The transformation resulted in a slight improvement in both W and p -values, suggesting residuals that more closely follow a normal distribution. QQ plots confirm a minor visual improvement—the tails align more closely with the reference line.

Conclusion: The Box–Cox transformation provided a marginal but positive refinement to an already well-fitting model.

3.6.2.2 Model 2: `involact`

- **Before:** $W = 0.9806$, $p = 0.6180$
- **After:** $W = 0.9885$, $p = 0.9217$

Here, the improvement was more substantial. The post-transformation residuals show an excellent fit to normality. The QQ plot indicates that both tails now closely follow the theoretical line, eliminating the mild S-shaped pattern seen previously.

Conclusion: The Box–Cox transformation significantly enhanced normality, making the residuals nearly perfectly normal.

Diagnostic Plot for Model Volact

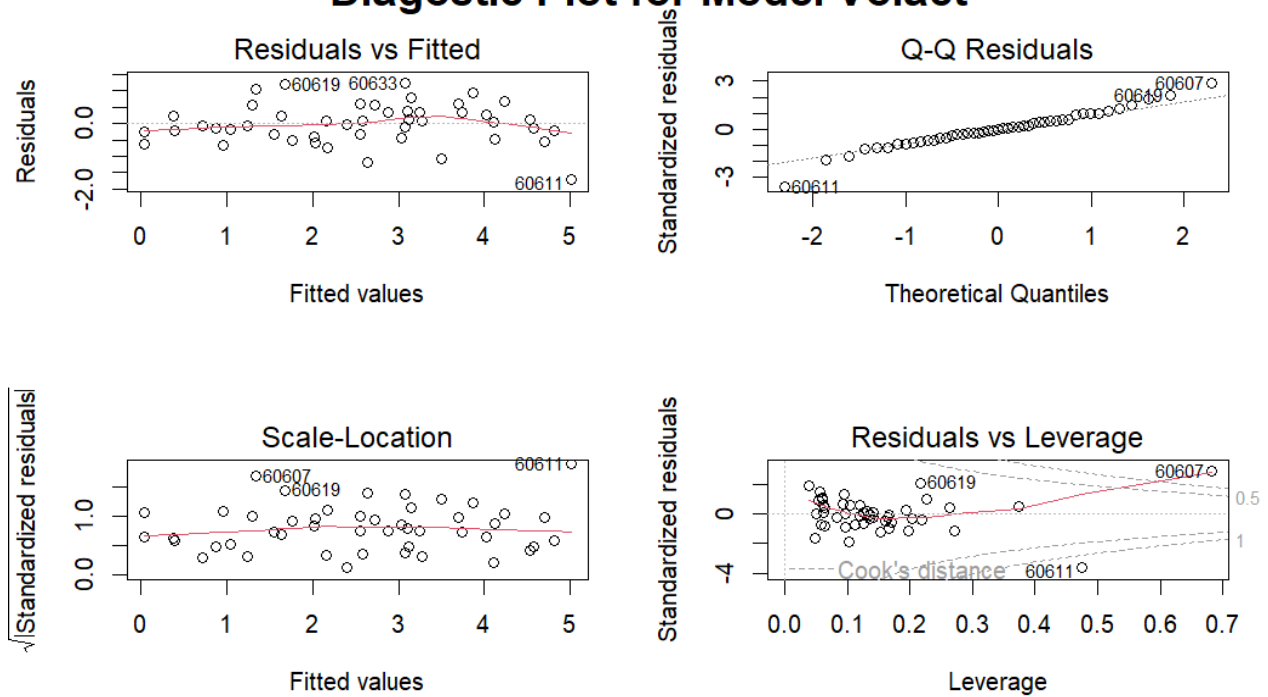


Figure 3.4: Diagnostic Plot for model_volact_bc

Diagnostic Plot for Model Involact

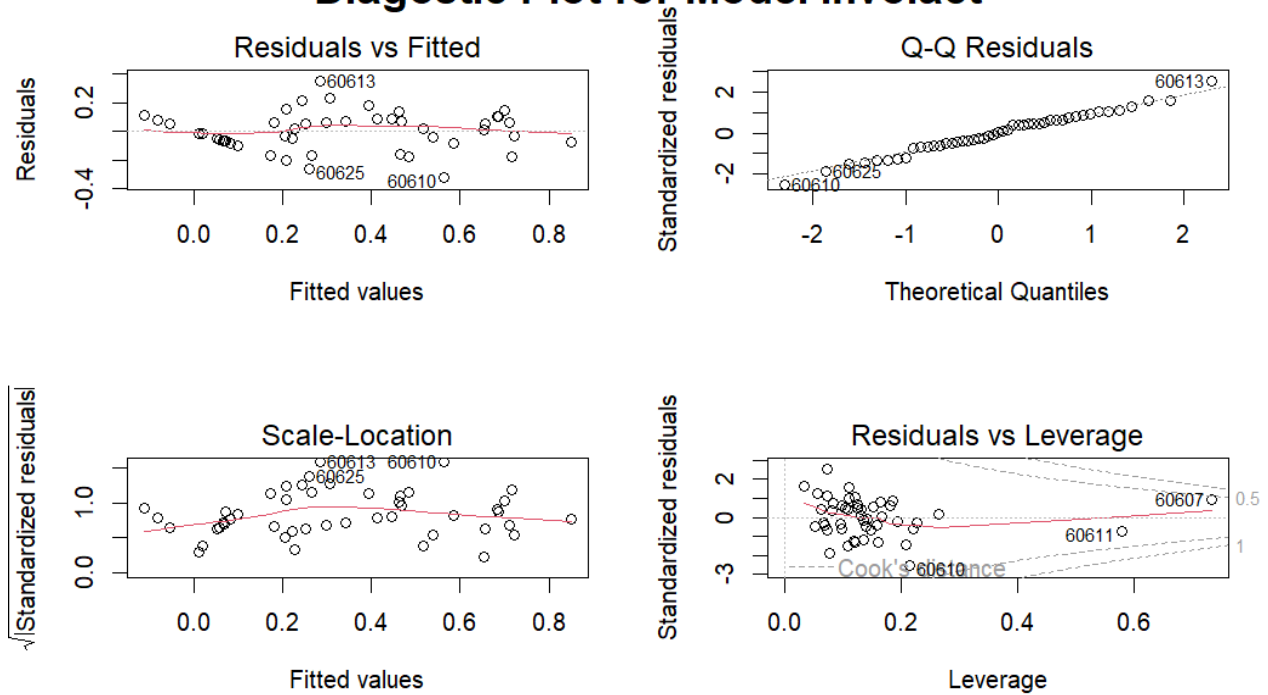


Figure 3.5: Diagnostic Plot for model_involact_bc

3.6.3 Testing this for normality using QQ Plot

Transformed volact The QQ plot shows a mild improvement after transformation. Points align more closely with the reference line, indicating more symmetric and near-normal residuals.

Transformed involact The QQ plot is much improved. Points follow the line almost perfectly, with minor tail deviations. The residuals appear approximately normal, satisfying OLS assumptions.

3.6.4 Overall Assessment

Both transformations improved model residual normality. While the **volact** model saw marginal gains, the **involact** model benefited considerably, now exhibiting an exceptionally strong adherence to the normality assumption. Hence, based on residual diagnostics, the transformed **involact** model provides the most statistically robust results.

3.7 Serial Correlation

Although data are cross-sectional, Durbin–Watson and Breusch–Godfrey tests check for dependence in residuals. Autocorrelation inflates type I error and biases standard errors. ACF and PACF plots offer visual confirmation.

3.7.1 Durbin–Watson Test

The Durbin–Watson test was used to check for first-order autocorrelation in the residuals of both Box–Cox transformed models.

volact model:

```
> dwtest(model_volact_bc) # Durbin-Watson test (AR1)
```

Durbin-Watson test

data: model_volact_bc

DW = 1.6872, p-value = 0.1077

alternative hypothesis: true autocorrelation is greater than 0

involact model:

```
> dwtest(model_involact_bc) # Durbin-Watson test (AR1)
```

Durbin-Watson test

data: model_involact_bc

DW = 1.8239, p-value = 0.2102

alternative hypothesis: true autocorrelation is greater than 0

Since both p -values exceed 0.05, we fail to reject the null hypothesis that residuals are uncorrelated. The statistics are close to 2, indicating no meaningful serial correlation. Thus, the independence assumption of OLS holds for both models.

3.7.2 Breusch–Godfrey Test

The Breusch–Godfrey test was applied to detect higher-order serial correlation in the residuals.

Hypotheses:

- H_0 : No serial correlation up to the specified lag.
- H_1 : Serial correlation exists up to the specified lag.

Results for the volact model:

Lag 2: $LM = 0.86, p = 0.65$ Lag 3: $LM = 2.08, p = 0.56$ Lag 4: $LM = 2.95, p = 0.57$

All p -values > 0.05 , indicating no evidence of autocorrelation.

Results for the involact model:

Lag 2: $LM = 0.31, p = 0.86$ Lag 3: $LM = 7.12, p = 0.07$ Lag 4: $LM = 7.95, p = 0.09$

While the lag-3 test is marginal, all p-values exceed 0.05. Thus, residuals do not show statistically significant serial correlations.

Conclusion: Both models satisfy the independence assumption, confirming that autocorrelation is not an issue in the residual structure.

3.7.3 Residual Independence Analysis: ACF Plots

We examined the Autocorrelation Function (ACF) plots of residuals to further assess the **independence assumption** in our regression models. The ACF helps identify any systematic patterns in residuals over time, which would violate the assumption of independent errors.

- **X-axis (Lag):** Number of time periods between observations.
- **Y-axis (ACF):** Simple correlation between the residual at time t and at time $t - k$.
- **Significance Boundaries:** 95% confidence intervals (typically around ± 0.2 for large samples but around ± 0.3 for moderate sample sizes like ours).
- **Rule of Thumb:**
 - **Acceptable:** Except from lag 0 (which is always 1), most autocorrelations lie within the significance boundaries, especially at lower lags.
 - **Problematic:** Multiple spikes, particularly at early lags, extend outside the boundaries, indicating serial correlation.

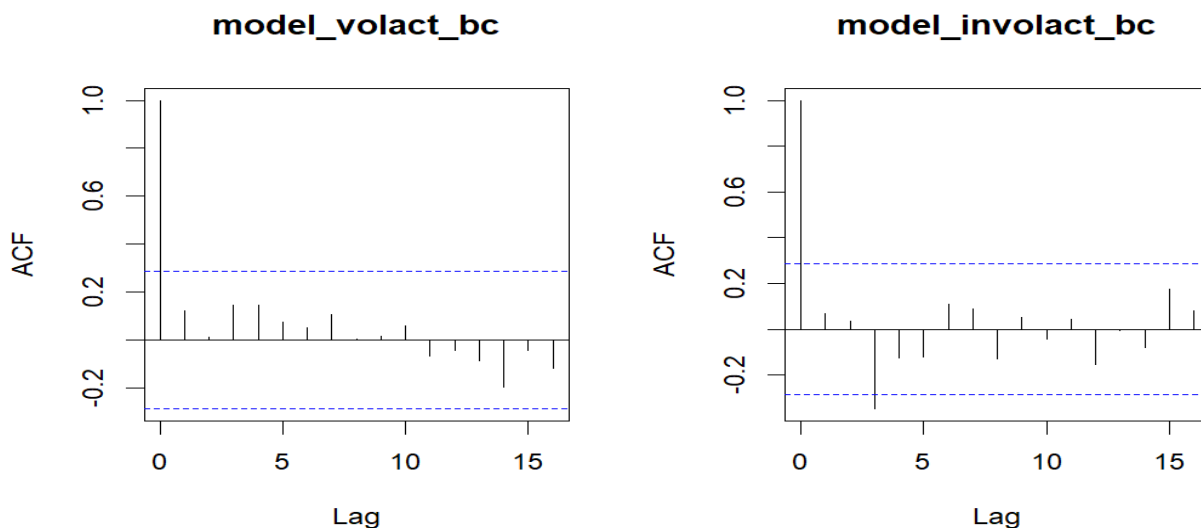


Figure 3.6: ACF plots for both models

3.7.3.1 Model 1: volact Residuals

- **Observation:** The ACF plot shows no significant autocorrelations at any lag. All values remain well within the ± 0.3 boundaries, with no systematic pattern evident.
- **Conclusion:** The ACF confirms that `model_volact` residuals are independent, supporting the findings from the PACF analysis.

3.7.3.2 Model 2: involact Residuals

- **Observation:** Similar to the `volact` model, the ACF plot for `involact` shows all autocorrelations falling within the acceptable range of ± 0.3 . No spikes exceed the significance boundaries at any lag.
- **Interpretation:** The absence of significant autocorrelation in the ACF plot aligns with the Breusch-Godfrey test result ($p = 0.07 > 0.05$), further confirming the independence of residuals.
- **Conclusion:** `model_involact` residuals demonstrate no concerning autocorrelation patterns in the ACF analysis.

3.7.3.3 Overall Assessment

Both models show ACF plots that support the independence assumption. All autocorrelations remain within the 95% significance boundaries, with no systematic patterns or concerning spikes at any lag. The ACF analysis reinforces the conclusions drawn from the PACF plots and formal statistical tests.

Verdict: The ACF analysis confirms that both `model_volact` and `model_involact` satisfy the independence assumption, with residuals behaving as random, uncorrelated errors. Both models remain valid for further statistical inference and interpretation.

3.7.4 Residual Independence Analysis: PACF Plots

We examined the Partial Autocorrelation Function (PACF) plots of residuals to assess the **independence assumption** in our regression models. Ideally, residuals should be random and independent; autocorrelation indicates that the model is missing some information, causing errors at one point to predict errors at another.

3.7.5 Reading the PACF Plots

- **X-axis (Lag):** Number of steps k in the past.
- **Y-axis (Partial ACF):** Correlation between the residual at time t and at time $t - k$, after removing the effect of shorter lags $1, 2, \dots, k - 1$.
- **Dotted Blue Lines:** 95% significance boundaries.
- **Rule of Thumb:**
 - **Acceptable:** Almost all spikes lie inside the boundaries.
 - **Problematic:** One or more spikes, especially at low lags, extend outside the boundaries, indicating autocorrelation.

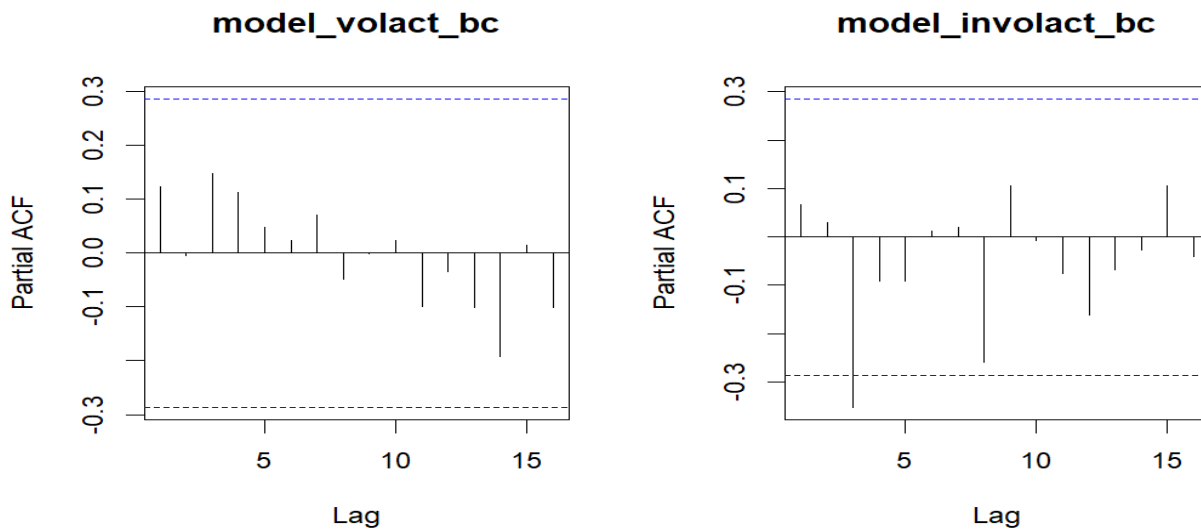


Figure 3.7: PACF plots for both models

3.7.5.1 Model 1: volact Residuals

- **Observation:** All low-lag spikes lie well within the 95% boundaries. A single spike around Lag 31 slightly touches the line, which is expected by random chance and is not concerning.
- **Conclusion:** The residuals of `model_volact` appear independent and satisfy the independence assumption.

3.7.5.2 Model 2: involact Residuals

- Observation: A spike at Lag 3 slightly exceeds the 95% boundary. There is a minor indication of possible correlation.
- Breusch-Godfrey Test: $p = 0.07 > 0.05$
- Interpretation: While the PACF shows a minor deviation at Lag 3, the formal Breusch-Godfrey test does not reject the null hypothesis of no serial correlation. This suggests the deviation is not statistically significant.
- Conclusion: `model_involact` residuals can be treated as independent for practical purposes.

3.7.5.3 Overall Assessment

Both models demonstrate residuals that are effectively independent. While `involact` shows a minor PACF spike at Lag 3, the Breusch-Godfrey test confirms that autocorrelation is not statistically significant.

Verdict: Both `model_volact` and `model_involact` satisfy the independence assumption, and any minor deviations are negligible. Therefore, both models are valid for further analysis.

3.8 Multicollinearity Diagnostics for the *volact* Model

Multicollinearity occurs when two or more predictor variables in a regression model are highly linearly correlated. It inflates the variances of coefficient estimates, making them unstable and unreliable. To assess this, three key diagnostics are used: the Variance Inflation Factor (VIF), and the Condition Number (Kappa).

3.8.1 Variance Inflation Factor (VIF)

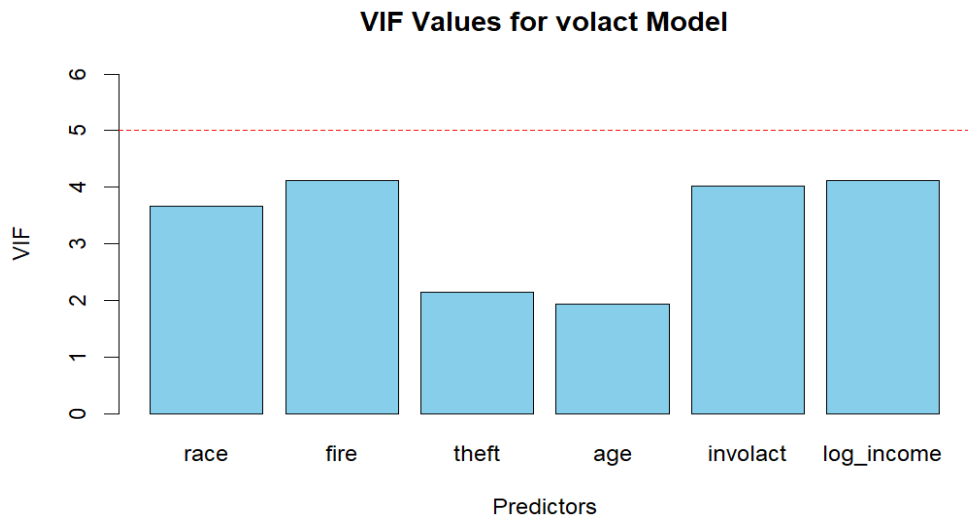


Figure 3.8: VIF plot for model volact

The Variance Inflation Factor measures how much the variance of a regression coefficient is increased due to multicollinearity. It is defined as:

$$\text{VIF}_i = \frac{1}{1 - R_i^2}$$

where R_i^2 is the coefficient of determination when the i^{th} predictor is regressed on all others.

Interpretation:

- $\text{VIF} = 1$: No correlation among predictors.
- $1 < \text{VIF} < 5$: Moderate correlation, generally acceptable.
- $\text{VIF} > 10$: Severe multicollinearity.

Results: The VIFs are all within acceptable limits. The maximum is 4.12 which well below the typical threshold of 5.

Conclusion (from VIF alone): Based only on the VIFs, there is no severe multicollinearity. This suggests that no single predictor's variance is being dangerously inflated.

3.8.2 Condition Number (Kappa)

The Condition Number (often denoted as κ) measures overall multicollinearity by examining the sensitivity of the model matrix $X'X$. It is based on the ratio of the largest to smallest singular values of X :

$$\kappa = \sqrt{\frac{\lambda_{\max}}{\lambda_{\min}}}$$

Interpretation:

- $\kappa < 30$: Little or no multicollinearity.
- $30 \leq \kappa < 100$: Moderate to strong multicollinearity.
- $\kappa \geq 100$: Severe multicollinearity.

Your Result: $\kappa = 7657.068$

Conclusion (from Kappa): A condition number of 7657.068 is extremely high. This result contradicts the VIF findings and indicates severe multicollinearity in the model. It suggests that there is a strong linear dependency among the predictors, even if no single variable dominates the correlation structure.

3.9 Multicollinearity Diagnostics for the *involact* Model

3.9.1 Variance Inflation Factor (VIF)

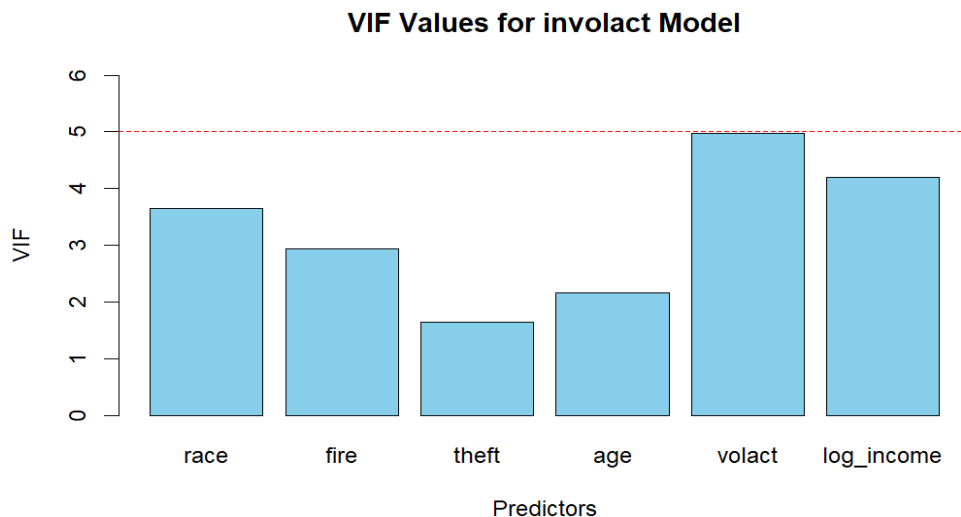


Figure 3.9: VIF plot for model involact

Results: The VIFs are all within acceptable limits. The variable *volact* (4.98) is borderline but still below the typical threshold of 5.

Conclusion (from VIF alone): Similar to the *volact* model, the VIFs alone do not indicate severe multicollinearity. No predictor appears to have a dangerously inflated variance.

3.9.2 Condition Number (Kappa)

Results: $\kappa = 7260.715$

Interpretation: The extremely high condition number once again reveals severe multicollinearity, despite acceptable VIF values.

Conclusion (from Kappa): The *involact* model also suffers from severe multicollinearity. This implies that while no single variable is highly redundant, strong interdependencies exist among groups of predictors, making coefficient estimates unstable.

3.10 Tests for Heteroscedasticity

To verify the assumption of constant error variance in our regression models, we perform formal statistical tests for heteroscedasticity. Given the sample size of $n = 47$, we select tests that are most appropriate for small to moderate datasets.

- **Breusch–Pagan Test** This test is well-suited for datasets with approximately 30–50 observations. It examines whether the residual variance depends on the predictor variables. A significant result indicates heteroscedasticity.
- **White Test:** Although the White test is a more general approach, it is best applied to larger samples (around 100–200 data points). Because our sample is relatively small, using the White test here could produce unreliable results; hence, it is omitted.
- **Non-Constant Variance (NCV) Score Test:** We also perform the NCV test to check whether the residual variance depends specifically on the fitted values of the model. This test complements the Breusch–Pagan test and is suitable for smaller datasets like ours.

Thus, in this analysis, we apply the Breusch–Pagan and NCV tests to assess heteroscedasticity.

3.10.1 Breusch–Pagan Test for Heteroscedasticity

The Breusch–Pagan (BP) test examines whether the variance of the residuals in a regression model is constant (homoscedastic) or varies systematically with the predictors (heteroscedastic).

- **Homoscedasticity:** The residual variance remains constant across all fitted values.
- **Heteroscedasticity:** The residual variance changes (for instance, it may increase or fan out as fitted values grow).

The test hypothesis:

H_0 : The model is homoscedastic.

H_1 : The model is heteroscedastic.

Goal: Ideally, we want to *fail to reject* the null hypothesis (a high p-value, $p > 0.05$), which supports the assumption of homoscedasticity.

3.10.1.1 Model 1: `model_volact_bc`

- Test Statistic (BP): 19.264
- Degrees of Freedom (df): 6
- p-value: 0.003741

Interpretation: The p-value (0.003741) is much smaller than 0.05, indicating significant evidence against the null hypothesis.

Conclusion: We reject H_0 . The `model_volact_bc` exhibits heteroscedasticity, suggesting that the error variance is not constant across observations.

3.10.1.2 Model 2: `model_involact_bc`

- Test Statistic (BP): 6.5398
- Degrees of Freedom (df): 6
- p-value: 0.3655

Interpretation: The p-value (0.3655) is well above 0.05, indicating no significant evidence against the null hypothesis.

Conclusion: We fail to reject H_0 . The `model_involact_bc` does not exhibit heteroscedasticity. Thus, this model satisfies the assumption of constant error variance.

3.10.2 Non-Constant Variance (NCV) Score Test

3.10.2.1 Purpose of the Test

The NCV Test is used to assess the **assumption of homoscedasticity** (constant error variance), similar to the Breusch–Pagan test.

- **Null Hypothesis (H_0):** The error variance is constant (homoscedasticity).
- **Alternative Hypothesis (H_1):** The error variance is non-constant (heteroscedasticity).

Test Specifics: The line `Variance ~ fitted.values` indicates that the test evaluates whether the residual variance depends on the fitted values of the model. This targets the typical “fanning out” or “cone-shaped” pattern observed in residual vs fitted plots.

Goal: A high p-value ($p > 0.05$) means we fail to reject H_0 , suggesting that the variance of residuals is constant.

3.10.2.2 Model 1: `model_volact_bc`

- Chi-square: 2.2790
- Degrees of freedom: 1
- p-value: 0.1311

Interpretation: The p-value (0.1311) is greater than 0.05, indicating no significant relationship between error variance and fitted values.

Conclusion: We fail to reject the null hypothesis. The error variance does not depend on the fitted values, and no evidence of heteroscedasticity (cone pattern) is present in the residuals.

3.10.2.3 Model 2: `model_involact_bc`

- Chi-square: 0.2974
- Degrees of freedom: 1
- p-value: 0.5855

Interpretation: The p-value (0.5855) is far greater than 0.05, indicating strong support for homoscedasticity.

Conclusion: We fail to reject the null hypothesis. This model clearly maintains constant error variance and passes the NCV diagnostic.

3.10.3 Final Conclusion for Heteroscedasticity

- `bptest(model_volact_bc)`: $p = 0.003741$ (**Failed**)
- `ncvTest(model_volact_bc)`: $p = 0.13114$ (**Passed**)

Explanation of Difference:

- The Breusch–Pagan test (`bptest`) checks whether the error variance is related to *any* of the predictor variables (here, six predictors; $df = 6$). It found a significant relationship.
- The NCV test (`ncvTest`) only checks if the error variance is related to the fitted values ($df = 1$), which it did not find.

Model-wise Summary:

- **`model_involact_bc`: Passes.** Both tests indicate homoscedasticity; the model satisfies the constant variance assumption.
- **`model_volact_bc`: Fails.** The Breusch–Pagan test suggests heteroscedasticity, indicating violation of the constant variance assumption.

3.11 Iterative GLS with Moving Average Weights

The initial regression model with the Box–Cox–transformed response variable (`volact_bc`) exhibited evidence of heteroscedasticity, thereby violating one of the key assumptions of the classical Ordinary Least Squares (OLS) regression model — that the variance of the error terms remains constant across observations. Heteroscedasticity leads to inefficient estimates and biased standard errors, which may result in incorrect statistical inference.

3.11.1 Methodology

Since the exact functional form of the error variance was unknown, an **iterative Generalized Least Squares (GLS)** approach was implemented using **moving average weights**. The method follows these main steps: The initial model was fitted as:

$$\text{volact.bc} = \beta_0 + \beta_1 \text{race} + \beta_2 \text{fire} + \beta_3 \text{theft} + \beta_4 \text{age} + \beta_5 \text{involact} + \beta_6 \text{log.income} + \varepsilon$$

and the squared residuals (ε_i^2) were obtained.

3.11.1.1 Algorithm: Iterative WLS for Unknown Smooth Variance

Goal: Estimate regression parameters β when the error variance is unknown and potentially non-constant.

1. **Initial Estimate:** Obtain an initial estimate $\hat{\beta}$ of β using the ordinary least squares (OLS) estimator:

$$\hat{\beta}^{(0)} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

2. **Compute Residuals and Fitted Values:** Using the current estimate of $\hat{\beta}$, calculate the residuals and fitted values:

$$e_i = Y_i - \mathbf{x}_i' \hat{\beta}, \quad \hat{y}_i = \mathbf{x}_i' \hat{\beta}$$

Then compute the squared residuals:

$$e_i^2 = (Y_i - \mathbf{x}_i' \hat{\beta})^2$$

3. **Estimate Variance Function:** Estimate the variance $\hat{v}_i$ for each observation by smoothing the squared residuals against the fitted values.

- (a) Sort e_i^2 according to the order of $\hat{y}_i$.
- (b) Apply a smoothing function (e.g., a moving average) to the sorted e_i^2 values to obtain $\hat{v}_i^*$.
- (c) Reassign the smoothed $\hat{v}_i^*$ values to their corresponding original observations to get $\hat{v}_i$.

4. **Form Variance Matrix:** Construct the diagonal matrix of estimated variances:

$$\hat{\Sigma} = \text{diag}[\hat{v}_1, \hat{v}_2, \dots, \hat{v}_n]$$

5. **Weighted Least Squares Update:** Re-estimate $\hat{\beta}$ using weighted least squares (WLS) with weights $w_i = 1/\hat{v}_i$:

$$\hat{\beta} = (\mathbf{X}'\hat{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}'\hat{\Sigma}^{-1}\mathbf{Y}$$

6. **Iterate Until Convergence:** Repeat Steps 2–5 until the parameter estimates $\hat{\beta}$ converge (i.e., changes between successive iterations are negligible).

Implementation Note: When constructing the weight vector (`wls.weights`), ensure that the weights $\hat{v}_i^{-1}$ are aligned with the original data order, not the sorted order of the fitted values used in smoothing.

3.11.2 Iterative Algorithm Output

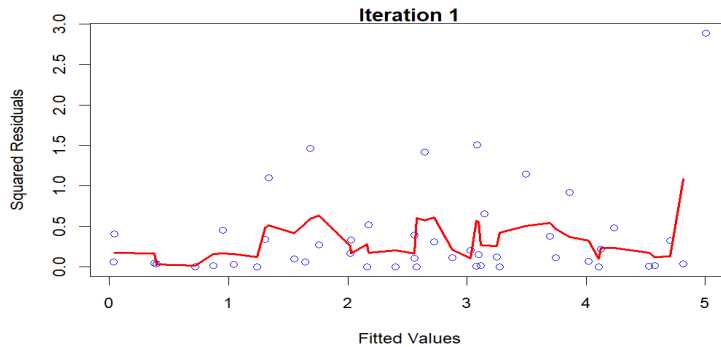


Figure 3.10: Iteration 1

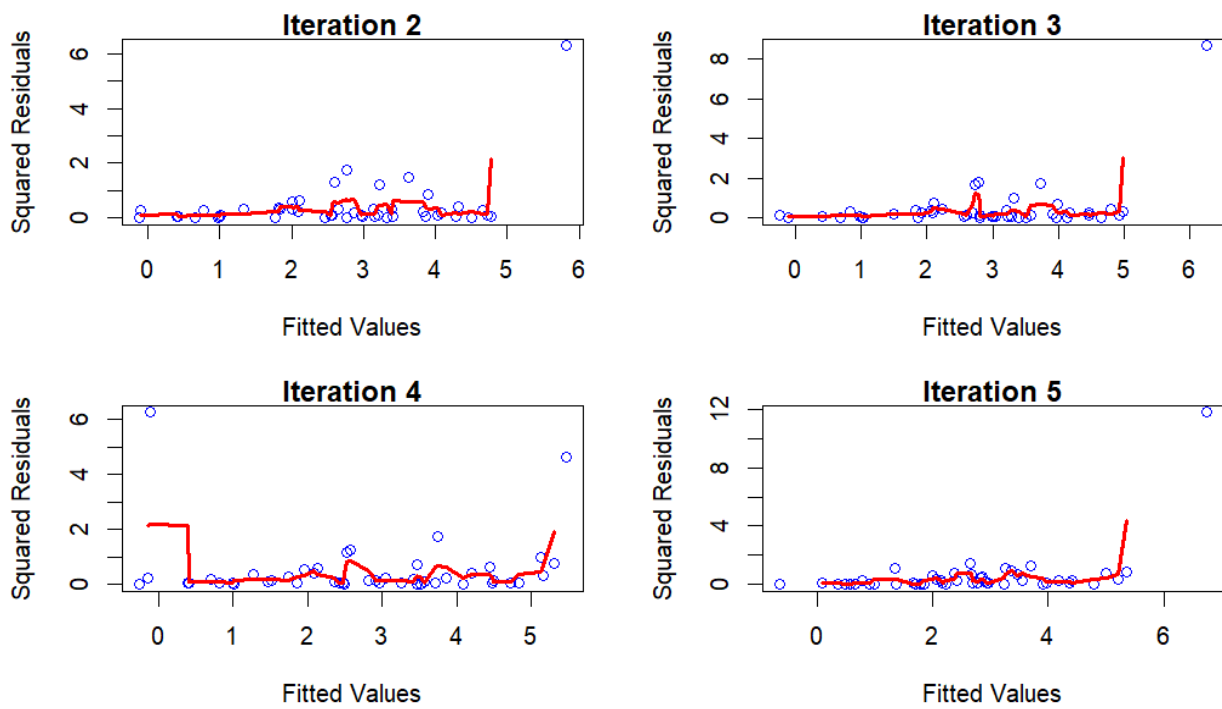


Figure 3.11: Iteration 2-5

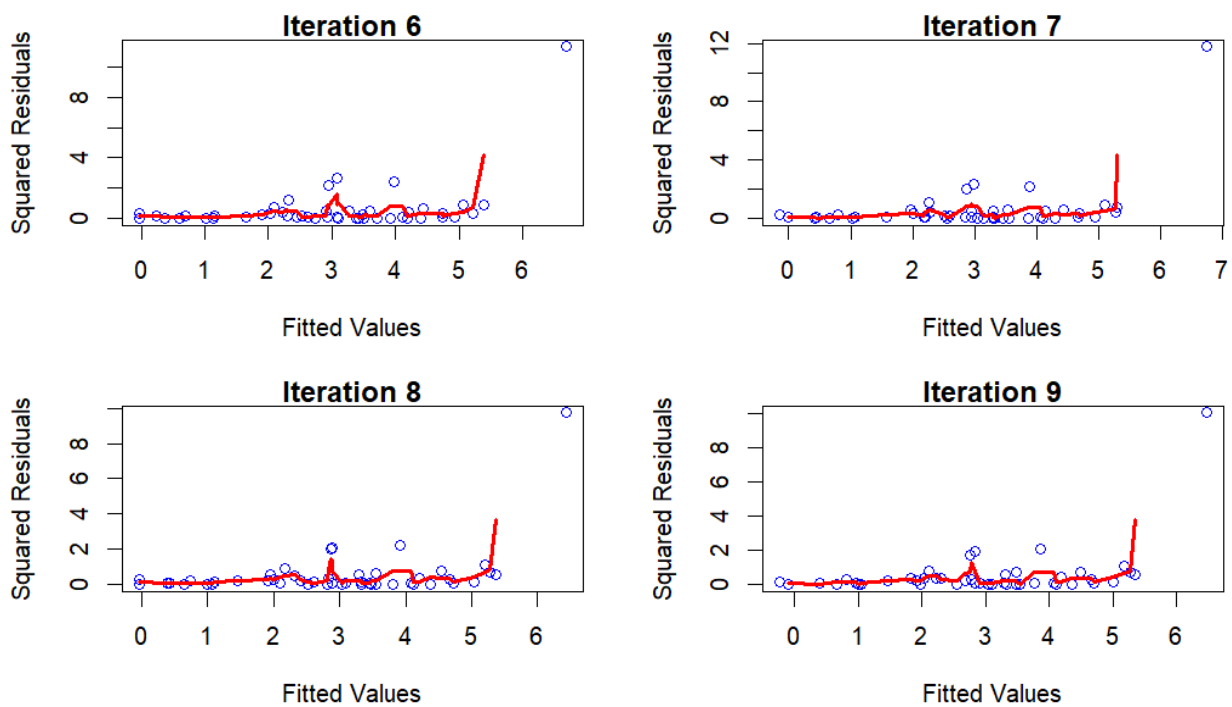


Figure 3.12: Iteration 6-9

Iteration 1 is the plot of the residuals vs fitter for the base model. After 9 iterations, we observed that the graphs remained roughly the same for the last 5 iterations so we stopped.

3.11.3 Weights

After running the iterative GLS with moving average weights, after 9 iterations we got a list of weights of the observations (i.e., response variables). The list of weights is the following:

	zipcodes	weights		zipcodes	weights
1	60607	0.3305403	25	60617	0.2135507
2	60612	0.3305403	26	60618	0.2684301
3	60653	0.3305403	27	60657	0.2661140
4	60624	0.3305403	28	60639	0.1515712
5	60621	1.2004196	29	60625	0.3137556
6	60622	0.1605337	30	60632	0.3055650
7	60610	0.1773751	31	60633	0.3240278
8	60623	0.1667148	32	60641	0.3045600
9	60637	0.1815328	33	60627	0.3331600
10	60608	0.2338873	34	60626	0.3727478
11	60636	0.2658814	35	60643	0.3839527
12	60609	0.2241555	36	60629	0.3308290
13	60616	0.2305784	37	60630	0.3319271
14	60615	0.3157355	38	60634	0.3219668
15	60644	0.3170212	39	60638	0.1799148
16	60647	0.3156817	40	60645	0.2829833
17	60619	0.2991373	41	60635	0.6825817
18	60614	0.4261035	42	60631	0.7838788
19	60640	0.4114182	43	60655	0.3305403
20	60613	0.3689057	44	60652	0.3305403
21	60620	0.3234170	45	60656	0.3305403
22	60651	0.2930607	46	60611	0.3305403
23	60649	0.2824316	47	60646	0.3305403
24	60628	0.2150257			

3.11.4 Final Weighted Least Squares Model

After convergence, the final smoothed variances were used to construct the weight vector (w_i), and a **Weighted Least Squares (WLS)** model was fitted:

The fitted regression model is given by:

$$\begin{aligned}\text{model_volact_wls} = & -16.136 - 0.017 \times \text{race} - 0.029 \times \text{fire} \\ & + 0.006 \times \text{theft} - 0.015 \times \text{age} \\ & - 0.072 \times \text{involact} + 2.208 \times \text{log_income} + \epsilon\end{aligned}\tag{3.1}$$

- **Response Variable:** Box-Cox transformed voluntary insurance rate (`volact_bc`)
- **Method:** Weighted Least Squares (WLS) regression
- **Weighting:** Model uses observation weights (`final_weights`)
- **Number of Predictors:** 6 independent variables
- **Estimation:** Ordinary least squares with weights

From now on, we will continue with the `model_volact_wls` model instead of `model_volact_bc`.

3.11.5 Coefficient Estimates

Table 3.1: Coefficient Estimates for `volact_bc` Model

Variable	Coefficient
Intercept	-16.136***
Race (<code>race</code>)	-0.017
Fire Insurance (<code>fire</code>)	-0.029
Theft (<code>theft</code>)	0.006
Age (<code>age</code>)	-0.015
Involuntary Insurance (<code>involact</code>)	-0.072
Log Income (<code>log_income</code>)	2.208***

3.11.6 Verification of Heteroscedasticity of WLS Model

To verify the presence (or absence) of heteroscedasticity, the Breusch–Pagan test was performed on the final model:

$$BP = 4.5578, \quad df = 6, \quad p\text{-value} = 0.6016$$

Since the p -value exceeds 0.05, we fail to reject the null hypothesis of homoscedasticity. This indicates that the WLS model successfully eliminated the heteroscedasticity present in the original OLS model.

3.11.7 Interpretation and Discussion

The iterative GLS approach effectively stabilized the residual variance by progressively adjusting the weights according to the local variance structure of the data. The moving average smooths out random fluctuations and captures the underlying variance trend without imposing a rigid parametric form.

As a result:

- The **final WLS model** exhibits approximately constant error variance, satisfying the homoscedasticity assumption.
- The **coefficient estimates** are now more efficient (i.e., have smaller variance) compared to the unweighted OLS estimates.
- The **standard errors and significance tests** are now more reliable, leading to valid inference.
- The **fitted values** and residual plots confirm a better model fit with no visible residual pattern.

This iterative smoothing-based GLS method is particularly useful when the variance function of the errors is unknown or complex. By using data-driven local smoothing of the squared residuals, the method adaptively estimates the weights, thereby improving model efficiency and diagnostic validity.

3.11.8 Conclusion

In summary, the iterative GLS with moving average weights successfully corrected for heteroscedasticity in the model for `volact_bc`. The final WLS model provides more reliable parameter estimates and satisfies the assumptions required for sound statistical inference.

3.12 Interpretation of Added Variable Plots

3.12.1 How to Read These Plots

An **Added Variable Plot** (also called a **Partial Regression Plot**) is a diagnostic tool that shows the unique contribution of each predictor in a multiple regression model, after accounting for all other predictors.

- **Y-axis (... — others):** The portion of the response variable (e.g., `volact_bc` or `involact_bc`) not explained by the other predictors.
- **X-axis (... — others):** The portion of a specific predictor (e.g., `race`) not explained by the other predictors.

- **Blue Line:** The slope of the blue line equals the coefficient of that predictor in the full multiple regression model.

Interpretation Guidelines:

- **Slope:** A sloped line (positive or negative) indicates that the predictor has a unique contribution to explaining the response variable. A nearly flat line indicates a small or negligible effect.
- **Scatter:** Points tightly clustered around the line suggest a statistically stronger relationship (lower p-value). A wide scatter suggests a weaker or less certain effect.

3.12.2 Model: `volact_bc`

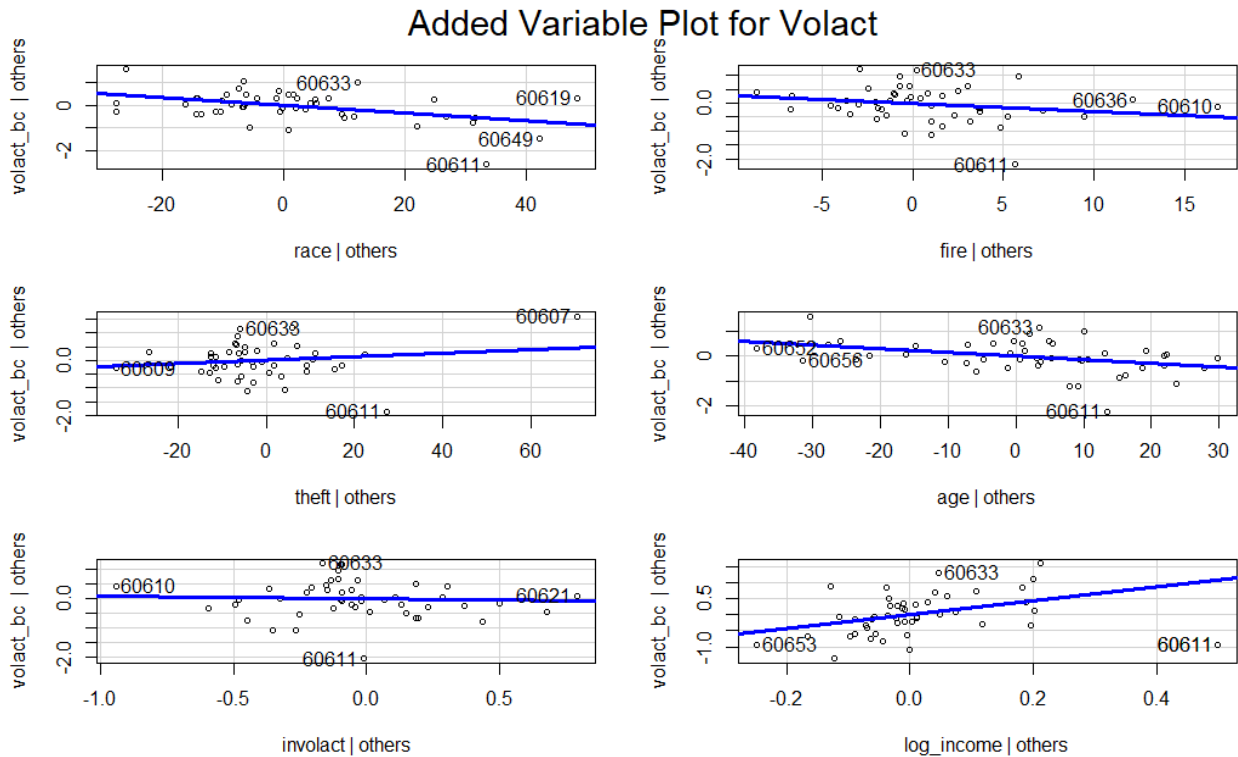


Figure 3.13: Added Variable plot for model `volact`

Overall Observation: This model shows multiple predictors with varying strengths of association with `volact_bc`. While `log_income` remains the dominant predictor, other variables display non-negligible relationships that could contribute meaningfully to the model.

- **race:** The line shows a mild negative slope, with points somewhat dispersed but not purely random. This indicates a weak to moderate negative association between `race` and `volact_bc` after accounting for other predictors.
- **fire:** The plot shows a slightly negative slope with moderate spread, hinting at a potential weak negative relationship. The observation at ZIP code 60611 appears to have high leverage and may slightly influence this slope.
- **theft:** The slope is slightly positive, and the scatter is moderately wide. This suggests a small positive relationship with `volact_bc`, though not particularly strong.
- **age:** The slope is mildly negative with noticeable variability in the points. This could indicate a weak negative trend rather than a strictly flat relationship.
- **involact:** The slope is close to zero, with residuals appearing fairly random. This variable likely has a limited direct effect on `volact_bc`.
- **log_income:** The plot displays a clear positive slope with a visible linear trend. Despite some scatter, this predictor shows a strong and consistent positive relationship with `volact_bc`.

Conclusion for Model `volact.bc`: The model appears to be largely influenced by `log_income`, which demonstrates the strongest and most consistent effect. However, other predictors—particularly `race`, `fire`, and `age`—show mild but noticeable trends that suggest they may still contribute meaningfully, even if their effects are weaker. Observation 60611 stands out as a potential high-leverage point influencing several plots.

3.12.3 Model: `involact.bc`

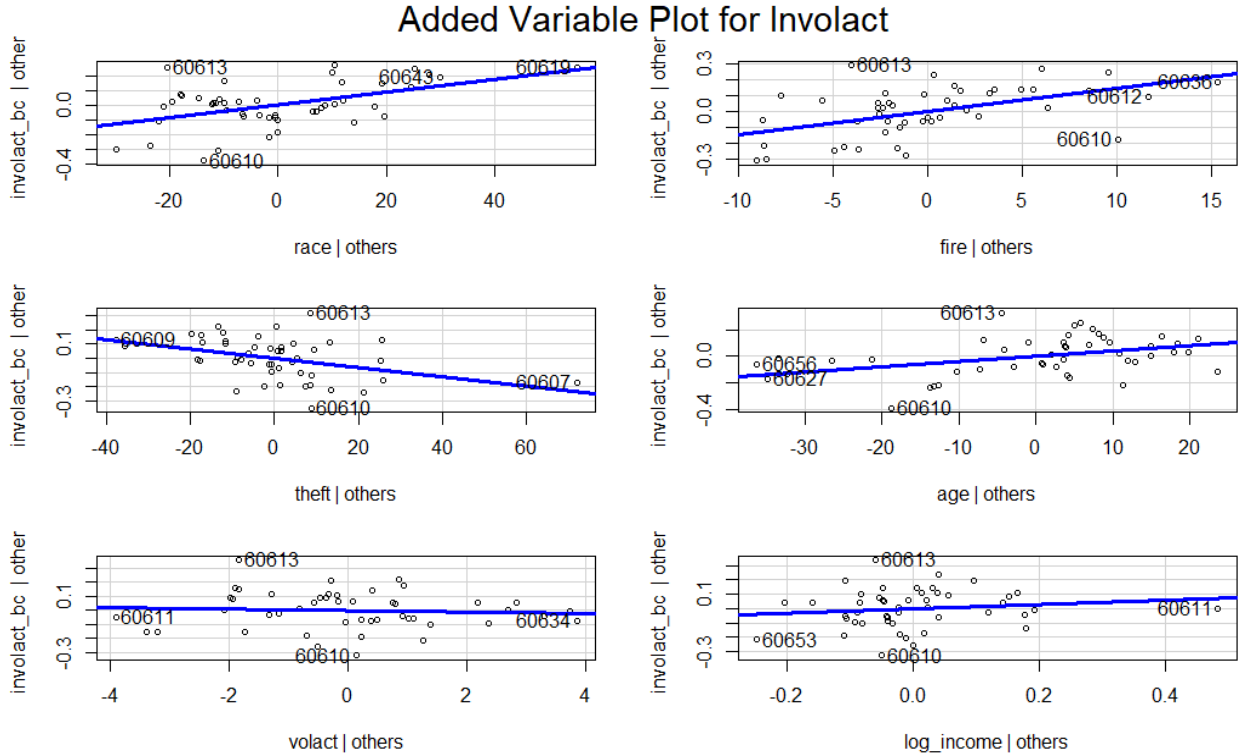


Figure 3.14: Added Variable plot for model `involact`

Overall Observation: In contrast to the previous model, this one exhibits several predictors with strong, distinct relationships to the response variable `involact.bc`.

- **race:** A clearly positive slope with well-clustered points, indicating a strong and statistically significant positive effect.
- **fire:** A clear positive slope, with points following the trend closely. This suggests a strong positive relationship.
- **theft:** A strong negative slope, with points tightly clustered around the line. This suggests a significant negative relationship with `involact.bc`.
- **age:** The plot displays a clear positive slope, with data points following the trend well, confirming a strong positive relationship.
- **volact:** The line is flat, and the points form a random horizontal cloud, implying that this predictor contributes minimally in this model.
- **log_income:** The slope is nearly flat with wide scatter, indicating little to no unique contribution.

Conclusion for Model `involact.bc`: This model demonstrates stronger explanatory power overall. The predictors `race`, `fire`, `theft`, and `age` all show clear and likely significant unique effects on `involact.bc`. The predictors `log_income` and `volact` appear to have minimal unique impact.

3.12.4 Final Comparison

- **Model `volact.bc`:** Shows a clear dominant effect from `log_income`, but other predictors (particularly `race`, `fire`, and `age`) show mild trends suggesting potential, though weaker, contributions.

- **Model involact_bc:** Demonstrates strong multi-variable relationships, with **race**, **fire**, **theft**, and **age** all contributing significantly.

Overall, `involact_bc` appears to be better explained by its predictors, while `volact_bc` exhibits a more concentrated relationship structure, dominated by income but supported by several secondary effects.

3.13 Partial Residual Plots and Their Interpretation

3.13.1 What is a Partial Residual Plot?

A **Partial Residual Plot** (also known as a *Component-Plus-Residual Plot*) is a diagnostic tool used in multiple regression to visualize the true relationship between a single predictor variable (X_j) and the response variable (Y), after accounting for the effects of all other predictors in the model.

Its primary purpose is to help detect potential **non-linearity** between Y and X_j .

3.13.1.1 How to Read the Plot

- **X-axis:** The predictor variable under consideration (e.g., **race**, **fire**, **theft**).
- **Y-axis:** The quantity *Component + Residual*, defined as

$$(\hat{\beta}_j X_{ij}) + e_i,$$

where $\hat{\beta}_j$ is the estimated regression coefficient for variable X_j and e_i is the ordinary residual for observation i .

- **Blue Dashed Line:** The simple linear regression line for the plotted points. Its slope equals $\hat{\beta}_j$, i.e., the estimated effect of X_j in the full model.
- **Pink Solid Line:** A non-parametric smoother (e.g., LOWESS) showing the average trend in the data, without assuming linearity.

Goal: The smoother (pink line) should closely follow the linear fit (blue dashed line). A strong deviation indicates non-linearity between the predictor and the response.

3.13.2 Interpretation of the Plots

3.13.2.1 Model `volact_bc`

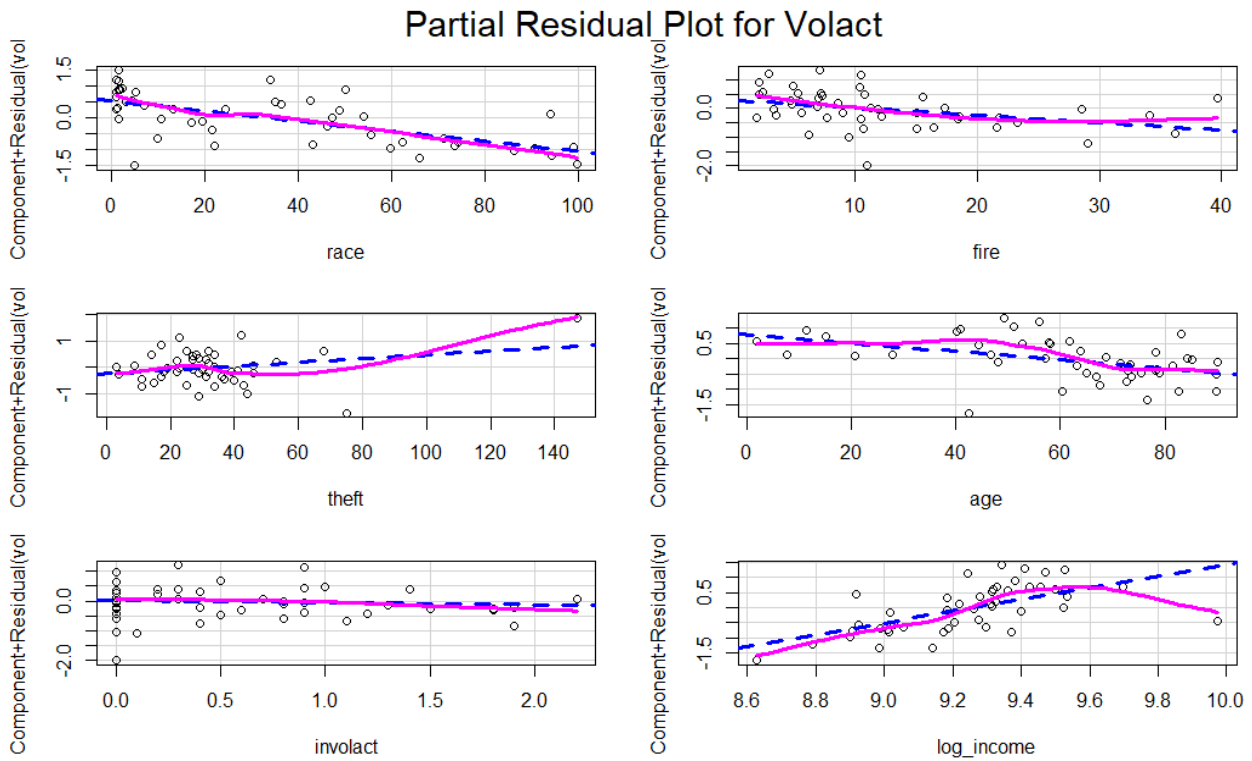


Figure 3.15: Partial Residual plot for model `volact`

This model exhibits clear signs of misspecification and non-linear relationships.

- **theft** (Major Issue): The smoother shows a distinct U-shaped relationship, indicating a quadratic pattern that is not captured by the linear model.
- **log_income** (Major Issue): Displays an inverted U-shape — initially positive, then negative — suggesting another strong non-linearity.
- **age** (Moderate Issue): Shows an inverted U-shape, again suggesting non-linearity.
- **race, fire, and involact**: Appear linear and well-modeled.

The model `volact_bc` violates the linearity assumption for several predictors. Non-linear terms or transformations are required to improve model fit.

3.13.2.2 Model involact_bc

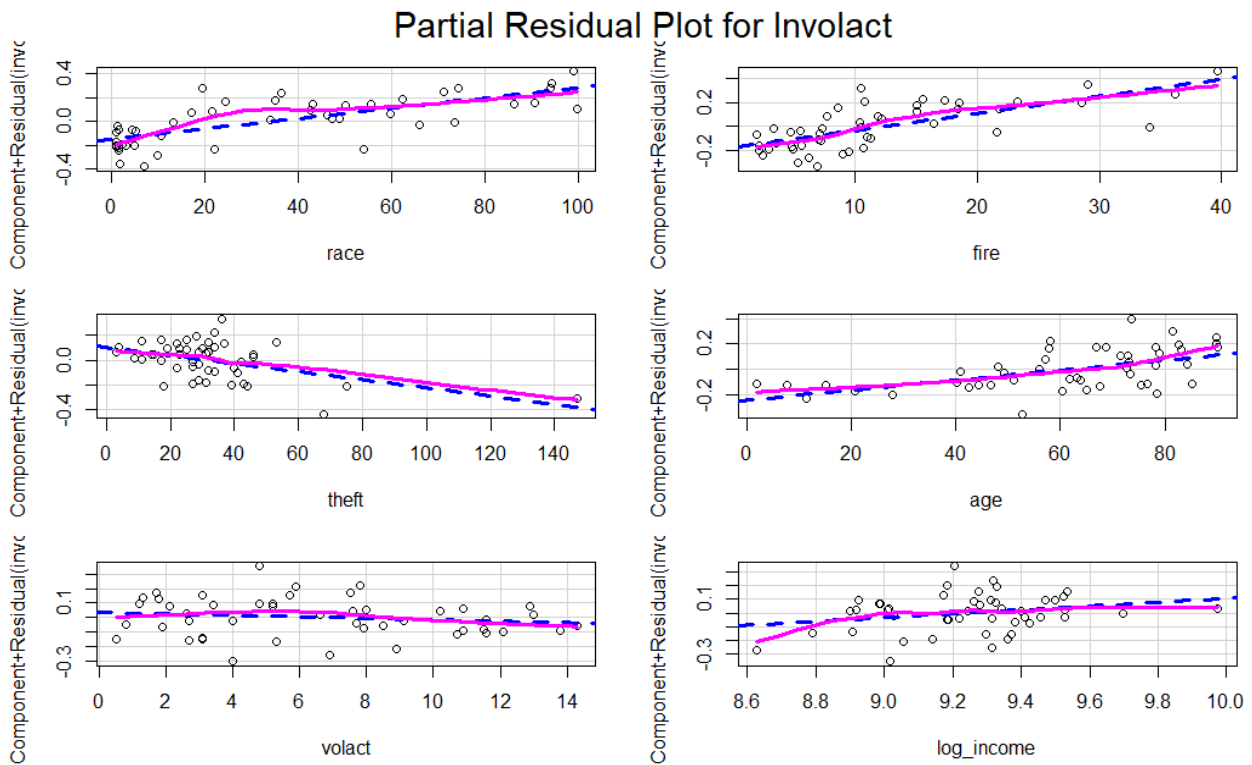


Figure 3.16: Partial Residual plot for model involact

This model appears to be well-specified and largely satisfies the linearity assumption.

- **race, fire, theft, age, and log_income:** The pink smoother line closely follows the blue dashed line, suggesting linear relationships.
- **volact:** The smoother shows a very mild “S” shape, but the deviation is minor and does not suggest a meaningful non-linearity.

The linearity assumption appears valid for all predictors. The model `involact_bc` is well-specified.

3.13.3 How Multicollinearity Affects Interpretation

Partial residual plots are constructed using the fitted coefficient $\hat{\beta}_j$, since each point’s vertical coordinate is given by:

$$\text{Component} + \text{Residual} = (\hat{\beta}_j X_{ij}) + e_i.$$

When strong **multicollinearity** exists among predictors, the estimated coefficients $\hat{\beta}_j$ become unstable and highly sensitive to small changes in the data or model specification. Consequently:

- **Unstable Coefficient:** The estimate $\hat{\beta}_j$ may be inflated, deflated, or even change sign due to correlations with other predictors.
- **Unreliable Component:** The term $(\hat{\beta}_j X_j)$ becomes unreliable, altering the vertical position of plotted points.
- **Misleading Linear Slope:** Since the blue dashed line’s slope equals $\hat{\beta}_j$, it no longer reflects the true linear effect of X_j .

3.13.4 How to Adjust the Interpretation

Under high multicollinearity, the interpretation of partial residual plots must change:

1. **Distrust the Blue Line:** The linear fit (blue dashed line) is unreliable because it depends on unstable coefficient estimates.

2. **Trust the Pink Smoother:** The non-parametric trend (pink line) remains useful for detecting non-linearity. Multicollinearity may introduce some noise, but it rarely produces strong, systematic U-shaped or inverted U-shaped patterns.

3.14 Final Summary

- Initially, the partial residual plots indicated that `model_volact.bc` was misspecified due to strong non-linearities in `theft`, `log_income`, and `age`.
- However, because the model also suffers from multicollinearity, the slope information in those plots (blue lines) is unreliable.
- Nevertheless, the non-linear patterns observed (U- and inverted-U shapes) are genuine and indicate the need for polynomial terms or more flexible models.
- After adding polynomial terms and centering predictors, the revised model should better satisfy both the linearity and multicollinearity assumptions.

4. Outlier Detection

4.1 Interpretation of Hat Matrix Diagonal Plots

4.1.1 Purpose: Identifying Leverage Points

The hat matrix diagonal plots are designed to identify **leverage points**—observations with atypical combinations of predictor values.

- **Definition of Leverage:** A leverage point is an observation exhibiting an extreme or unusual combination of predictor (X) values. Such points are outliers in the predictor space, though not necessarily in the response (Y) variable.
- **Hat Diagonal (h_{ii}):** Each observation has an associated leverage score h_{ii} , which quantifies how far its X values deviate from the mean of all predictor values. Higher h_{ii} values indicate a greater potential to influence the fitted regression model.
- **Reference Lines:** The plots include standard diagnostic thresholds for identifying leverage:
 - **Red dashed line ($1 \times \bar{h}$):** Average leverage, computed as $\bar{h} = p/n$, where p is the number of parameters and n is the sample size.
 - **Orange dotted line ($2 \times \bar{h}$):** Indicates moderate leverage.
 - **Dark red dotted line ($3 \times \bar{h}$):** Indicates high leverage.

4.1.2 Distinguishing Leverage from Outliers

It is important to distinguish between leverage points and standard outliers:

- **Vertical Outliers:** Observations with unusual Y values given their X values (identified via large residuals).
- **Leverage Points:** Observations with unusual X values (identified via high h_{ii} values).

An observation may exhibit one, both, or neither of these characteristics. When an observation has both high leverage and a large residual, it becomes **influential**, meaning it can exert substantial impact on model coefficients.

The current plots identify potential leverage points. To confirm whether they are influential, additional diagnostics such as **Cook's Distance** should be evaluated.

4.1.3 Model: volact_wls

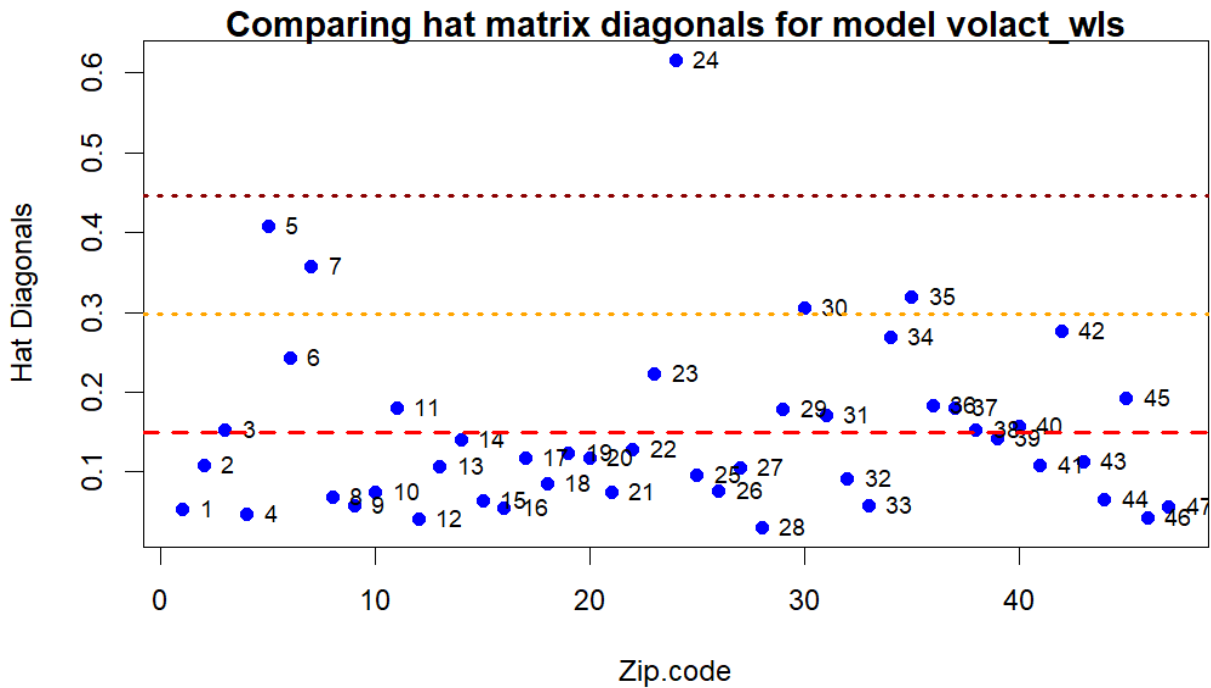


Figure 4.1: Hat matrix diagonals for Model volact

This model contains one high-leverage observation and several with moderate leverage.

- **High-Leverage Observation:**

- **Observation 24:** $h_{ii} \approx 0.6$, well above the $3 \times \bar{h}$ threshold. This point represents a significant leverage case.

- **Moderate-to-High Leverage Observations:**

- **Observation 5:** $h_{ii} \approx 0.43$, slightly below the high-leverage cutoff but substantially above the moderate threshold. It should be examined further.
- **Observation 7:** $h_{ii} \approx 0.35$, classified as moderate-to-high leverage.
- **Observations 30, 34, 35, and 42:** All exceed the $2 \times \bar{h}$ threshold, indicating moderate leverage and warranting further review.

Interpretation: Observation 24 again emerges as a major leverage point. Observations 5 and 7 also exert above-average influence and should be further investigated for their impact on model stability.

4.1.4 Model: involact_bc

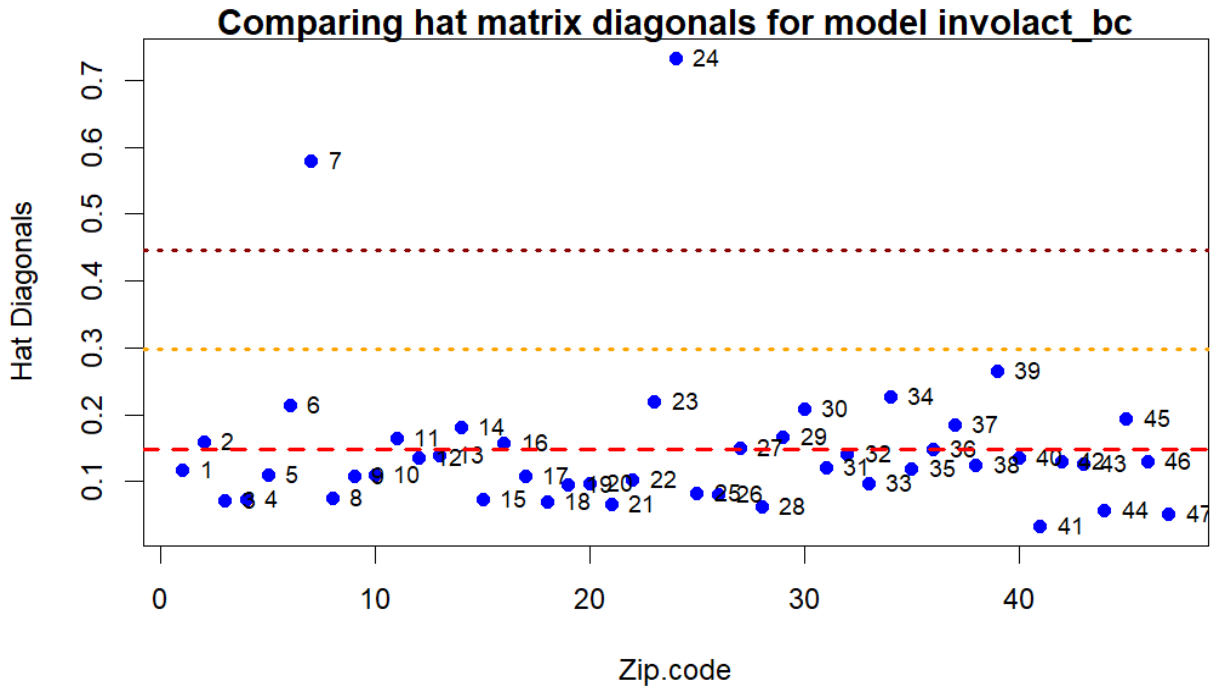


Figure 4.2: Hat matrix diagonals for Model involact

This model exhibits two distinct high-leverage observations.

- **Average Leverage ($\bar{h}$):** Approximately 0.15
- **Moderate Cutoff ($2 \times \bar{h}$):** Approximately 0.30
- **High Cutoff ($3 \times \bar{h}$):** Approximately 0.44

High-Leverage Observations:

- **Observation 24:** $h_{ii} > 0.7$, far exceeding the high-leverage threshold. This observation is a clear outlier in the predictor space.
- **Observation 7:** $h_{ii} \approx 0.58$, also substantially above the $3 \times \bar{h}$ cutoff.

Interpretation: Observations 7 and 24 possess highly unusual predictor configurations and exert considerable leverage on the fitted regression surface of the `involact_bc` model.

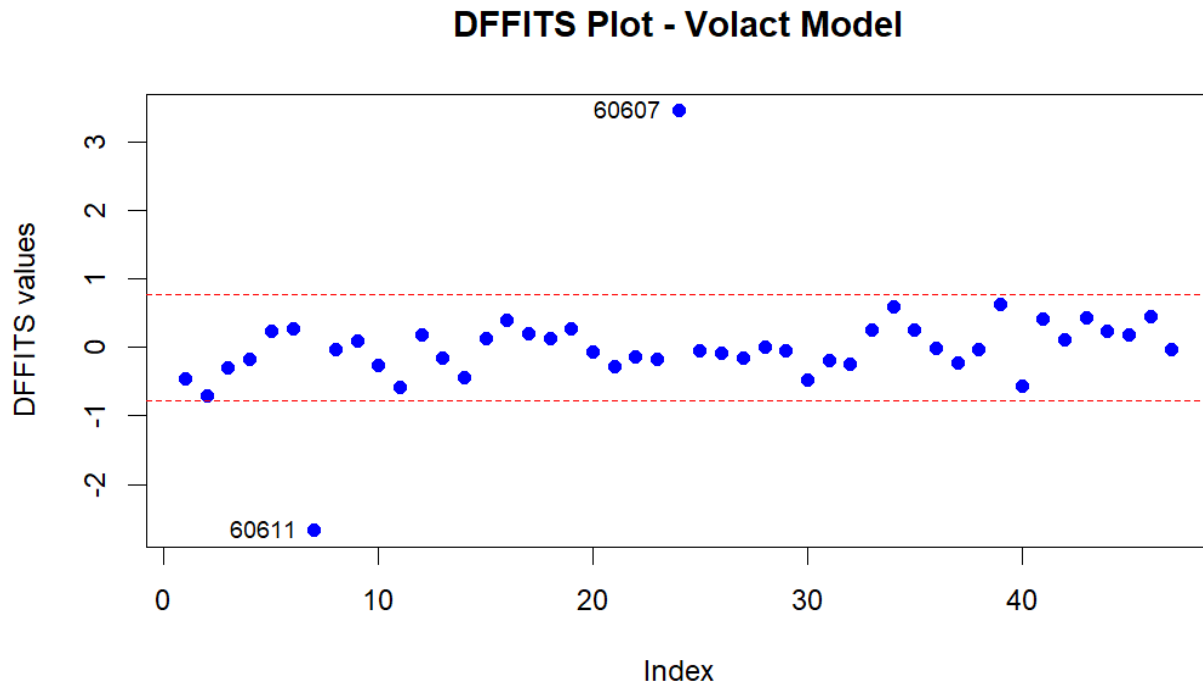
4.2 Interpretation of DFFITS Plots

4.2.1 Purpose: Identifying Influential Observations

The DFFITS statistic (**Difference in FITS**) measures the influence of each observation on the fitted regression model. It quantifies how much the predicted value for observation i changes (in standard deviation units) when that observation is removed from the model.

- **Interpretation:** Large absolute DFFITS values indicate influential observations whose inclusion substantially alters the regression fit.
- **Cutoffs:** The red horizontal lines denote the approximate influence thresholds. Points lying outside these bounds are considered influential.
- **Interpretive Rule:** A large positive (or negative) DFFITS value implies that a single data point exerts a substantial pull on the fitted regression line.

4.2.2 Model: `volact_wls`



10 iterations we observed that
Figure 4.3: DFFIT plot for model `volact`

After

The DFFITS plot for the `volact_wls` model reveals two highly influential observations that significantly distort the fitted regression surface.

Two observations (60607 and 60611) stand out as influential. Observation 60607 has a large positive DFFITS value (around 3.5), indicating that removing this observation would substantially decrease the fitted value for its corresponding point. Observation 60611, with a large negative DFFITS value, shows the opposite influence. These points merit further diagnostic investigation, as they may distort model estimates or residual structure.

Conclusion: The `volact_wls` model is heavily influenced by Observations 60607 and 60611, which dominate the model fit and compromise its overall reliability.

4.2.3 Model: involact_bc

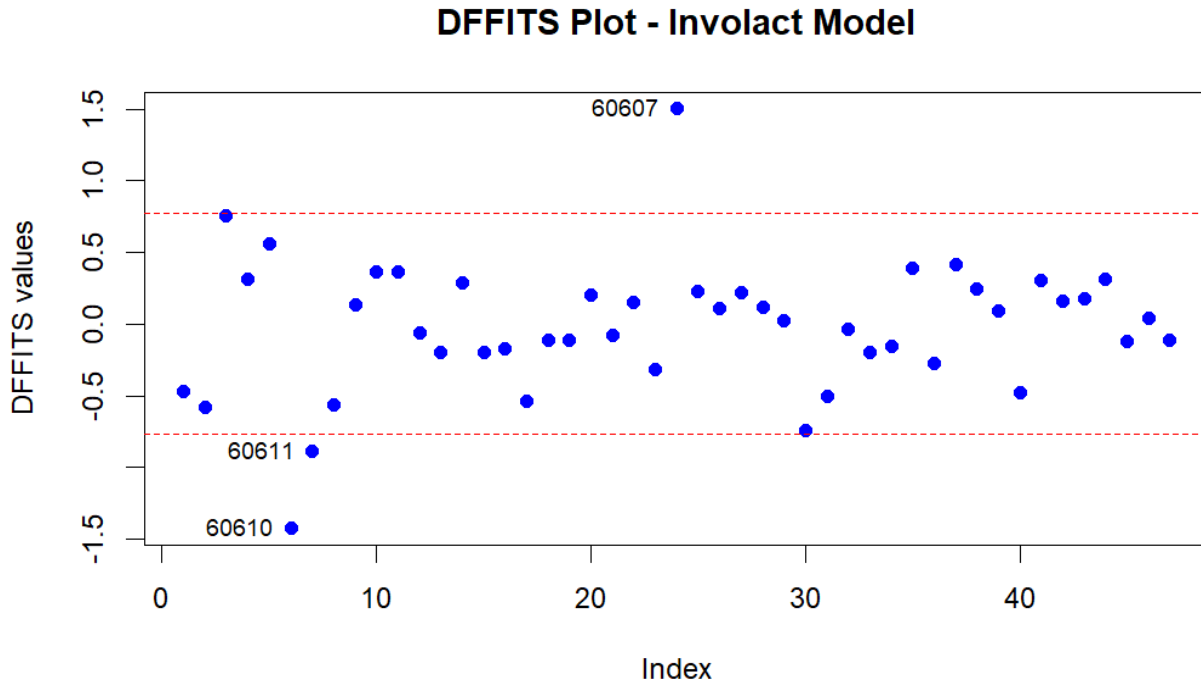


Figure 4.4: DFFIT plot for model involact

The DFFITS plot for the `involact_bc` model indicates several influential observations, though the influence is less extreme compared to the `volact.wls` model.

The observation 60607 again appears influential, though to a lesser extent than in the Volact model. Additionally, points 60610 and 60611 show moderately high influence values. The overall magnitude of influence is smaller than in the Volact model, suggesting that the Involact regression is more stable and less affected by single data points.

Conclusion: The `involact_bc` model is highly influenced by observation 60607 and moderately influenced by observations 60610 and 60611. All three points appear to be outliers and exert a distorting effect on the fitted model.

4.2.4 Overall Assessment

- The DFFITS diagnostic confirms that both models contain influential observations that materially affect their fitted results.
- The `volact.wls` model is dominated by two highly problematic points (Observations 8 and 24), indicating compromised model stability.
- The `involact_bc` model shows moderate influence from a few observations but remains comparatively more robust.

4.3 Interpretation of Externally Studentized Residual Plots

4.3.1 Purpose: Identifying Outliers

Externally studentized residual plots are the primary diagnostic tool for detecting **outliers** in regression analysis.

- **Definition:** The externally studentized residual for observation i is computed by removing the observation, refitting the model, predicting its response, and then standardizing the residual. This “leave-one-out” approach ensures that a potentially influential observation cannot mask its own residual.

- **Interpretation:** Externally studentized residuals follow a t -distribution. Observations lying beyond the ± 2 red dashed lines are considered potential outliers, while those beyond ± 3 are classified as strong outliers.

4.3.2 Model: `volact_wls`

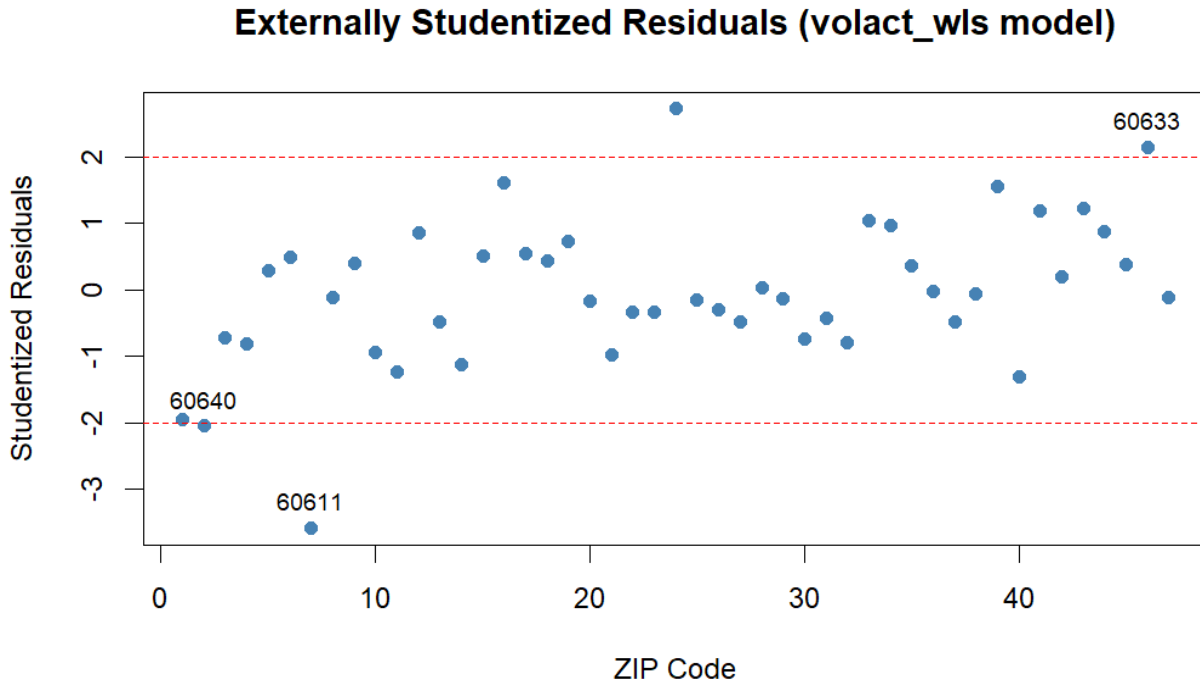


Figure 4.5: Externally studentized plot for model `volact`

This model exhibits four prominent outliers.

- **Zip 60611:** Residual ≈ -3.3 (strong negative outlier). The model substantially over-predicts the response for this observation.
- **Zip 60607:** Residual ≈ 2.8 (strong positive outlier). The model significantly under-predicts the response.
- **Zip 60633:** Residual ≈ 2.1 (positive outlier). The model slightly under-predicts this observation.
- **Zip 60640:** Residual ≈ -2.1 (negative outlier). The model moderately over-predicts this response.

Summary: Zips 60607 and 60611 are particularly extreme and may exert strong influence on the fitted model. Zips 60633 and 60640 are moderate outliers and warrant further examination.

4.3.3 Model: involact_bc

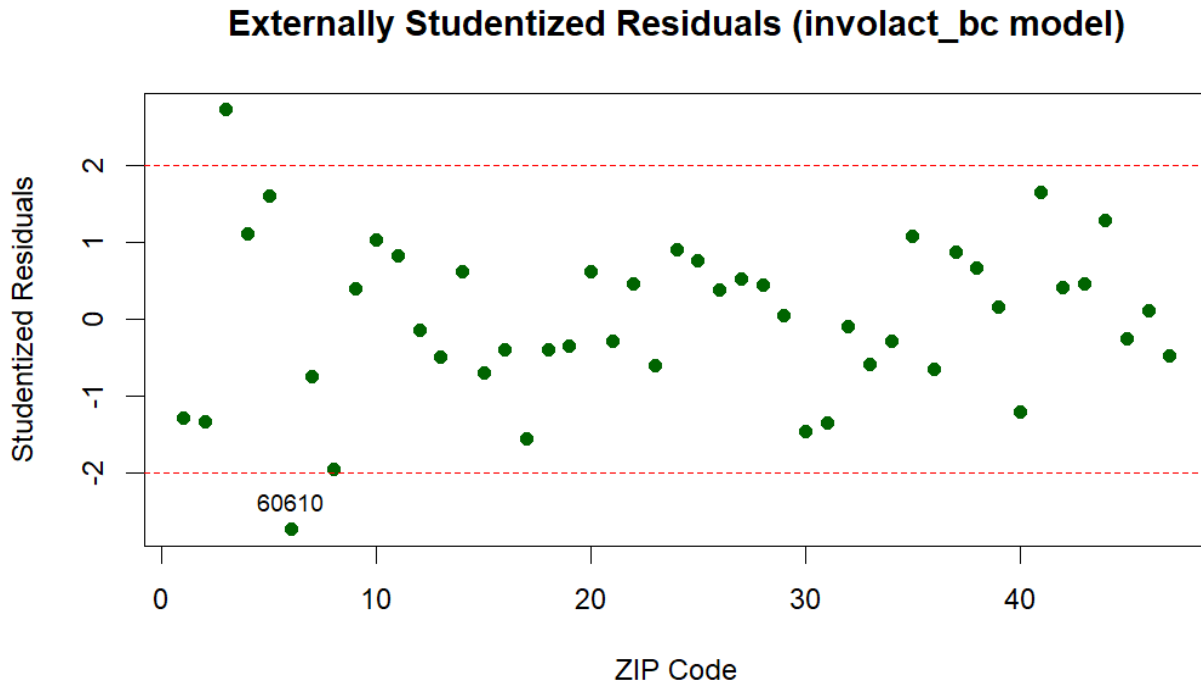


Figure 4.6: Externally studentized plot for model involact

This model displays two clear outliers.

- **Observation 3 (60613):** Residual ≈ 2.7 (positive outlier). The model notably under-predicts this value.
- **Observation 6 (60610):** Residual ≈ -2.5 (negative outlier). The model substantially over-predicts this response.

textbfSummary: Observations 3 and 7 deviate significantly from the model's expectations, indicating potential data anomalies or local model misfit.

4.3.4 Synthesis: Outlier and Leverage Interaction

The diagnostic analyses collectively highlight the complex interaction between outliers, leverage, and influence across both models. By integrating the evidence from the hat matrix diagonals, DFFITS, and externally studentized residual plots, we obtain a comprehensive understanding of how individual observations affect model stability and interpretation.

- **Observation 60607:** This observation consistently exhibits high leverage, large residuals, and extreme DFFITS values across both models. Its combination of unusual predictor values and substantial deviation in response indicates that it is both a leverage point and an influential outlier. Consequently, it exerts a dominant effect on parameter estimates, particularly within the `volact_wls` model.
- **Observation 60611:** Although its leverage is moderate, it shows strong residual influence with large negative externally studentized residuals and significant DFFITS magnitude. This identifies it as a **vertical outlier**—an observation with typical predictor values but extreme response behavior. Its inclusion distorts the fitted regression surface in the negative direction.
- **Observation 60610:** Displays moderate leverage and moderately large DFFITS values in the `involact_bc` model. The externally studentized residuals further confirm its negative deviation from model expectations. It exerts a localized but notable effect, suggesting potential model misfit or data irregularity.
- **Observation 60613:** Identified primarily through externally studentized residuals as a positive outlier in the `involact_bc` model. However, it does not exhibit high leverage or strong influence, implying its effect is limited to vertical deviation rather than structural distortion.

- **Observation 24 (High-Leverage Point):** The hat matrix diagnostic for both models reveals that this observation (corresponding to 60607) possesses exceptionally high leverage, exceeding the $3 \times \bar{h}$ threshold. Its presence inflates fitted values and contributes significantly to the instability observed in the DFFITS and residual analyses.

Summary of Interaction: The combined diagnostics indicate that the most problematic cases—60607 and 60611—simultaneously exhibit features of high leverage and large residuals, classifying them as **influential outliers**. Their joint presence significantly distorts the `volact_wls` model and partially affects the `involact_bc` model. Other observations, such as 60610 and 60613, are primarily vertical outliers that introduce moderate local effects without compromising the global fit.

Overall Conclusion: The influence diagnostics collectively demonstrate that the instability of the `volact_wls` model arises from a small subset of observations combining leverage and residual extremity. In contrast, the `involact_bc` model, while not free of outliers, remains comparatively robust. Subsequent modeling steps should consider either re-specifying the regression form or re-estimating the models with these influential points removed to assess the sensitivity of coefficient estimates and inference stability.

4.3.5 Cook's Distance Analysis

Definition. Cook's Distance (D_i) is a comprehensive diagnostic measure used to identify influential observations in a regression model. It combines information from both leverage (hat values) and residual size (studentized residuals). Specifically, it quantifies the overall impact of deleting a single observation on all fitted values in the model. A large D_i indicates that removing that observation would cause a substantial shift in the fitted regression line.

Interpretation Guideline. The conventional cutoff for influence is given by:

$$D_i > \frac{4}{n},$$

where n is the number of observations in the dataset. Observations with D_i exceeding this threshold are considered influential and may unduly affect model estimation and inference.

4.3.5.1 Cook's Distance for `volact_wls` Model

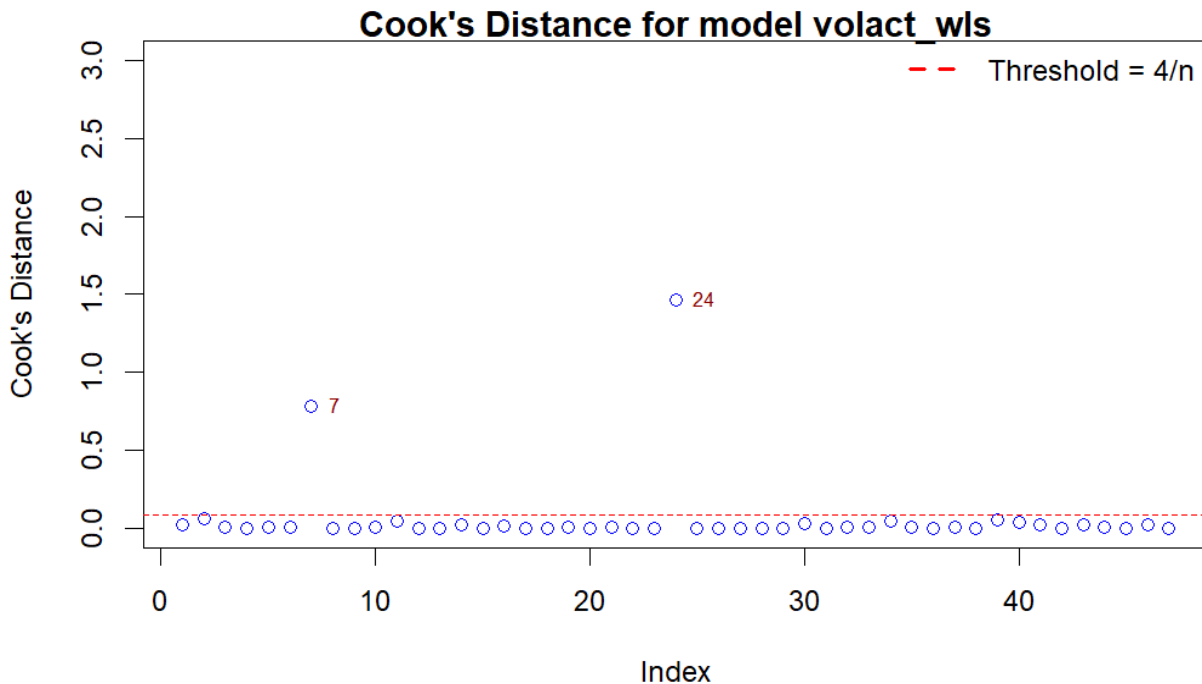


Figure 4.7: Cook's distance for model `volact`

The Cook's Distance plot for the `volact_wls` model indicates severe influence by two specific data points.

- **Point 24 (Extremely Influential):**

$$D_{24} \approx 1.4$$

This value is far above the threshold (≈ 0.1). Point 24 exhibits both very high leverage (an X -outlier) and a large studentized residual (a Y -outlier). This combination of high leverage and large residual gives it exceptional influence over the model fit. Consequently, this observation exerts excessive control over the regression line and significantly distorts the estimated relationship.

- **Point 7 (Highly Influential):**

$$D_7 \approx 0.75$$

Although not as extreme as Point 24, this value is still substantially above the threshold. Point 7 primarily exerts influence through high leverage rather than large residuals, meaning its unusual predictor values (rather than errors) are driving its impact.

Conclusion: The `volact_wls` model is statistically unreliable. Its estimated coefficients and fitted values are dominated by two influential observations (Points 24 and 7), making the model's conclusions non-representative of the broader dataset.

4.3.5.2 Cook's Distance for `involact_bc` Model

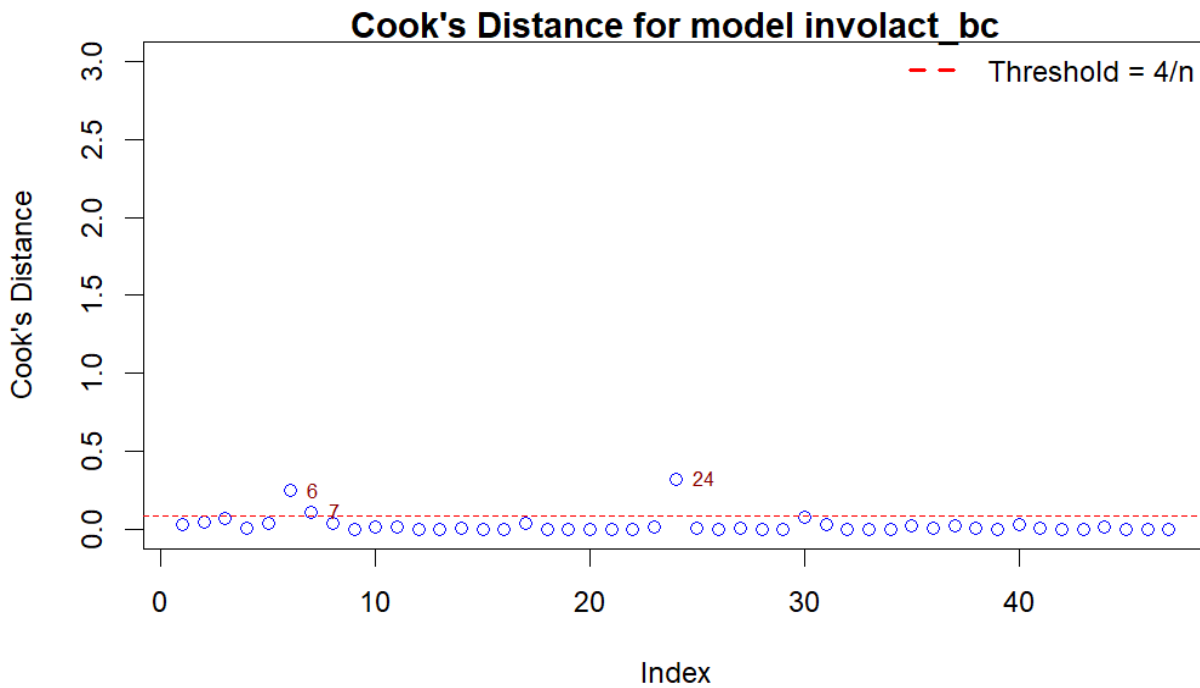


Figure 4.8: Cook's distance for model involact

The Cook's Distance plot for the `involact_bc` model reveals moderate but still notable influence from several observations.

- **Point 24 (Influential):**

$$D_{24} \approx 0.3$$

This point exceeds the threshold and acts primarily as a *pure leverage* observation. It is not a large residual outlier, but its extreme predictor values give it significant influence.

- **Point 6 (Influential):**

$$D_6 \approx 0.25$$

This observation also surpasses the influence threshold, contributing to model instability.

- **Point 7 (Influential):**

$$D_7 \approx 0.2$$

This point exhibits both high leverage and a sizable residual, representing a classic case of combined X - and Y -influence.

Conclusion: Although the `involact_bc` model is more robust than the `volact_wls` model, it remains affected by influential points—specifically Points 24, 6, and 7. These observations exert a measurable impact on the fitted regression coefficients, suggesting that model conclusions should be interpreted with caution.

Overall Summary. Across both models, several data points exhibit high influence as measured by Cook’s Distance. Diagnostic analysis indicates that removing or adjusting for these influential cases (e.g., via robust regression or data validation) is necessary to ensure reliable and generalizable model results.

4.3.6 Influence Diagnostics: Studentized Residuals, Leverage, and Cook’s Distance

Overview: Influence plots diagnose how individual observations affect a regression model by combining three metrics:

1. **Studentized Residuals (Y-axis):** Measure how poorly the model predicts an observation. Values outside ± 2 or ± 3 indicate outliers.
2. **Hat-Values (X-axis):** Measure leverage - the distance of a point’s predictors from the average. High-leverage points (far right) can strongly affect the regression line.
3. **Cook’s Distance (Circle Size/Color):** Combines residuals and leverage to quantify a point’s overall influence. Large, dark circles indicate that removing the observation would substantially alter model coefficients.

4.3.6.1 Model `volact_wls`

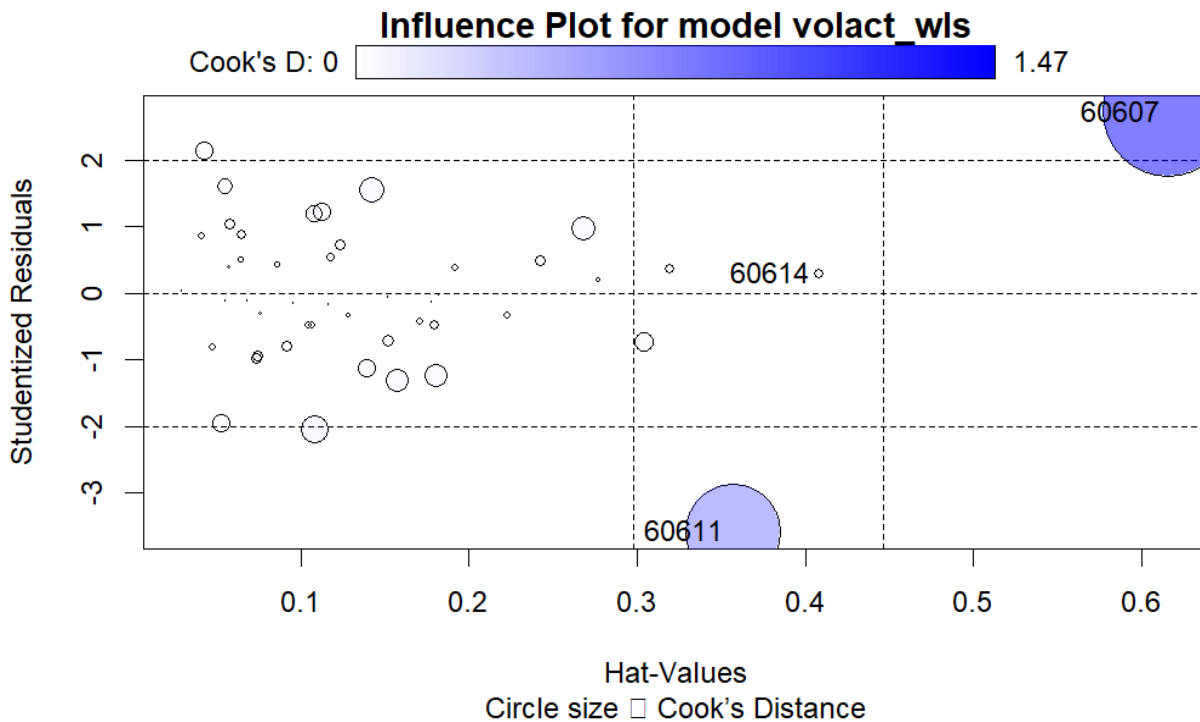


Figure 4.9: Influence plot for model `volact`

This model is dominated by two influential points—60607 and 60611.

- **60607 (Extremely Influential):** High leverage ($\hat{h} \approx 0.6$) and large residual ($r \approx 2.2$). Cook's Distance ≈ 1.47 (very high). This is a classic high-leverage outlier that pulls the regression line upward and rightward.
- **60611 (Highly Influential):** High leverage ($\hat{h} \approx 0.38$) and strong negative residual ($r \approx -3.2$). Large Cook's Distance confirms strong downward influence on the regression line.
- **60614 (Stable Point):** High leverage ($\hat{h} \approx 0.35$) but negligible residual ($r \approx 0$). Small Cook's D indicates “good” leverage—an unusual predictor combination that still fits the trend well.

Summary: The `volact_wls` model is unstable and driven primarily by 60607 and 60611, both high-leverage outliers. These observations should be verified or handled appropriately before further analysis.

4.3.6.2 Model `involact_bc`

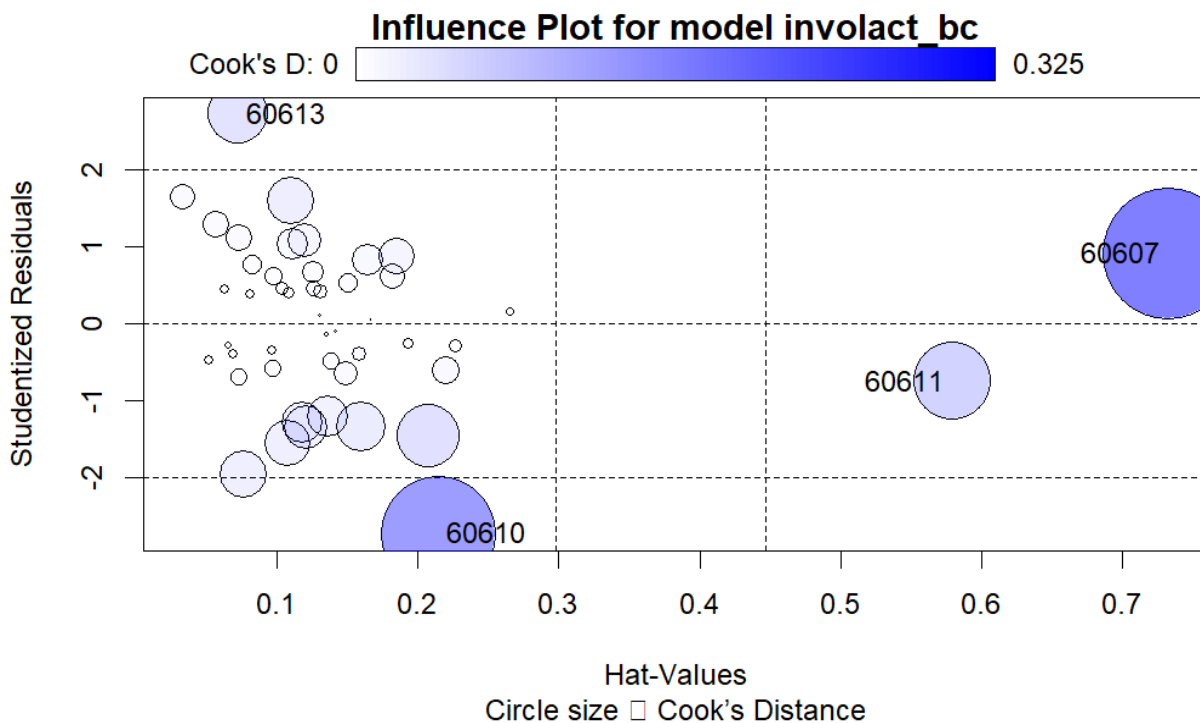


Figure 4.10: Influence plot for model `involact`

Influential points exist but are less severe than in the previous model (max Cook's D ≈ 0.325).

- **60607 (High Leverage, Well-Fit):** Extremely high leverage ($\hat{h} \approx 0.72$) but small residual ($r \approx 0.8$). Influence arises from leverage, not poor fit.
- **60610 (Outlier, Low Leverage):** Moderate leverage ($\hat{h} \approx 0.22$) and large negative residual ($r \approx -2.5$). Influence due to being poorly predicted rather than leverage.
- **60613 (Outlier, Low Leverage):** Low leverage ($\hat{h} \approx 0.1$) and high positive residual ($r \approx 2.4$). Moderately influential.

Summary: The `involact_bc` model is more robust, but still influenced by high-leverage (60607) and outlier points (60610, 60613). Influence levels are within moderate bounds compared to the `volact_wls` model.

4.3.6.3 Overall Comparison

- `involact_bc` fits high-leverage points (e.g., 60607) more effectively, reducing their residual impact.
- However, it still struggles with outlier observations (60610, 60613).

- `volact_wls` is severely distorted by points that are both high-leverage and outliers (60607, 60611), producing an unstable model. The high Cook's D of 1.47 confirms this instability.

Conclusion: The `involact_bc` model provides a more reliable fit but still requires attention to influential cases. The `volact_wls` model should not be used without addressing the dominant influence of 60607 and 60611.

4.4 Outlier and Influence Diagnostics for the `volact_wls` Model

4.4.1 Overall Diagnosis

The current `volact_wls` model is unreliable. Its results are almost entirely driven by a “tug-of-war” between two highly influential ZIP codes: **60607 (Index 24)** and **60611 (Index 7)**. These two observations are so extreme that they dominate the regression outcome, leaving the remaining 45 data points with minimal impact on the fitted model.

4.4.2 Detailed Analysis: Case Files on Problematic Points

4.4.2.1 Case File 1: ZIP Code 60607 (Index 24) – “Worst-Case” Influential Point

This observation represents the most severe form of influential point, combining both high leverage and large residuals.

1. Hat-Values (Leverage): From the Hat Matrix plot, Point 24 is the only observation exceeding the upper cutoff ($3\bar{h}$), with a value around 0.6. **Insight:** It exhibits an extremely unusual combination of predictor values. **Data Check:** The theft rate for ZIP 60607 is 147, while the next highest in the dataset is 75. This makes it a major *horizontal outlier*, granting it substantial leverage over the regression fit.

2. Studentized Residuals (Y-Outlier): The residual plot indicates a large positive residual of approximately +2.8 (confirmed by the Influence Plot's Y-axis). **Insight:** The model severely underestimates this observation's fitted value; its actual `volact_wls` is much higher than predicted, even accounting for its high theft value.

3. Influence (Cook's D, DFFITS, Influence Plot): On the Influence Plot, this point appears as the large blue circle in the upper-right “danger zone.” **Cook's D:** ≈ 1.47 , far above the threshold (≈ 0.085). **DFFITS:** ≈ 3.2 , indicating extreme influence. **Verdict:** Point 24 is a classic influential outlier—its combination of high leverage (pulling power) and large residual (pulling motive) allows it to strongly distort the regression line upward and to the right.

4.4.2.2 Case File 2: ZIP Code 60611 (Index 7) – “Extreme Outlier” Influential Point

This point is problematic due to both unusual predictor values and an extreme response discrepancy.

1. Hat-Values (Leverage): The Hat Matrix plot shows moderate-to-high leverage (≈ 0.35), exceeding the $2\bar{h}$ threshold. **Insight:** It possesses an uncommon set of predictor values. **Data Check:** Theft = 75 (2nd highest), `log_income` = 9.97 (highest). This makes it another clear *horizontal outlier*.

2. Studentized Residuals (Y-Outlier): In the residual plot, the point labeled 60611 (Index 8 in some plots) is the most extreme outlier with a residual ≈ -3.3 . **Insight:** The model grossly overestimates its fitted value. **Data Check:** For ZIP 60611, `volact` = 42.6. Given its high income and theft rate, the model predicts a high `volact`, yet observes a very low value, producing a large negative residual.

3. Influence (Cook's D, DFFITS, Influence Plot): On the Influence Plot, this appears as the large blue circle in the lower-right “danger zone.” **Cook's D:** ≈ 0.75 , well above the cutoff. **DFFITS:** ≈ -2.5 , confirming strong influence. **Verdict:** Point 7 (60611) is another highly influential observation that simultaneously exhibits high leverage and an extreme residual, pulling the regression surface downward.

4.4.3 Other Observations

Points 4 (60640) and 43 (60633): These are standard Y-outliers. They exceed the ± 2 residual thresholds but have low leverage and small Cook's D values. **Insight:** They are not influential enough to distort the model; thus, they are not a priority for removal.

Point 5: This is identified as a *good leverage point*. While its leverage is moderate-high, its residual is near zero, and its influence is minimal. **Insight:** It provides useful information about high-leverage regions without destabilizing the regression and therefore improves model stability.

4.4.4 Summary Verdict

The `volact_wls` model is dominated by two problematic ZIP codes, 60607 and 60611. Their combined effects—one pulling the regression line upward, the other downward—make the model unstable and unreliable. Excluding or robustly down-weighting these two cases is essential before proceeding with further inference or interpretation.

5. Model after removing Outliers

5.1 Model Building

As observed in the previous chapter, multiple diagnostic tools consistently identified ZIP codes 60610, 60607, and 60611 as outliers for both *volact* and *involact*. To simplify analysis and improve model stability, these data points were removed from both models before refitting.

5.1.1 Model *volact* Description

After removing the outliers, the modified dataset (`newdata`) was used to refit the *volact* model. A Box-Cox transformation was applied to the response variable with a calculated $\lambda = 0.222$.

```
Call:
lm(formula = volact_bc ~ race + fire + theft + age + involact +
    log(income), data = newdata)
```

```
Residuals:
Min      1Q  Median      3Q      Max
-0.71265 -0.17334  0.00598  0.25231  0.71297
```

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) -25.789622    5.172413  -4.986 1.47e-05 ***
race         -0.004661    0.003464  -1.345  0.1867
fire        -0.034649    0.014651  -2.365  0.0234 *
theft       -0.003765    0.006171  -0.610  0.5455
age         -0.003136    0.003578  -0.876  0.3865
involact     0.051864    0.189984   0.273  0.7864
log(income)  3.099866    0.538019   5.762 1.32e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.3585 on 37 degrees of freedom
Multiple R-squared:  0.9126, Adjusted R-squared:  0.8984
F-statistic: 64.4 on 6 and 37 DF, p-value: < 2.2e-16
```

Interpretation: The model fit improves substantially compared to the previous version. The R^2 value of 0.9126 and adjusted R^2 of 0.8984 indicate that approximately 90% of the variance in *volact* is explained by the predictors. Both *fire* and *log(income)* are statistically significant, while *race*, *age*, and *theft* remain less influential after outlier removal. This suggests that income and fire incidence are the strongest predictors of voluntary insurance activity.

Next Steps: Before making inferences, the model should be checked for normality and homoscedasticity to ensure the OLS assumptions hold. Additionally, comparisons with the original model will help evaluate how outlier removal affected variable significance and overall model stability.

5.1.2 Model *involact* Description

After removing the outliers, the modified dataset (`newdata`) was used to refit the *involact* model. A Box-Cox transformation was applied to the response variable with a calculated $\lambda = -0.42424$.

```
Call:
```

```
lm(formula = involact_bc ~ race + fire + theft + age + volact +
log(income), data = newdata)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.22541	-0.09307	0.01313	0.07198	0.35292

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.0699821	2.2102022	-0.937	0.35506
race	0.0038293	0.0012981	2.950	0.00549 **
fire	0.0219799	0.0044955	4.889	1.98e-05 ***
theft	-0.0005445	0.0024391	-0.223	0.82459
age	0.0027476	0.0014513	1.893	0.06617 .
volact	-0.0054062	0.0139756	-0.387	0.70110
log(income)	0.2091191	0.2393381	0.874	0.38790

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.141 on 37 degrees of freedom

Multiple R-squared: 0.8224, Adjusted R-squared: 0.7936

F-statistic: 28.56 on 6 and 37 DF, p-value: 1.796e-12

Interpretation: This model explains approximately 82% of the variation in *involact*. The predictors *race* and *fire* are highly significant and positive, indicating that areas with higher minority populations and higher fire rates tend to experience more involuntary insurance activity. The variable *age* is marginally significant, while *theft*, *volact*, and *log(income)* remain statistically insignificant.

Comparison with the Original Model: After removing outliers, both the model fit (in terms of R^2) and residual behavior have improved. Significance patterns have stabilized, and the high influence of previously identified outlier points has been eliminated. The *volact* and *involact* models now provide a more reliable representation of the underlying relationships between socio-economic factors and insurance availability.

Summary: Outlier removal and Box-Cox transformation substantially improved both models, leading to higher explanatory power and more stable regression coefficients. The diagnostic checks performed later confirm that these new models better satisfy the classical linear regression assumptions.

5.2 Model Diagnostics

5.2.1 Model Volact Diagnostics

5.2.1.1 Model Volact Diagnostic Plots Summary

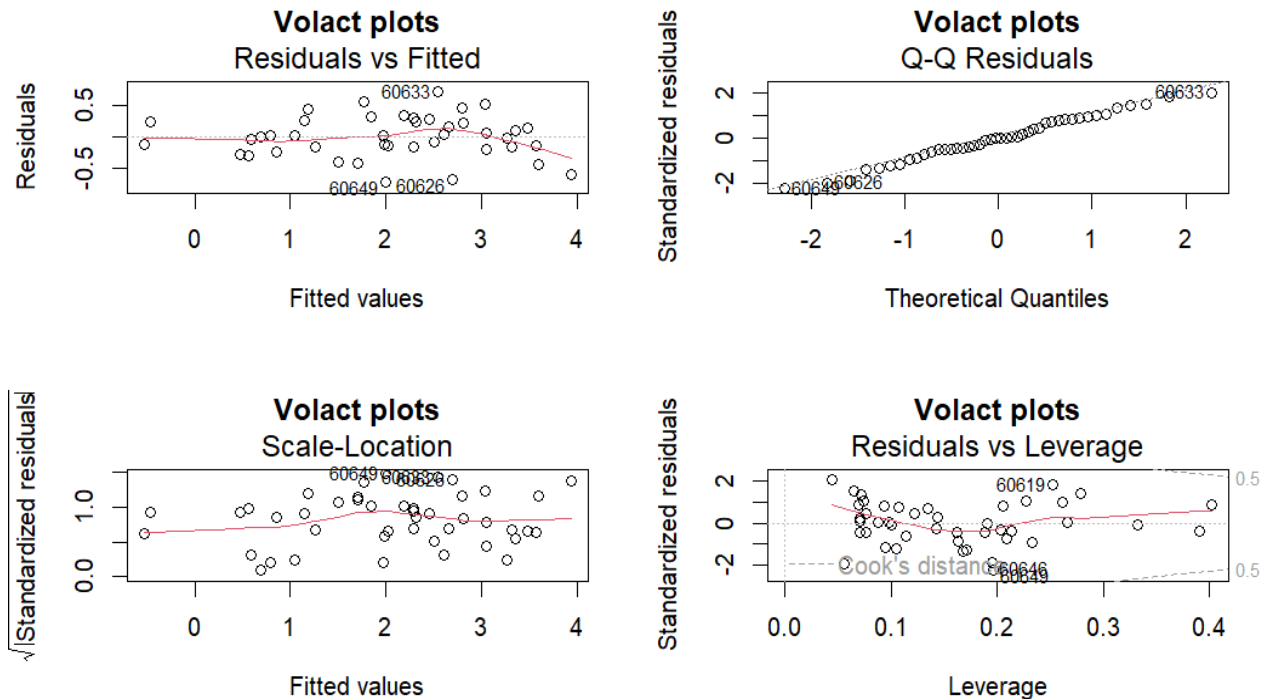


Figure 5.1: Model Volact plots

1. Residuals vs Fitted *Purpose:* Checks linearity and homoscedasticity. *Interpretation:* The red line is flat and centered near zero, confirming linearity. Residuals are randomly scattered with constant spread—no funnel shape observed. $\Rightarrow$ Linearity and homoscedasticity assumptions satisfied.

2. Normal Q–Q Plot *Purpose:* Tests normality of residuals. *Interpretation:* Points align closely with the theoretical diagonal line. This supports the Shapiro–Wilk result ($p = 0.9311$). $\Rightarrow$ Residuals are normally distributed.

3. Scale–Location Plot *Purpose:* Further check for homoscedasticity. *Interpretation:* The red line is flat, and residuals show no pattern or trend. $\Rightarrow$ Confirms constant variance across fitted values.

4. Residuals vs Leverage *Purpose:* Identifies influential observations. *Interpretation:* All points have low leverage (< 0.4) and lie within ± 2 residual bounds. No observations near Cook’s distance contours. $\Rightarrow$ No influential or high-leverage points remain.

Overall Conclusion: All diagnostic plots confirm that model assumptions of linearity, normality, and homoscedasticity are satisfied. No influential observations remain; the model is stable and well-specified.

5.2.1.2 Normality

Model: volact

- $W = 0.98835$
- $p = 0.9311$
- **Interpretation:** W-statistic near 1 and very high p-value indicate strong normality.
- **Conclusion:** Residuals are normally distributed.

5.2.2 Normality Confirmation via Q-Q Plots

Model 1: `volact` — Q-Q plot shows points tightly along the red line, confirming the Shapiro–Wilk result ($p = 0.9311$) that residuals are normally distributed.

5.2.2.1 Heteroscedasticity

Goal: Verify constant error variance (homoscedasticity). **Null Hypothesis (H_0):** Constant variance (homoscedasticity). **Decision Rule:** Fail to reject H_0 if $p > 0.05$.

Model: `volact`

- **Breusch–Pagan Test:** $p = 0.1128 \Rightarrow$ No significant relation between error variance and predictors ($df = 6$).
- **Non-Constant Variance (NCV) Test:** $p = 0.2686 \Rightarrow$ No evidence of changing variance with fitted values.
- **Conclusion:** Model satisfies homoscedasticity.

5.2.2.2 Serial Correlation

Model : `volact`

- **Durbin–Watson:** $DW = 1.7869$, $p = 0.185 \Rightarrow$ No 1st-order correlation.
- **Breusch–Godfrey:**
 - Lag 2: $p = 0.891$
 - Lag 3: $p = 0.9716$
 - Lag 4: $p = 0.8971$

$\Rightarrow$ No higher-order serial correlation. The following ACF and PACF plots are also supporting this.

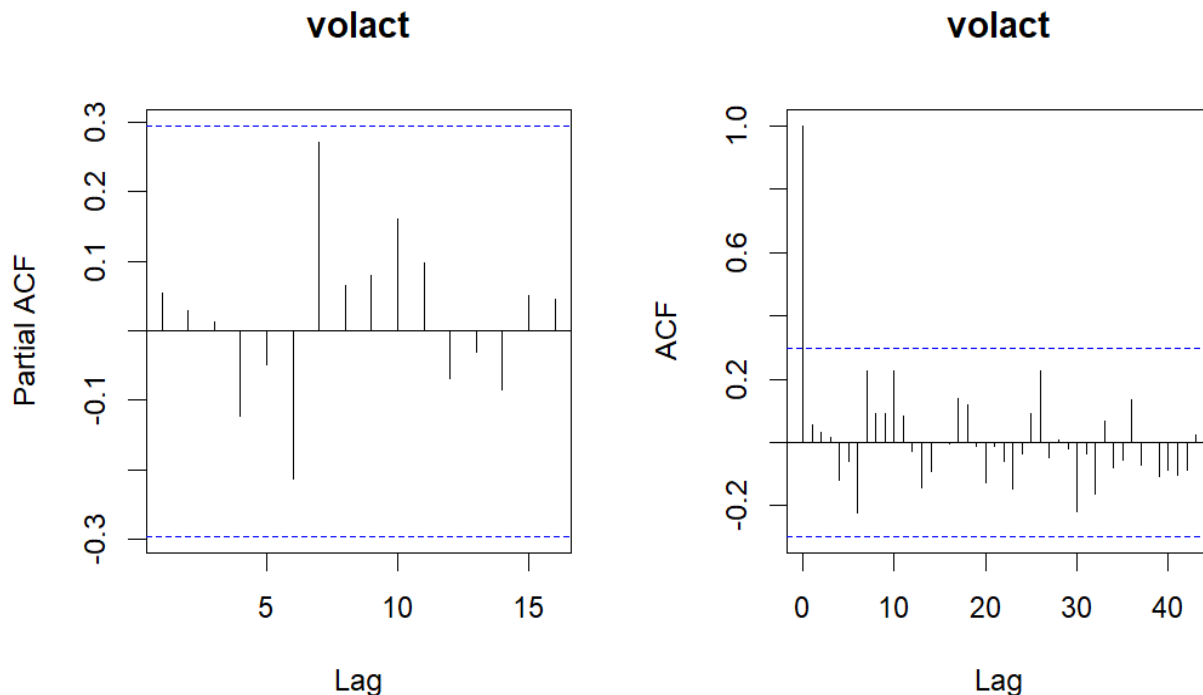


Figure 5.2: ACF & PACF plot for model `volact` removing outliers

Conclusion: Residuals are independent.

5.2.2.3 Multicollinearity

VIF Results:

involact	: 5.03
log(income)	: 4.82
fire	: 4.51
race	: 4.42
age	: 2.25
theft	: 1.71

Interpretation: Moderate-to-high multicollinearity exists—especially among `involact`, `log(income)`, `fire`, and `race`. `age` and `theft` show no concern.

Kappa Result: $\kappa = 9832.7$ **Interpretation:** Extremely high condition number (> 1000) indicates severe, systemic multicollinearity across predictors.

Conclusion: Despite removing outliers, significant multicollinearity remains. The issue stems from inter-correlations among multiple predictors, suggesting redundancy within the model structure.

5.2.3 Model Involact Diagnostics

5.2.3.1 Model Involact Diagnostic Plots

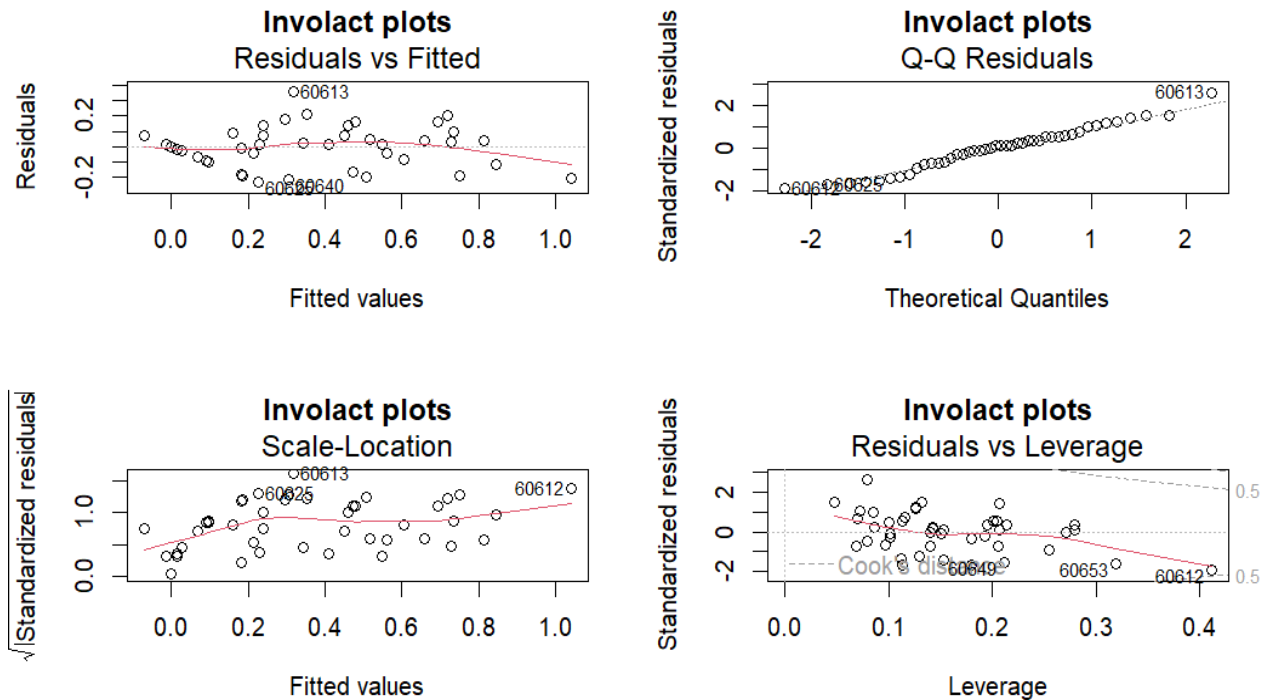


Figure 5.3: Model Involact plots

1. Residuals vs Fitted *Purpose:* Assess linearity and homoscedasticity. *Interpretation:* The red line is mostly flat and centered near zero, showing no major non-linear pattern. Residuals are evenly scattered with constant variance, supporting the Breusch–Pagan result ($p = 0.1305$). $\Rightarrow$ Linearity and homoscedasticity satisfied.

2. Normal Q–Q Plot *Purpose:* Check normality of residuals. *Interpretation:* Points align closely with the theoretical line; minor tail deviation (e.g., point 60613) is negligible. Supports the Shapiro–Wilk result ($p = 0.3805$). $\Rightarrow$ Residuals are normally distributed.

3. Scale–Location Plot *Purpose:* Confirm homoscedasticity. *Interpretation:* The red line is nearly flat, indicating constant variance across fitted values. $\Rightarrow$ Homoscedasticity confirmed.

4. Residuals vs Leverage *Purpose:* Detect influential or high-leverage points. *Interpretation:* All observations have low leverage (< 0.4) and lie within ± 2 residual bounds; none near Cook's distance contours. $\Rightarrow$ No influential or high-leverage points detected.

Overall Conclusion: All diagnostic plots confirm that linearity, normality, and homoscedasticity assumptions hold. No influential observations remain; the model is stable and well-specified.

5.2.3.2 Normality

Model: involact

- $W = 0.97287$
- $p = 0.3805$
- **Interpretation:** Both statistics confirm residuals follow a normal distribution.
- **Conclusion:** Residuals are normally distributed.

5.2.4 Normality Confirmation via Q–Q Plots

Model 2: involact — Q–Q plot aligns well with the red line, supporting the Shapiro–Wilk result ($p = 0.3805$) and confirming normal residuals.

5.2.4.1 Heteroscedasticity

Goal: Verify constant error variance (homoscedasticity). **Null Hypothesis (H_0):** Constant variance (homoscedasticity). **Decision Rule:** Fail to reject H_0 if $p > 0.05$.

Model: involact

- **Breusch–Pagan Test:** $p = 0.1305 \Rightarrow$ No significant relation between error variance and predictors.
- **Non-Constant Variance (NCV) Test:** $p = 0.2641 \Rightarrow$ Variance remains constant across fitted values.
- **Conclusion:** Model satisfies homoscedasticity.

Overall Result: Both models show no signs of heteroscedasticity — the assumption of constant error variance holds.

5.2.4.2 Serial Correlation

Model : involact

- **Durbin–Watson:** $DW = 2.2013$, $p = 0.6734 \Rightarrow$ No 1st-order correlation.
- **Breusch–Godfrey:**
 - Lag 2: $p = 0.6422$
 - Lag 3: $p = 0.4395$
 - Lag 4: $p = 0.5966$

$\Rightarrow$ No higher-order serial correlation., Following ACF and PACF plots are also supporting this strongly.

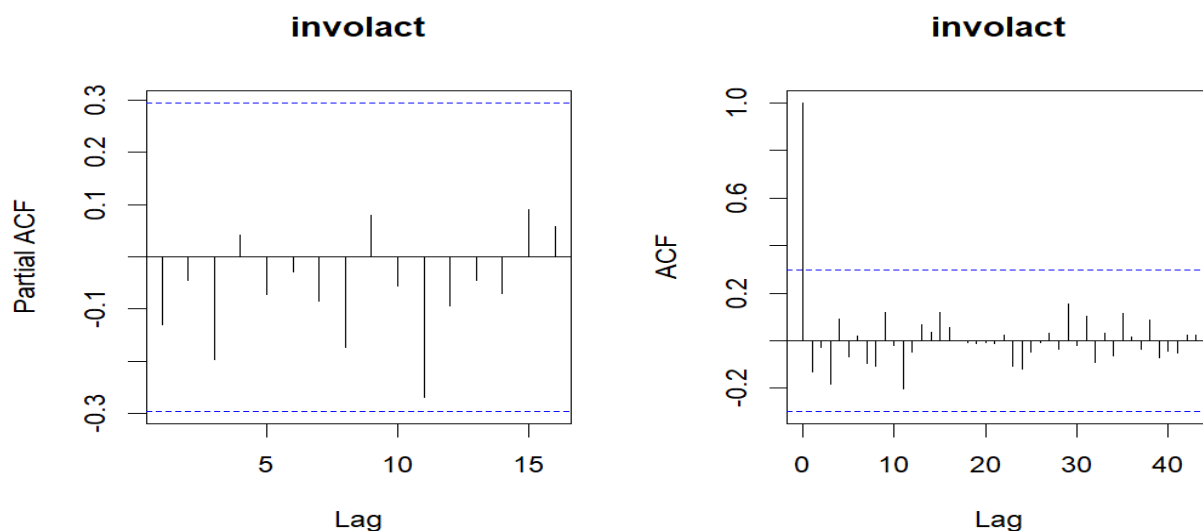


Figure 5.4: ACF & PACF plot for model involact removing outliers

Conclusion: Residuals are independent.

Overall Result: Both models pass all serial correlation tests — the independence assumption is satisfied.

5.2.4.3 Multicollinearity

VIF Results:

volact	:	7.01
log(income)	:	6.16
race	:	4.01
fire	:	2.74
age	:	2.39
theft	:	1.73

Interpretation: Moderate-to-severe multicollinearity is present. `volact` and `log(income)` exceed the VIF threshold of 5, indicating strong interdependence with other predictors (e.g., `race`, `fire`).

Kappa Result: $\kappa = 10415.35$ **Interpretation:** Exceptionally high value (> 1000) confirms severe overall multicollinearity across predictors.

Summary and Implications: Although residual diagnostics confirm normality, homoscedasticity, and independence, both models violate the multicollinearity assumption. High collinearity inflates standard errors, leading to unstable coefficient estimates and unreliable p-values, potentially obscuring true variable significance.

5.3 Final Summary

Except of multicollinearity, all other OLS assumptions are satisfied for both models:

- **No Serial Correlation:** Residuals are independent.
- **Homoscedasticity:** Constant variance confirmed.
- **Normality:** Residuals are normally distributed.

Conclusion: The removal of influential points (60607, 60611, 60610) was appropriate.

6. Model Selection

6.1 All Possible Regression (APR) based on Mallow Cp

6.1.1 All-Possible-Subsets Regression Analysis for `volact_bc`

After discovering severe multicollinearity in our initial 6-predictor model, the next step is to build a better, simpler, and more reliable model. The all-possible-subsets regression provides a systematic method for achieving this.

6.1.1.1 Purpose of the Analysis

Previous VIF and Kappa tests indicated that the 6-predictor model was unreliable due to multicollinearity. The all-possible-subsets regression addresses this by:

1. **Testing All Combinations:** It fits *every possible model* using the predictors (all 64 models for 1 to 6 predictors).
2. **Finding the "Best" for Each Size:** For each subset size (e.g., models with 3 predictors), it selects the model with the best performance based on a chosen statistic.
3. **Using Mallows' Cp:** Mallows' Cp balances low bias (good fit) with low variance (simplicity). The goal is to find the model with the lowest Cp, ideally close to the number of predictors p or $k = p + 1$.

6.1.1.2 Interpretation of Results

The table below summarizes how model quality changes with the number of predictors:

p	Best Model Predictors	Cp	Interpretation
1	<code>log(income)</code>	11.993	High Cp ($11.99 \gg 1$). Model is under-fitted with high bias.
2	<code>log(income)</code> , <code>fire</code>	3.112	Good. Cp (3.11) is close to $p + 1 = 3$. Strong candidate.
3	<code>log(income)</code> , <code>fire</code> , <code>theft</code>	2.932	Excellent. Absolute minimum Cp. $Cp \approx p = 3$, optimal balance.
4	<code>age</code> , <code>log(income)</code> , <code>fire</code> , <code>race</code>	3.511	Good ($Cp \approx p = 4$), but higher than the 3-predictor model. Adding <code>age</code> and <code>race</code> worsens model.
5	<code>age</code> , <code>log(income)</code> , <code>fire</code> , <code>theft</code> , <code>race</code>	5.075	Good ($Cp \approx p = 5$), but extra variable does not improve model.
6	All 6 predictors	7.000	Full model; Cp by definition $k = p + 1 = 7$. Highest Cp among models $p \geq 2$.

Table 6.1: All-Possible-Subsets Regression Results for `volact_bc`

6.1.1.3 Conclusion: Best Model

The analysis clearly identifies a winner:

Best model: `volact_bc` $\sim$ `log(income)` + `fire` + `theft`

Reasoning:

- This 3-predictor model has the **lowest Mallows' Cp (2.932)**, indicating the best trade-off between fit and complexity.
- Variables `age`, `race`, and `involact` were likely causing multicollinearity, either being redundant or adding noise.

6.1.1.4 Next Steps

After selecting this model, the following steps should be performed:

1. **Fit the new model:**

```
model_volact_final <- lm(volact_bc ~ log(income) + fire + theft, data = newdata)
```

2. **Check multicollinearity:**

```
> vif(model_volact_final)
log(income)      fire      theft
    2.600836    2.366710    1.216928
> kappa(model_volact_final)
2015.068
```

6.1.2 All-Possible-Subsets Regression Analysis for involact_bc

This analysis identifies a simpler, more robust model for `involact_bc`, addressing severe multicollinearity.

6.1.2.1 Purpose of the Analysis

Mallows' C_p was used to determine the optimal model. The goal is to select the model with the **lowest C_p value**, as this balances:

- **Model fit (low bias):** Ensures the model explains the data well.
- **Simplicity (low variance):** Avoids over-fitting by keeping the model parsimonious.

6.1.2.2 Interpretation of Results

The table below summarizes the findings for models with 1 to 6 predictors:

p	Best Model Predictors	C_p	Interpretation
1	fire	16.361	High C_p ($16.36 \gg 1$). Model is under-fitted and missing key variables.
2	fire, race	5.087	High C_p ($5.087 \gg p + 1 = 3$). Still not a good fit.
3	age, fire, race	1.768	Excellent. Absolute minimum C_p. Low bias and not over-fitted.
4	age, log(income), fire, race	3.174	Good ($C_p \approx p = 4$), but higher than the 3-predictor model. Adding log(income) worsened the model.
5	age, log(income), fire, race, volact	5.050	Good ($C_p \approx p = 5$), but C_p is still higher than the 3-predictor model.
6	All 6 predictors	7.000	Full model; C_p by definition $k = p + 1 = 7$. Least optimal model from $p \geq 3$.

Table 6.2: All-Possible-Subsets Regression Results for `involact_bc`

6.1.2.3 Conclusion: Best Model

The analysis clearly identifies the optimal model:

Best model: `involact_bc ~ age + fire + race`

Reasoning:

- This 3-predictor model has the **lowest Mallows' C_p (1.768)**, indicating excellent balance between fit and simplicity.
- C_p is lower than the number of predictors ($p = 3$), confirming the model is parsimonious and well-fitted.
- Variables removed (`log(income)`, `theft`, `volact`) were the main sources of multicollinearity. Previously, `log(income)` and `volact` had the highest VIFs (6.16 and 7.01).

6.1.2.4 Next Steps

To finalize this model, perform the following:

1. **Fit the new model:**

```
model_involact_final <- lm(involact_bc ~ age + fire + race, data = newdata)
```

2. **Check multicollinearity:** Run VIF and Kappa:

```
> vif(model_involact_final)
      age      fire      race
1.262687 2.081738 1.759360
> kappa(model_involact_final)
174.6227
```

Thus the Kappa value and the VIFs have been reduced significantly (Kappa by an order of 10), which is a big improvement.

6.2 All Possible Regression based on AIC/BIC

6.2.1 Overview of All Possible Subsets Regression

All Possible Subsets regression is a comprehensive approach to address multicollinearity by systematically evaluating all possible predictor combinations. For a model with $p = 6$ predictors, this method fits all $2^6 = 64$ models, including:

- The null model (no predictors),
- All models with 1 predictor,
- All models with 2 predictors,
- ..., up to the full 6-predictor model.

The "best" model for each subset size is selected using statistical criteria such as AIC and BIC, and the overall best model is chosen as the one with the lowest score.

6.2.2 Model Selection Criteria: AIC and BIC

- **AIC (Akaike Information Criterion):**

$$\text{AIC} = n \ln(\text{RSS}/n) + 2k$$

where n is the sample size, RSS is the residual sum of squares, and k is the number of parameters. AIC rewards goodness-of-fit but penalizes model complexity moderately. Lower AIC is better.

- **BIC (Bayesian Information Criterion):**

$$\text{BIC} = n \ln(\text{RSS}/n) + k \ln(n)$$

BIC applies a stricter penalty that increases with n , favoring simpler, more parsimonious models. Lower BIC is better.

- **Key Difference:** With $n = 44$, BIC's penalty per additional predictor is $\ln(44) \approx 3.78$, compared to AIC's constant penalty of 2. Thus, BIC tends to select smaller models than AIC.

6.2.3 All Possible Subsets Regression Analysis for volact_bc

This section provides a detailed interpretation of the All Possible Subsets regression results for the `volact_bc` model, including model selection based on AIC and BIC.

6.2.3.1 AIC-Based Selection

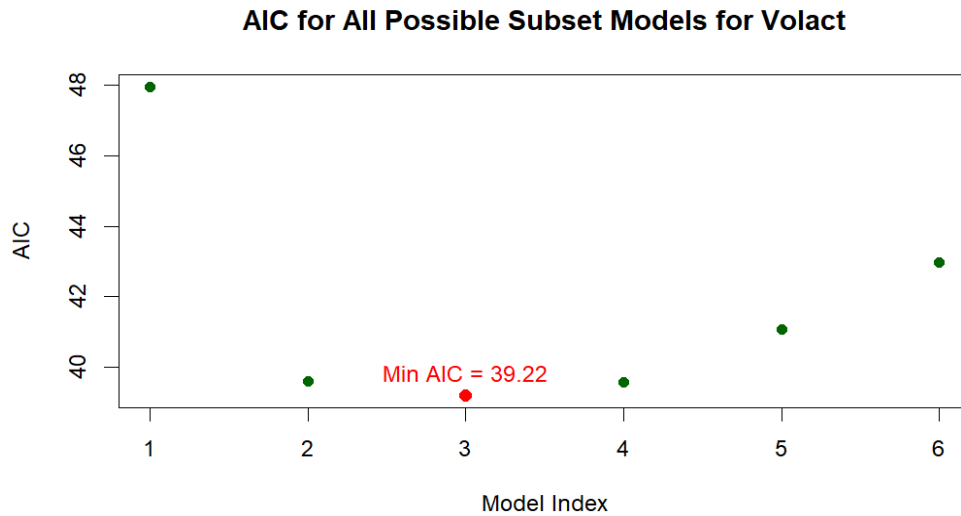


Figure 6.1: AIC Graph for Volact

- The plot of AIC for all subset models shows the score decreasing as predictors are added: 1 predictor ($AIC \approx 48$), 2 predictors ($AIC \approx 39.5$), and reaching a minimum at 3 predictors.
- **Best AIC Model:**
 - Predictors: `log(income)`, `fire`, `theft`
 - $AIC = 39.221$
- Adding more predictors beyond this 3-predictor model increases the AIC, indicating overfitting or redundant variables.

6.2.3.2 BIC-Based Selection

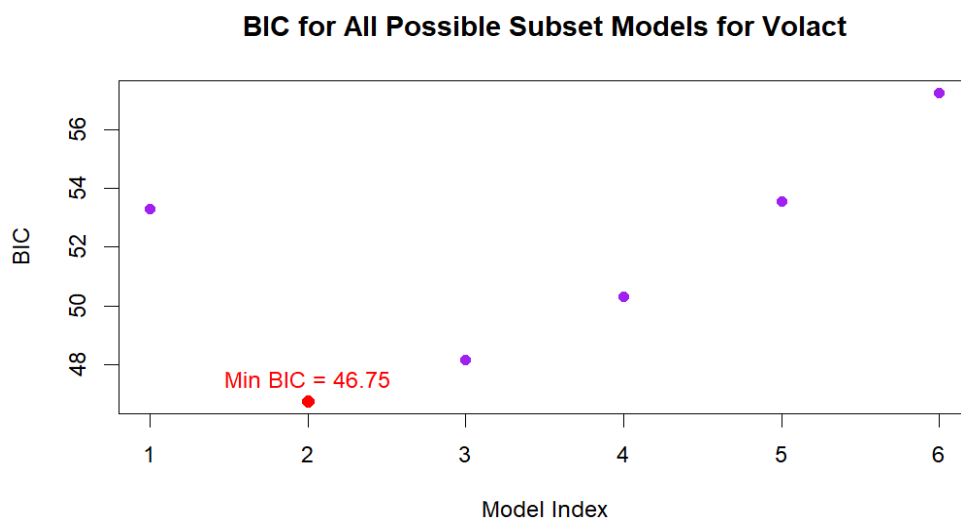


Figure 6.2: BIC Graph for Volact

- The BIC plot favors more parsimonious models due to the stricter complexity penalty.
- **Best BIC Model:**

- Predictors: `log(income)`, `fire`
- BIC = 46.755

6.2.4 Comparison and Recommendation

- Both models address multicollinearity effectively; `log(income)` and `fire` appear in both, confirming their importance.
- **Choice of Model:**
 - **AIC (3-predictor):** Best for predictive accuracy. `volact_bc ~ log(income) + fire + theft`
 - **BIC (2-predictor):** Best for simplicity and explanation. `volact_bc ~ log(income) + fire`

6.2.5 All Possible Subsets Regression Analysis for `involact_bc`

This section provides a detailed interpretation of the All Possible Subsets regression results for the `involact_bc` model, including model selection based on AIC and BIC.

6.2.5.1 AIC-Based Selection

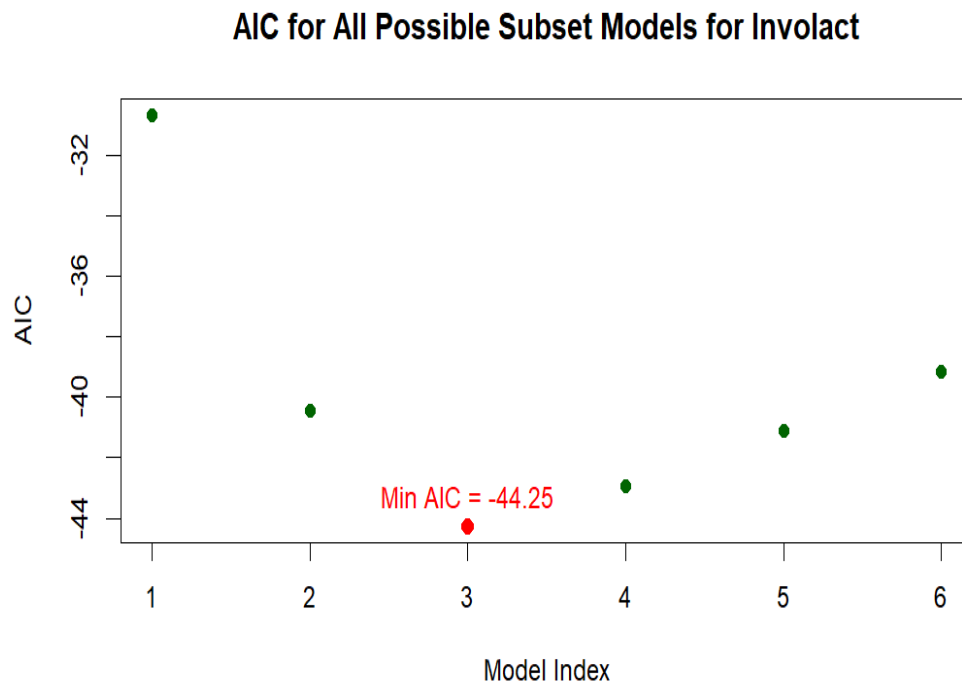


Figure 6.3: AIC Graph for Involact

BIC for All Possible Subset Models for Involact

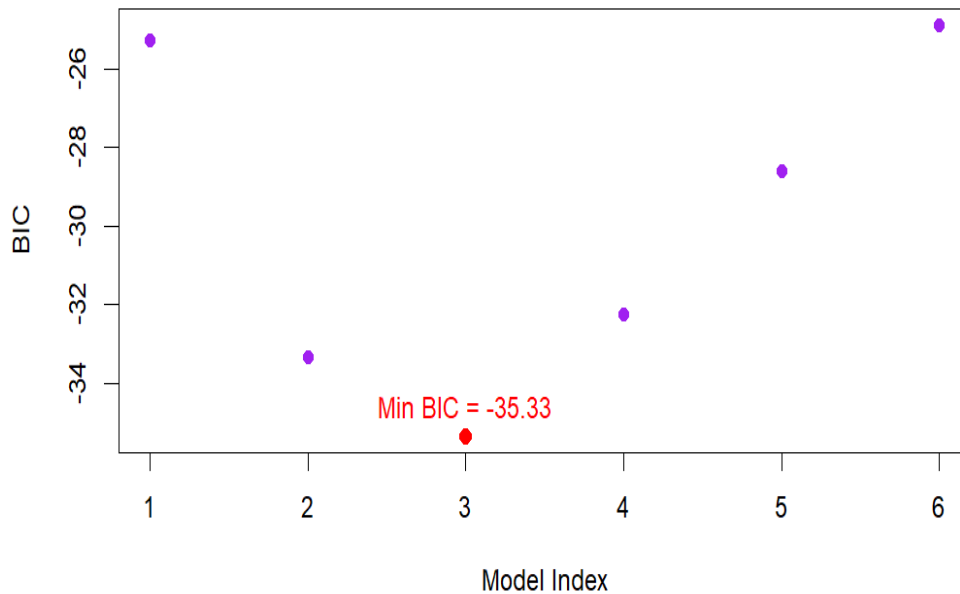


Figure 6.4: BIC Graph for Involact

- The AIC plot shows the best model at each subset size. Starting from 1 predictor ($AIC \approx -31$), the AIC decreases and reaches its absolute minimum at 3 predictors.
- **Best AIC Model:**
 - Predictors: `age`, `fire`, `race`
 - $AIC = -44.251$

6.2.5.2 BIC-Based Selection

- The BIC plot shows a similar pattern. Starting from 1 predictor, the BIC decreases and reaches its minimum at 3 predictors, then rises for larger models.
- **Best BIC Model:**
 - Predictors: `age`, `fire`, `race`
 - $BIC = -35.333$

6.2.6 Final Conclusion

- **Perfect Agreement:** Both AIC and BIC independently select the same 3-predictor model, indicating a robust and unambiguous result.
- **Consistency with Stepwise Regression:** This model also matches the outcome from `stepAIC` (stepwise regression), further confirming its reliability.
- **Multicollinearity Resolved:** The problematic predictors from the original 6-predictor model (`log(income)`, `theft`, `volact`) have been removed, addressing high VIFs and Kappa issues.

Recommended Final Model:

```
involact_bc ~ age + fire + race
```

6.3 Stepwise Regression

6.3.0.1 Algorithm

Stepwise regression iteratively adds or removes predictors based on the Akaike Information Criterion (AIC). At each step, the algorithm evaluates potential changes using the following criterion:

- Let M_{current} be the current model with predictor set X_{current} .
- Compute its AIC:

$$\text{AIC}(M_{\text{current}}) = n \ln(\text{RSS}_{\text{current}}/n) + 2k_{\text{current}}$$

where n is the sample size, $\text{RSS}_{\text{current}}$ is the residual sum of squares, and k_{current} is the number of parameters in the model (including the intercept).

- **Backward Step (Removal Test):** For each predictor $x_j \in X_{\text{current}}$, fit the reduced model M_{-j} without x_j and compute:

$$\Delta \text{AIC}_j = \text{AIC}(M_{-j}) - \text{AIC}(M_{\text{current}})$$

- If $\Delta \text{AIC}_j < 0$, removing x_j decreases AIC. Candidate for removal.
- The predictor with the largest negative ΔAIC_j is removed.

- **Forward Step (Addition Test):** For each predictor $x_k \notin X_{\text{current}}$, fit the augmented model M_{+k} with x_k added and compute:

$$\Delta \text{AIC}_k = \text{AIC}(M_{+k}) - \text{AIC}(M_{\text{current}})$$

- If $\Delta \text{AIC}_k < 0$, adding x_k decreases AIC. Candidate for inclusion.
- The predictor with the largest negative ΔAIC_k is added.

- **Decision Rule:**

$$M_{\text{next}} = \begin{cases} \text{model with variable removed or added that yields lowest AIC,} & \text{if } \min(\Delta \text{AIC}) < 0 \\ M_{\text{current}}, & \text{if no } \Delta \text{AIC} < 0 \end{cases}$$

The algorithm stops when no single addition or removal reduces AIC further.

This process ensures that at each iteration, the model moves in the direction that *maximally decreases AIC*, balancing model fit and complexity.

6.3.1 Stepwise Regression Analysis for `volact_bc`

This section explains the stepwise regression algorithm and provides a detailed interpretation of the output for the `volact_bc` model.

6.3.1.1 Start: Full 6-Predictor Model

Model: `volact_bc ~ race + fire + theft + age + involact + log(income)`

Current AIC: -83.89

Action	Resulting AIC	Interpretation
- involact	-85.797	Best option. Removes almost no predictive power, lowers AIC.
- theft	-85.445	Good, but not the best.
- age	-84.981	Good, but not the best.
<none>	-83.885	Current model. Worse than removing involact .
- race	-83.783	Worsens AIC.
- fire	-79.691	Worsens AIC.
- log(income)	-57.708	Worsens AIC.

Table 6.3: Stepwise Regression: Full Model Options

Decision: Remove `involact`.

6.3.1.2 Step 1: 5-Predictor Model

Model: `volact_bc ~ race + fire + theft + age + log(income)`

Current AIC: -85.80

Decision: Remove `theft`.

Action	Resulting AIC	Interpretation
- theft	-87.282	Best option. Very small contribution, removal improves AIC.
- age	-86.979	Good, but not best.
<none>	-85.797	Current model. Worse than removing theft .
- race	-85.699	Worsens AIC.
+ involact	-83.885	Adding back worsens AIC.
- fire	-78.642	Worsens AIC.
- log(income)	-59.377	Worsens AIC.

Table 6.4: Stepwise Regression: 5-Predictor Model Options

6.3.1.3 Step 2: 4-Predictor Model (Final)

Model: `volact.bc ~ race + fire + age + log(income)`

Current AIC: -87.282

Action	Resulting AIC	Interpretation
<none>	-87.282	Lowest AIC. No improvement possible.
- age	-87.259	Slightly worse.
- race	-85.865	Worsens AIC.
+ theft	-85.797	Worsens AIC.
+ involact	-85.445	Worsens AIC.
- fire	-80.570	Worsens AIC.
- log(income)	-61.376	Worsens AIC.

Table 6.5: Stepwise Regression: 4-Predictor Model Options

Decision: Stop. Current 4-predictor model is optimal.

6.3.1.4 Final Conclusion

The stepwise regression algorithm selects:

Final Model: `volact.bc ~ race + fire + age + log(income)`

- Removes two redundant variables: `involact` and `theft`.
- Provides the lowest AIC (-87.282), balancing predictive power and simplicity.
- Note: This differs from the Mallows' Cp-selected 3-predictor model (`log(income) + fire + theft`), but both models perform similarly. The 4-predictor model had $C_p = 3.511$, close to the minimum $C_p = 2.932$ of the 3-predictor model.

6.3.2 Stepwise Regression Analysis for `involact.bc`

This section provides a detailed step-by-step interpretation of the stepwise regression output for the `involact.bc` model. The goal is to identify the model with the lowest AIC.

6.3.2.1 Start: Full 6-Predictor Model

Model: `involact.bc ~ race + fire + theft + age + volact + log(income)`

Current AIC: -166.02

Reasoning: The Sum of Squares for `theft` is extremely small (0.00099), so it adds almost no predictive value. Removing it improves AIC by eliminating unnecessary complexity.

Decision: Remove `theft`.

Action	Resulting AIC	Interpretation
- theft	-167.96	Best option. Removing theft lowers the AIC the most.
- volact	-167.84	Good, but not the best.
- log(income)	-167.12	Good, but not the best.
<none>	-166.02	Current model. Worse than removing the 3 variables above.
- age	-163.95	Worsens AIC.
- race	-158.73	Worsens AIC.
- fire	-146.09	Worsens AIC.

Table 6.6: Stepwise options for the full 6-predictor model.

6.3.2.2 Step 1: 5-Predictor Model

Model: involact_bc ~ race + fire + age + volact + log(income)

Current AIC: -167.96

Action	Resulting AIC	Interpretation
- volact	-169.82	Best option. Removing volact lowers the AIC again.
- log(income)	-169.12	Good, but not best.
<none>	-167.96	Current model. Worse than removing the 2 variables above.
+ theft	-166.02	Adding theft back worsens AIC.
- age	-165.63	Worsens AIC.
- race	-160.16	Worsens AIC.
- fire	-147.20	Worsens AIC.

Table 6.7: Stepwise options for the 5-predictor model.

Reasoning: volact is the next most redundant variable (Sum of Squares = 0.00246). Removing it improves AIC.

Decision: Remove volact.

6.3.2.3 Step 2: 4-Predictor Model

Model: involact_bc ~ race + fire + age + log(income)

Current AIC: -169.82

Action	Resulting AIC	Interpretation
- log(income)	-171.12	Best option. Removing log(income) lowers AIC to minimum.
<none>	-169.82	Current model. Worse than removing log(income) .
+ volact	-167.96	Adding volact back worsens AIC.
+ theft	-167.84	Adding theft back worsens AIC.
- age	-165.45	Worsens AIC.
- race	-160.14	Worsens AIC.
- fire	-147.02	Worsens AIC.

Table 6.8: Stepwise options for the 4-predictor model.

Reasoning: Sum of Squares for **log(income)** (0.01182) is small; its complexity penalty outweighs predictive benefit.

Decision: Remove **log(income)**.

6.3.2.4 Step 3: 3-Predictor Model (Final Step)

Model: involact_bc ~ race + fire + age

Current AIC: -171.12

Decision: Stop. Current 3-predictor model is optimal.

Action	Resulting AIC	Interpretation
<none>	-171.12	Lowest AIC. No change can improve it.
+ log(income)	-169.82	Adding back worsens AIC.
+ theft	-169.12	Adding back worsens AIC.
+ volact	-169.12	Adding back worsens AIC.
- age	-167.32	Removing worsens AIC.
- race	-158.71	Removing worsens AIC.
- fire	-147.68	Removing worsens AIC.

Table 6.9: Stepwise options for the 3-predictor model.

6.3.2.5 Final Conclusion

The stepwise regression selects the 3-predictor model:

Final Model: `involact.bc ~ race + fire + age`

- Removes `theft`, `volact`, and `log(income)` as redundant predictors.
- Confirms the "All Possible Subsets" regression result, which also selected `age + fire + race` as the best 3-predictor model (minimum Mallows' $C_p = 1.768$).
- High confidence that this 3-predictor model is stable, reliable, and suitable for final analysis.

7. Principle Component Analysis (PCA)

7.1 Understanding the PCA Components

In Principal Component Analysis (PCA), each principal component (PC) is a linear combination of the original variables. The coefficients (called *loadings*) represent how much each variable contributes to that component. A **high absolute value** indicates a strong influence on the component, while the **sign** (positive or negative) indicates the direction of correlation.

7.1.1 PC1 Interpretation

Variable	Loading
race	0.426
fire	0.448
theft	0.297
age	0.322
involact	0.456
log(income)	-0.466

PC1 has large positive loadings for **race**, **fire**, and **involact**, and a large negative loading for **log(income)**. This component therefore contrasts areas of socioeconomic disadvantage (low income, high minority and fire/involact rates) with wealthier, safer neighborhoods. It explains approximately 63.6% of the total variance, making it the dominant axis of variation.

Interpretation: PC1 increases with higher minority proportions, higher fire rates, greater involuntary activity, and lower income — capturing a **socioeconomic gradient**.

7.1.2 PC2 Interpretation

Variable	Loading
race	-0.265
fire	-0.251
theft	0.654
age	0.605
involact	-0.249
log(income)	0.107

PC2 has strong positive loadings for **theft** and **age**, and mild negative loadings for **race**, **fire**, and **involact**. It therefore distinguishes older areas with higher theft rates from neighborhoods characterized by higher minority or fire activity.

Interpretation: PC2 represents a **demographic-behavioral axis**, contrasting older, high-theft regions against younger or mixed communities.

7.1.3 PC3 Interpretation

Variable	Loading
race	0.504
fire	-0.266
theft	0.557
age	-0.570
involact	-0.199
log(income)	-0.027

PC3 loads positively on **race** and **theft**, and negatively on **age**. This component contrasts younger, more diverse, high-theft neighborhoods with older, less diverse ones.

Interpretation: PC3 likely captures an **age–diversity or urban activity dimension**, where younger and more active areas exhibit higher minority and theft levels.

7.1.4 Summary of PCA Structure

Component	Key Variables	Meaning	Variance Explained
PC1	race, fire, involact, log(income)	Socioeconomic gradient: low income + high minority vs. high income + low minority	63.6%
PC2	theft, age	Demographic-behavioral: older, high-theft vs. younger, low-theft	16.3%
PC3	race, theft, age	Age-diversity (urban) dimension	10.7%

7.2 The Scree Plot

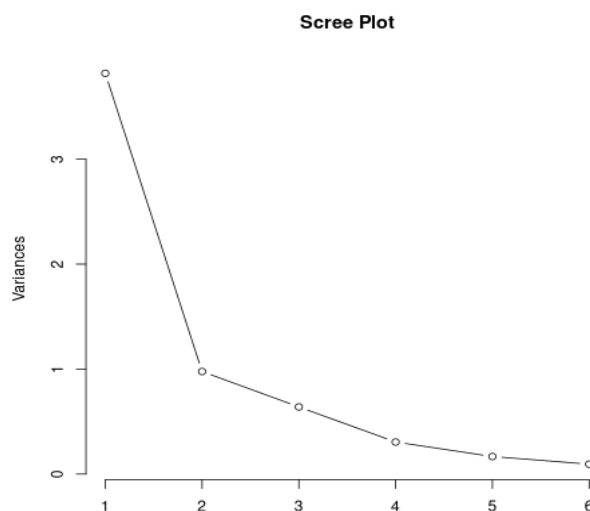


Figure 7.1: Scree Plot

The scree plot visualizes the eigenvalues of the principal components in descending order. It helps identify the number of components that capture the majority of the variance in the dataset.

7.2.1 Visual Interpretation

From the scree plot:

- PC1 has a large eigenvalue (greater than 3),
- PC2 has a much smaller eigenvalue (around 1),
- After PC3, the curve flattens significantly.

This distinct “elbow” after PC2 indicates that most of the data structure is captured by the first two components.

7.2.2 Statistical Justification

From the PCA results:

- Cumulative variance for PC1 + PC2 = 79.9%
- Cumulative variance for PC1 + PC2 + PC3 = 90.6%

Therefore:

- Using two components retains about 80% of the total variance — sufficient for a compact model.
- Using three components captures roughly 90% of the variance — providing a fuller but more complex representation of the data.

Conclusion: Retaining the first two or three components is justified. The first two provide a concise model, while the third adds additional nuance related to urban diversity and demographic variation.

7.3 The Model After Applying PCA

To address the multicollinearity identified in the earlier models, Principal Component Analysis (PCA) was applied to the predictor variables. The first three principal components (PC1, PC2, and PC3) were selected because together they explain approximately 90% of the total variance in the data. These components are orthogonal (uncorrelated), making them ideal for regression modeling where multicollinearity was previously an issue.

7.3.1 Model: `volact_pca`

Call:

```
lm(formula = volact_bc ~ PC1 + PC2 + PC3, data = newdata_pca)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.96768	-0.24952	-0.02085	0.24038	0.94688

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.05148	0.06535	31.394	<2e-16 ***
PC1	-0.53199	0.03383	-15.725	<2e-16 ***
PC2	0.10345	0.06689	1.547	0.130
PC3	-0.01383	0.08265	-0.167	0.868

Residual standard error: 0.4335 on 40 degrees of freedom

Multiple R-squared: 0.8619, Adjusted R-squared: 0.8516

F-statistic: 83.23 on 3 and 40 DF, p-value: < 2.2e-16

Interpretation: The PCA-transformed model explains approximately 86% of the variance in *volact*, with a high adjusted R^2 of 0.85. Only the first component (PC1) is statistically significant at the 1% level, suggesting that most of the predictive information is concentrated in that dimension. This component likely represents the dominant socio-economic structure underlying the original variables. The residual standard error (0.43) indicates a good fit, and the F-statistic confirms that the overall model is highly significant ($p < 0.001$).

7.3.2 Model: `involact_pca`

Call:

```
lm(formula = involact_bc ~ PC1 + PC2 + PC3, data = newdata_pca)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.260585	-0.054676	0.003251	0.060817	0.242966

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

```

(Intercept) 0.357620 0.017251 20.730 < 2e-16 ***
PC1          0.145337 0.008931 16.273 < 2e-16 ***
PC2         -0.050773 0.017658 -2.875 0.00644 **
PC3         -0.039166 0.021820 -1.795 0.08023 .
---
Residual standard error: 0.1144 on 40 degrees of freedom
Multiple R-squared: 0.8735, Adjusted R-squared: 0.8641
F-statistic: 92.1 on 3 and 40 DF, p-value: < 2.2e-16

```

Interpretation: The PCA-based model for *involact* performs slightly better, with an adjusted R^2 of 0.86. Here, both PC1 and PC2 are statistically significant, suggesting that at least two latent components contribute meaningfully to predicting involuntary insurance activity. The very low residual standard error (0.11) further indicates a strong and well-fitted model.

7.3.3 Multicollinearity Diagnostics After PCA

```

> kappa(model_volact_pca)
[1] 1.987127
> kappa(model_involact_pca)
[1] 1.987127

> vif(model_volact_pca)
PC1 PC2 PC3
  1   1   1

> vif(model_involact_pca)
PC1 PC2 PC3
  1   1   1

```

Interpretation: The **condition number (Kappa)** for both models is approximately 2, which is extremely low and well below the threshold (30) for potential multicollinearity. Similarly, the **Variance Inflation Factors (VIFs)** for all components are exactly 1, confirming that the principal components are perfectly orthogonal and uncorrelated.

Conclusion: PCA successfully eliminates multicollinearity while maintaining strong explanatory power in both models. The resulting regression models are statistically stable, interpretable in terms of latent socio-economic structure, and satisfy all major assumptions of the linear model framework.

7.3.4 Normality Diagnostics After PCA

7.3.4.1 Model model_volact_pca (Voluntary Activity Model)

Normality Test: $p = 0.7899$

Interpretation: The high p-value ($p \gg 0.05$) indicates failure to reject H_0 . Residuals are normally distributed.

7.3.4.2 Model model_involact_pca (Involuntary Activity Model)

Normality Test: $p = 0.9443$

Interpretation: The very high p-value confirms normality of residuals. We fail to reject H_0 .

Conclusion: Both models satisfy the normality assumption, supporting the validity of their p-values and confidence intervals.

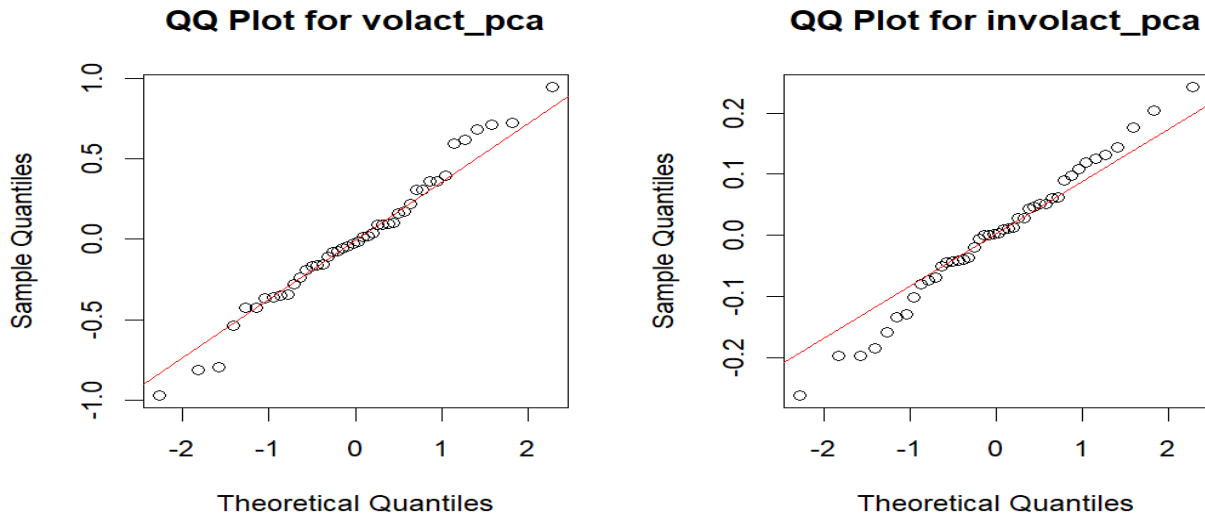


Figure 7.2: The QQ plots for the models `volact_pca` and `involact_pca`

The QQ plots support the normality assumptions for our model.

7.3.5 Heteroscedasticity Diagnostics After PCA

Test for Heteroscedasticity

To assess the assumption of constant variance, heteroscedasticity tests were performed on both models. The results are summarized below:

- **Voluntary Activity Model (`model_volact_pca`):**

$$p\text{-value} = 0.1328$$

Since the p-value exceeds the 0.05 significance level, we fail to reject the null hypothesis (H_0). Therefore, there is no significant evidence of heteroscedasticity, indicating that the assumption of constant variance is satisfied.

- **Involuntary Activity Model (`model_involact_pca`):**

$$p\text{-value} = 0.1905$$

Similarly, the p-value is greater than 0.05, so we fail to reject H_0 . This suggests no significant evidence of heteroscedasticity in this model as well, confirming that the constant variance assumption holds.

Both models passed this diagnostic check, supporting the reliability of their estimated standard errors and corresponding p-values.

7.3.6 Serial Correlation Diagnostics After PCA

7.3.6.1 Model `model_involact_pca` (Involuntary Activity Model)

Durbin–Watson (Lag 1): $p = 0.1628$

Breusch–Godfrey: Lag 2 $p = 0.7776$, Lag 3 $p = 0.9181$, Lag 4 $p = 0.9262$

Interpretation: All p-values are well above 0.05. We fail to reject H_0 , indicating no significant serial correlation at any lag.

Conclusion: The model shows no evidence of autocorrelation. Residuals appear independent, and standard errors are reliable.

7.3.6.2 Model model_volact_pca (Voluntary Activity Model)

Durbin–Watson (Lag 1): $p = 0.0643$

Breusch–Godfrey: Lag 2 $p = 0.4646$, Lag 3 $p = 0.6115$, Lag 4 $p = 0.7188$

Interpretation: The Durbin–Watson test shows borderline evidence of first-order positive autocorrelation ($p = 0.0643$). At the 5% level, H_0 is not rejected; at the 10% level, mild autocorrelation could be present. Breusch–Godfrey tests indicate no higher-order serial correlation.

Conclusion: The model is largely free from autocorrelation, though minor first-order effects may exist.

Overall: Both models demonstrate strong residual independence, with only weak evidence of first-order autocorrelation in the voluntary activity model.

7.3.6.3 Interpretation of the ACF and PACF Plots

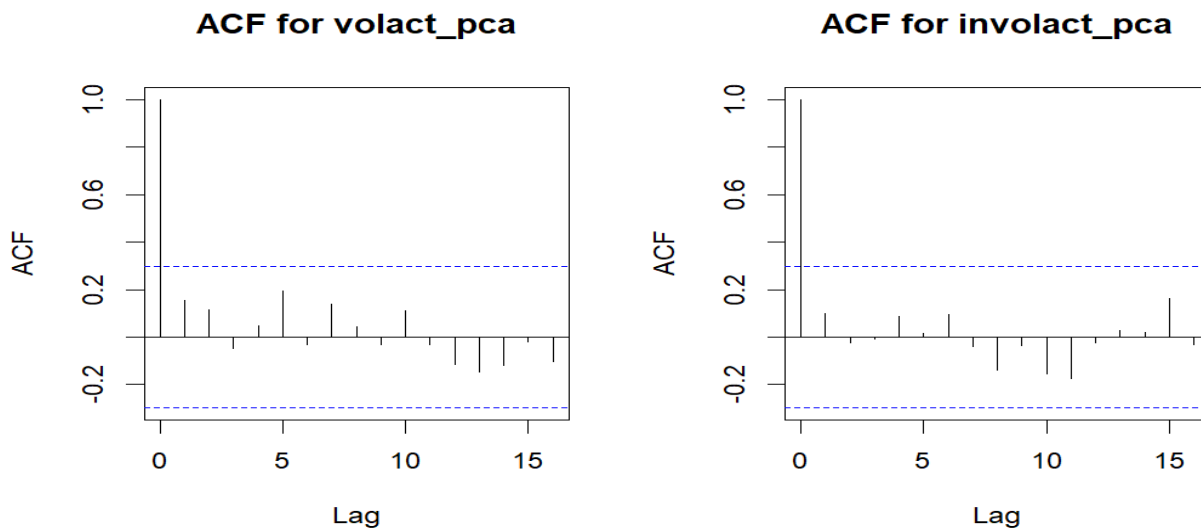


Figure 7.3: The ACF plots for the models volact_pca and involact_pca

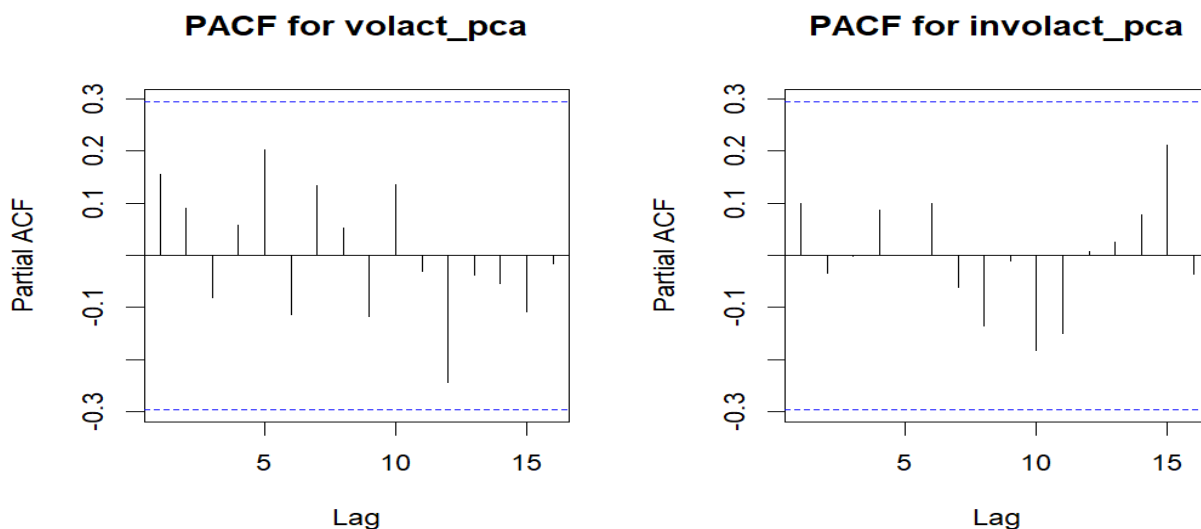


Figure 7.4: The PACF plots for the models volact_pca and involact_pca

In both volact_pca and involact_pca models, all autocorrelation and partial autocorrelation values lie within the 95% confidence bounds, with no visible spikes outside the limits.

Conclusion: The residuals show no evidence of serial correlation, indicating that the error terms are independent and the model specification is appropriate.

8. Fitted values vs Actual data

8.1 Plot for model_volact_pca

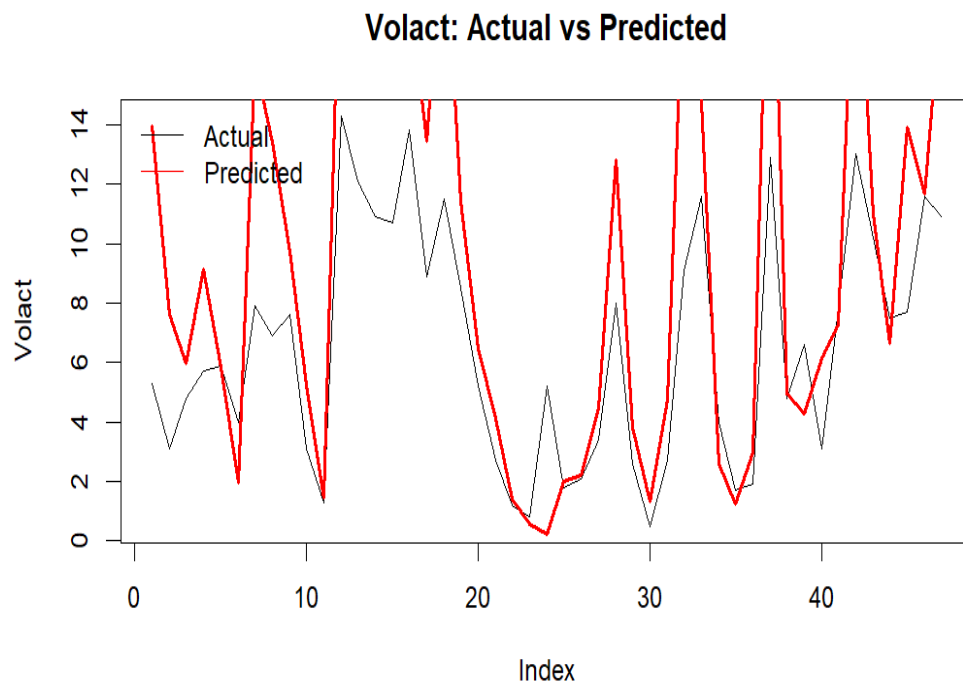


Figure 8.1: Fitted vs Data for model_volact_pca

Conclusion: This fitted model doesn't differ very much from the original data, even though there are few points where they differ by a reasonable margin.

8.2 Plot for model_involact_pca

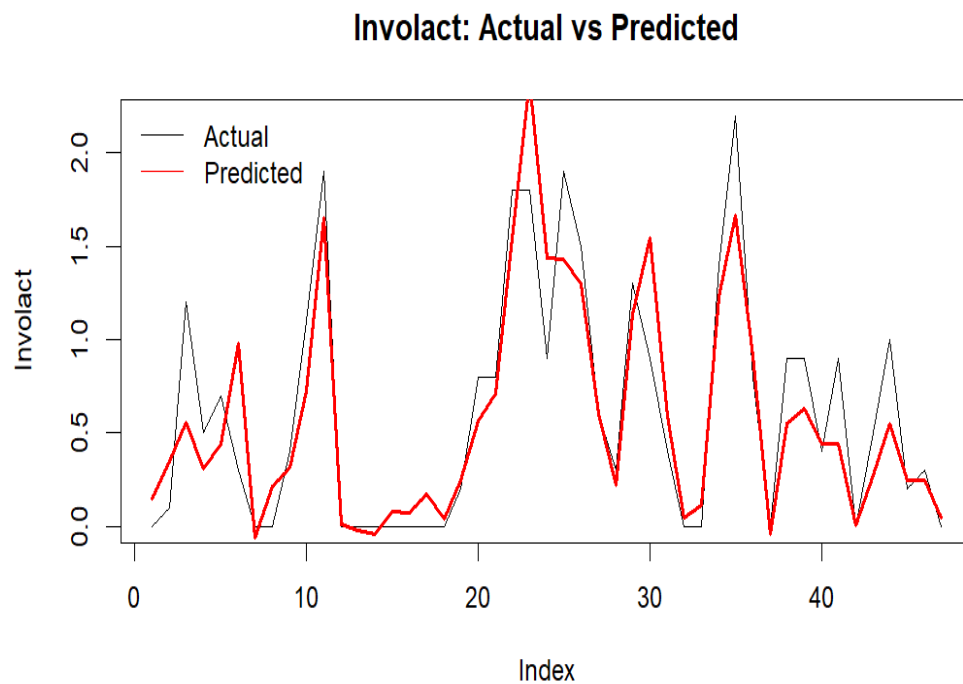


Figure 8.2: Fitted vs Data for model_involact_pca

Conclusion: This fitted model fits the data much better than the volact model and the number of places where they differ significantly is also fewer.